

```

import numpy as np # linear algebra
import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv)
import seaborn as sns
sns.set(style="whitegrid")
import matplotlib.pyplot as plt
from collections import Counter
%matplotlib inline

import os
for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))

import warnings
warnings.filterwarnings('ignore')

fifa19 = pd.read_csv(r'C:\Users\hp\Downloads\FIFA.csv', index_col=0)
fifa19

```

	ID	Name	Age	\
0	158023	L. Messi	31	
1	20801	Cristiano Ronaldo	33	
2	190871	Neymar Jr	26	
3	193080	De Gea	27	
4	192985	K. De Bruyne	27	
...	...	...	...	
18202	238813	J. Lundstram	19	
18203	243165	N. Christoffersson	19	
18204	241638	B. Worman	16	
18205	246268	D. Walker-Rice	17	
18206	246269	G. Nugent	16	

	Photo	Nationality	\
0	https://cdn.sofifa.org/players/4/19/158023.png	Argentina	
1	https://cdn.sofifa.org/players/4/19/20801.png	Portugal	
2	https://cdn.sofifa.org/players/4/19/190871.png	Brazil	
3	https://cdn.sofifa.org/players/4/19/193080.png	Spain	
4	https://cdn.sofifa.org/players/4/19/192985.png	Belgium	
...	...	...	
18202	https://cdn.sofifa.org/players/4/19/238813.png	England	
18203	https://cdn.sofifa.org/players/4/19/243165.png	Sweden	
18204	https://cdn.sofifa.org/players/4/19/241638.png	England	
18205	https://cdn.sofifa.org/players/4/19/246268.png	England	
18206	https://cdn.sofifa.org/players/4/19/246269.png	England	

	Flag	Overall	Potential	\
0	https://cdn.sofifa.org/flags/52.png	94	94	
1	https://cdn.sofifa.org/flags/38.png	94	94	
2	https://cdn.sofifa.org/flags/54.png	92	93	
3	https://cdn.sofifa.org/flags/45.png	91	93	

4	https://cdn.sofifa.org/flags/7.png	91	92
...	...	...	...
18202	https://cdn.sofifa.org/flags/14.png	47	65
18203	https://cdn.sofifa.org/flags/46.png	47	63
18204	https://cdn.sofifa.org/flags/14.png	47	67
18205	https://cdn.sofifa.org/flags/14.png	47	66
18206	https://cdn.sofifa.org/flags/14.png	46	66
Club		Club	
Logo \			
0	FC Barcelona		
	https://cdn.sofifa.org/teams/2/light/241.png		
1	Juventus		
	https://cdn.sofifa.org/teams/2/light/45.png		
2	Paris Saint-Germain		
	https://cdn.sofifa.org/teams/2/light/73.png		
3	Manchester United		
	https://cdn.sofifa.org/teams/2/light/11.png		
4	Manchester City		
	https://cdn.sofifa.org/teams/2/light/10.png		
...	...		
...			
18202	Crewe Alexandra		
	https://cdn.sofifa.org/teams/2/light/121.png		
18203	Trelleborgs FF		
	https://cdn.sofifa.org/teams/2/light/703.png		
18204	Cambridge United		
	https://cdn.sofifa.org/teams/2/light/1944.png		
18205	Tranmere Rovers		
	https://cdn.sofifa.org/teams/2/light/15048.png		
18206	Tranmere Rovers		
	https://cdn.sofifa.org/teams/2/light/15048.png		
... Composure Marking StandingTackle SlidingTackle			
GK	Diving \		
0	...	96.0	33.0
		28.0	26.0
		6.0	
1	...	95.0	28.0
		31.0	23.0
		7.0	
2	...	94.0	27.0
		24.0	33.0
		9.0	
3	...	68.0	15.0
		21.0	13.0
		90.0	
4	...	88.0	68.0
		58.0	51.0
		15.0	
...	...	...	...
18202	...	45.0	40.0
		48.0	47.0
		10.0	
18203	...	42.0	22.0
		15.0	19.0
		10.0	

18204	...	41.0	32.0	13.0	11.0	6.0
18205	...	46.0	20.0	25.0	27.0	14.0
18206	...	43.0	40.0	43.0	50.0	10.0

	GKHandling	GKKicking	GKPositioning	GKReflexes	Release	Clause
0	11.0	15.0	14.0	8.0		€226.5M
1	11.0	15.0	14.0	11.0		€127.1M
2	9.0	15.0	15.0	11.0		€228.1M
3	85.0	87.0	88.0	94.0		€138.6M
4	13.0	5.0	10.0	13.0		€196.4M
...	...	...	...	...		...
18202	13.0	7.0	8.0	9.0		€143K
18203	9.0	9.0	5.0	12.0		€113K
18204	5.0	10.0	6.0	13.0		€165K
18205	6.0	14.0	8.0	9.0		€143K
18206	15.0	9.0	12.0	9.0		€165K

[18207 rows x 88 columns]

fifa19.head()

	ID	Name	Age	\
0	158023	L. Messi	31	
1	20801	Cristiano Ronaldo	33	
2	190871	Neymar Jr	26	
3	193080	De Gea	27	
4	192985	K. De Bruyne	27	

	Photo	Nationality	\
0	https://cdn.sofifa.org/players/4/19/158023.png	Argentina	
1	https://cdn.sofifa.org/players/4/19/20801.png	Portugal	
2	https://cdn.sofifa.org/players/4/19/190871.png	Brazil	
3	https://cdn.sofifa.org/players/4/19/193080.png	Spain	
4	https://cdn.sofifa.org/players/4/19/192985.png	Belgium	

	Flag	Overall	Potential	\
0	https://cdn.sofifa.org/flags/52.png	94	94	
1	https://cdn.sofifa.org/flags/38.png	94	94	
2	https://cdn.sofifa.org/flags/54.png	92	93	
3	https://cdn.sofifa.org/flags/45.png	91	93	
4	https://cdn.sofifa.org/flags/7.png	91	92	

	Club	Club
Logo	...	\
0	FC Barcelona	
	https://cdn.sofifa.org/teams/2/light/241.png	...
1	Juventus	

```

https://cdn.sofifa.org/teams/2/light/45.png ...
2 Paris Saint-Germain
https://cdn.sofifa.org/teams/2/light/73.png ...
3 Manchester United
https://cdn.sofifa.org/teams/2/light/11.png ...
4 Manchester City
https://cdn.sofifa.org/teams/2/light/10.png ...

```

	Composure	Marking	StandingTackle	SlidingTackle	GKDividing
GKHandling \					
0	96.0	33.0	28.0	26.0	6.0
11.0					
1	95.0	28.0	31.0	23.0	7.0
11.0					
2	94.0	27.0	24.0	33.0	9.0
9.0					
3	68.0	15.0	21.0	13.0	90.0
85.0					
4	88.0	68.0	58.0	51.0	15.0
13.0					

	GK Kicking	GK Positioning	GK Reflexes	Release Clause
0	15.0	14.0	8.0	€226.5M
1	15.0	14.0	11.0	€127.1M
2	15.0	15.0	11.0	€228.1M
3	87.0	88.0	94.0	€138.6M
4	5.0	10.0	13.0	€196.4M

[5 rows x 88 columns]

```
fifa19.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
Index: 18207 entries, 0 to 18206
```

```
Data columns (total 88 columns):
```

#	Column	Non-Null Count	Dtype
0	ID	18207 non-null	int64
1	Name	18207 non-null	object
2	Age	18207 non-null	int64
3	Photo	18207 non-null	object
4	Nationality	18207 non-null	object
5	Flag	18207 non-null	object
6	Overall	18207 non-null	int64
7	Potential	18207 non-null	int64
8	Club	17966 non-null	object
9	Club Logo	18207 non-null	object
10	Value	18207 non-null	object
11	Wage	18207 non-null	object
12	Special	18207 non-null	int64

13	Preferred Foot	18159	non-null	object
14	International Reputation	18159	non-null	float64
15	Weak Foot	18159	non-null	float64
16	Skill Moves	18159	non-null	float64
17	Work Rate	18159	non-null	object
18	Body Type	18159	non-null	object
19	Real Face	18159	non-null	object
20	Position	18147	non-null	object
21	Jersey Number	18147	non-null	float64
22	Joined	16654	non-null	object
23	Loaned From	1264	non-null	object
24	Contract Valid Until	17918	non-null	object
25	Height	18159	non-null	object
26	Weight	18159	non-null	object
27	LS	16122	non-null	object
28	ST	16122	non-null	object
29	RS	16122	non-null	object
30	LW	16122	non-null	object
31	LF	16122	non-null	object
32	CF	16122	non-null	object
33	RF	16122	non-null	object
34	RW	16122	non-null	object
35	LAM	16122	non-null	object
36	CAM	16122	non-null	object
37	RAM	16122	non-null	object
38	LM	16122	non-null	object
39	LCM	16122	non-null	object
40	CM	16122	non-null	object
41	RCM	16122	non-null	object
42	RM	16122	non-null	object
43	LWB	16122	non-null	object
44	LDM	16122	non-null	object
45	CDM	16122	non-null	object
46	RDM	16122	non-null	object
47	RWB	16122	non-null	object
48	LB	16122	non-null	object
49	LCB	16122	non-null	object
50	CB	16122	non-null	object
51	RCB	16122	non-null	object
52	RB	16122	non-null	object
53	Crossing	18159	non-null	float64
54	Finishing	18159	non-null	float64
55	HeadingAccuracy	18159	non-null	float64
56	ShortPassing	18159	non-null	float64
57	Volleys	18159	non-null	float64
58	Dribbling	18159	non-null	float64
59	Curve	18159	non-null	float64
60	FKAccuracy	18159	non-null	float64
61	LongPassing	18159	non-null	float64

62	BallControl	18159	non-null	float64
63	Acceleration	18159	non-null	float64
64	SprintSpeed	18159	non-null	float64
65	Agility	18159	non-null	float64
66	Reactions	18159	non-null	float64
67	Balance	18159	non-null	float64
68	ShotPower	18159	non-null	float64
69	Jumping	18159	non-null	float64
70	Stamina	18159	non-null	float64
71	Strength	18159	non-null	float64
72	LongShots	18159	non-null	float64
73	Aggression	18159	non-null	float64
74	Interceptions	18159	non-null	float64
75	Positioning	18159	non-null	float64
76	Vision	18159	non-null	float64
77	Penalties	18159	non-null	float64
78	Composure	18159	non-null	float64
79	Marking	18159	non-null	float64
80	StandingTackle	18159	non-null	float64
81	SlidingTackle	18159	non-null	float64
82	GKDividing	18159	non-null	float64
83	GKHandling	18159	non-null	float64
84	GK Kicking	18159	non-null	float64
85	GK Positioning	18159	non-null	float64
86	GK Reflexes	18159	non-null	float64
87	Release Clause	16643	non-null	object

dtypes: float64(38), int64(5), object(45)

memory usage: 12.4+ MB

fifa19['Body Type'].value_counts()

Body Type

Normal	10595
Lean	6417
Stocky	1140
Messi	1
C. Ronaldo	1
Neymar	1
Courtois	1
PLAYER_BODY_TYPE_25	1
Shaqiri	1
Akinfenwa	1

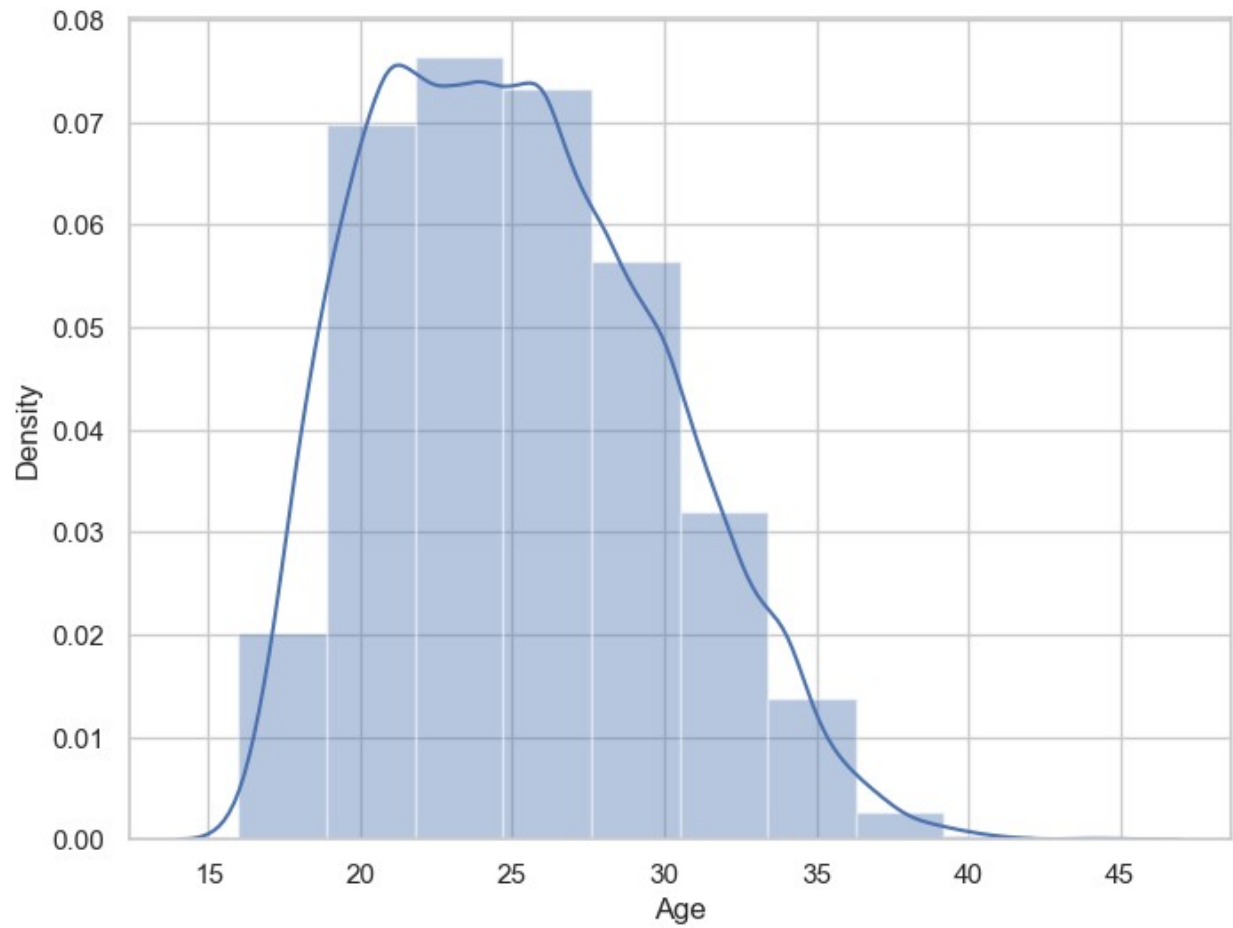
Name: count, dtype: int64

f, ax = plt.subplots(figsize=(8,6))

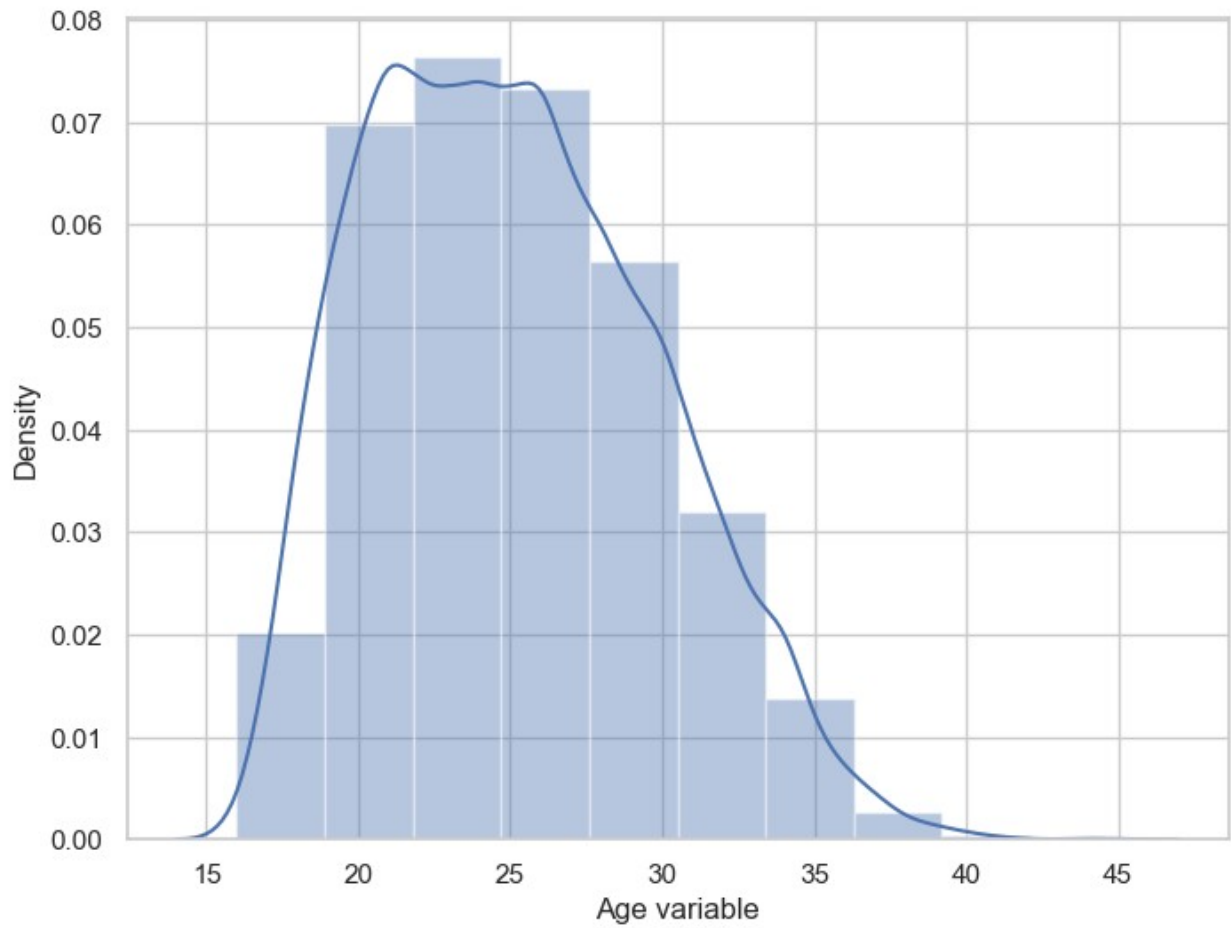
x = fifa19['Age']

ax = sns.distplot(x, bins=10)

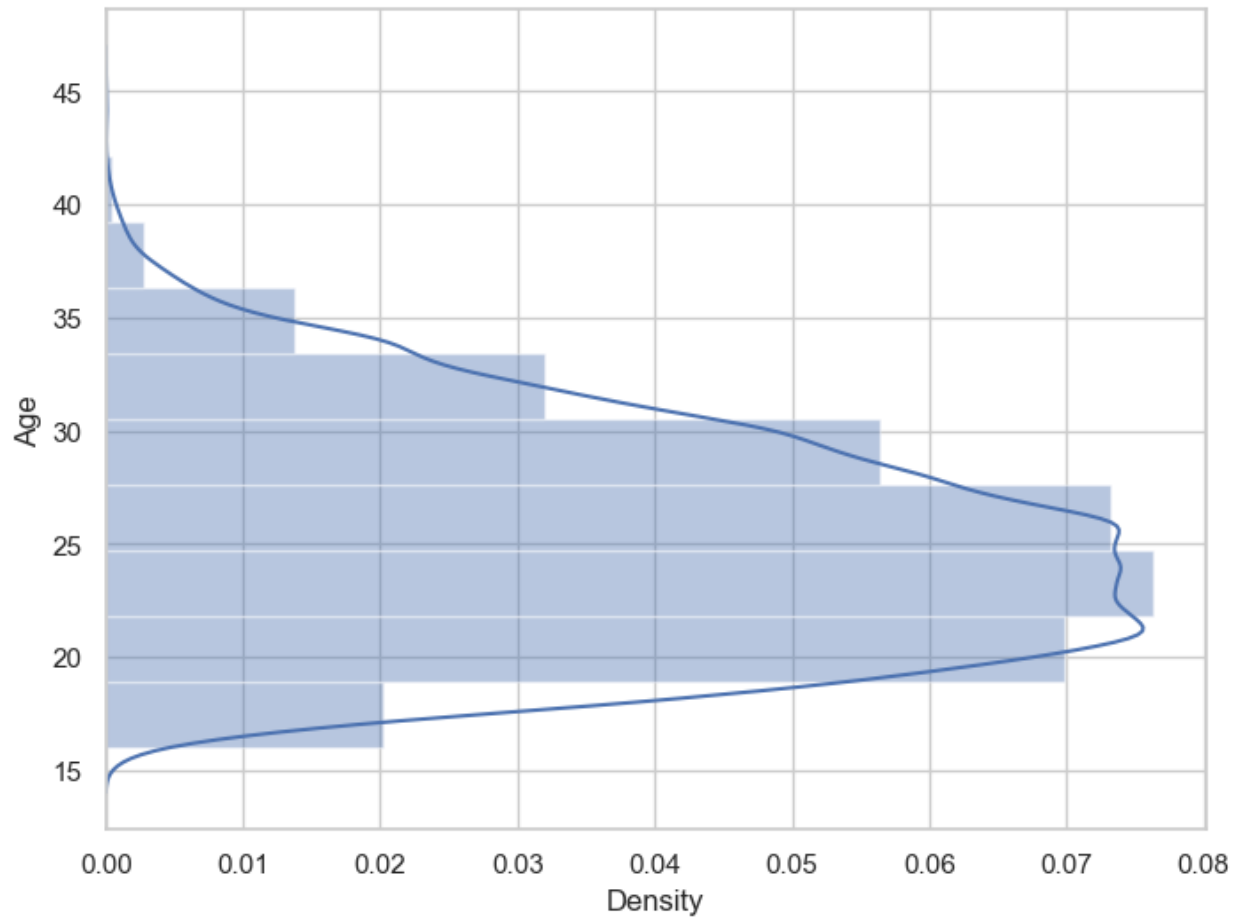
plt.show()



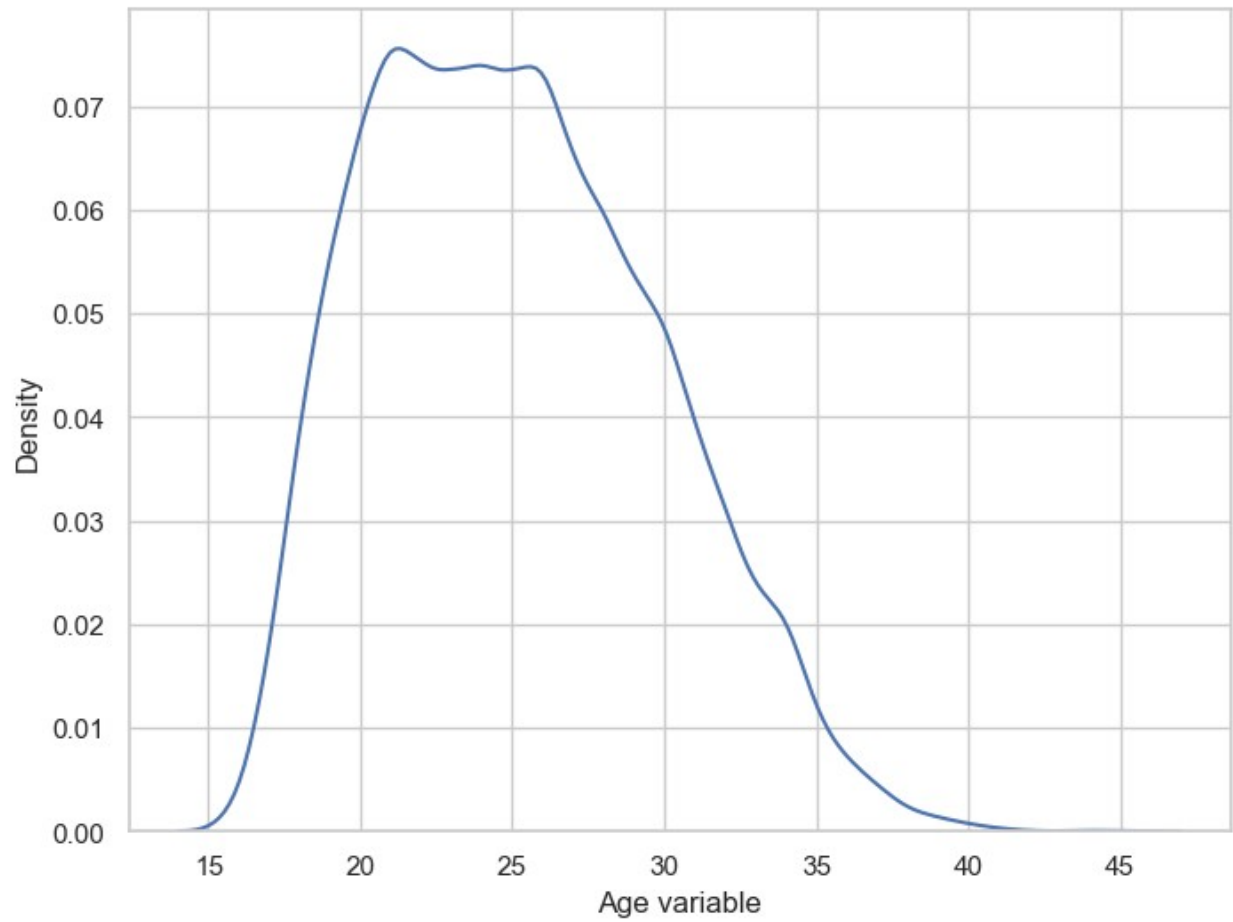
```
f, ax = plt.subplots(figsize=(8,6))
x = fifa19['Age']
x = pd.Series(x, name="Age variable")
ax = sns.distplot(x, bins=10)
plt.show()
```



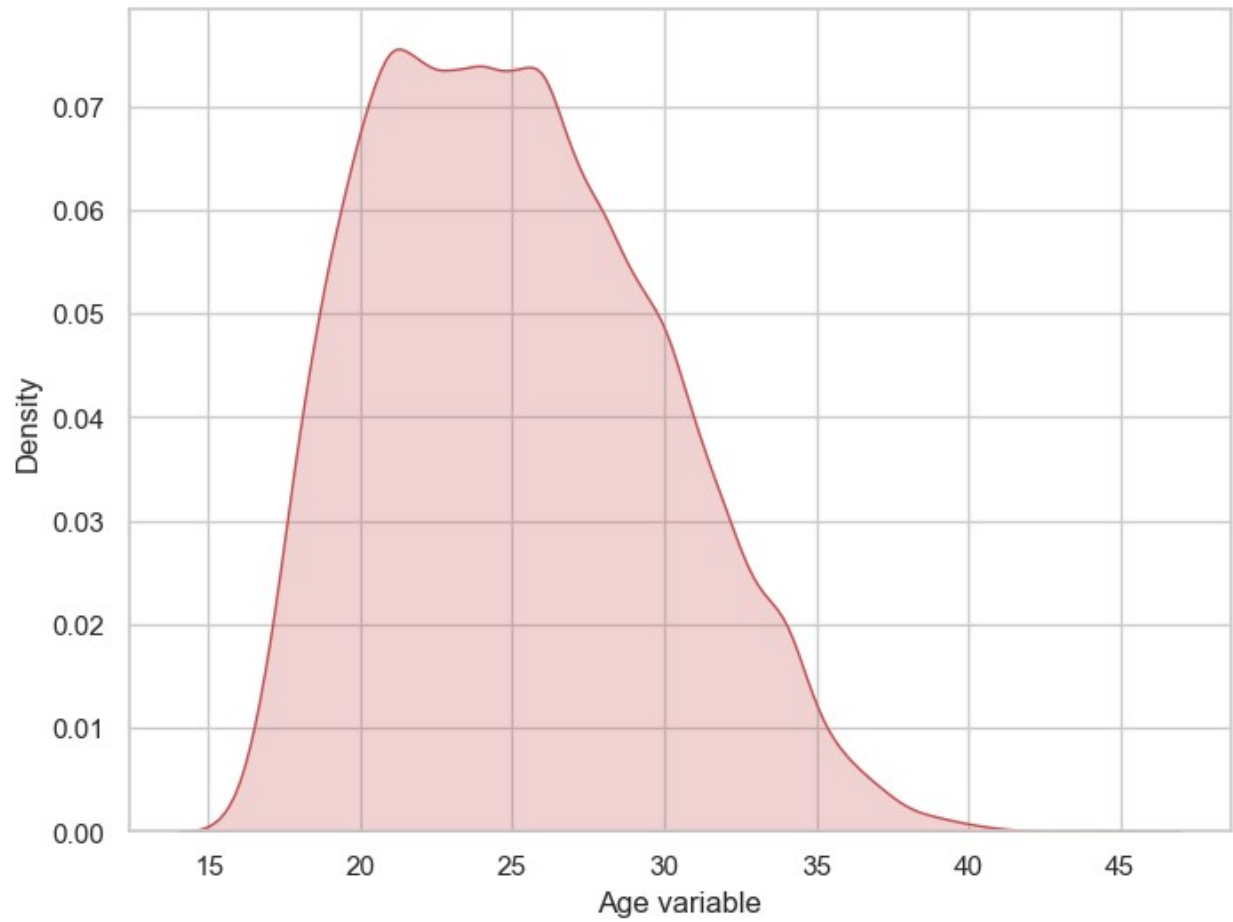
```
f, ax = plt.subplots(figsize=(8,6))
x = fifa19['Age']
ax = sns.distplot(x, bins=10, vertical = True)
plt.show()
```

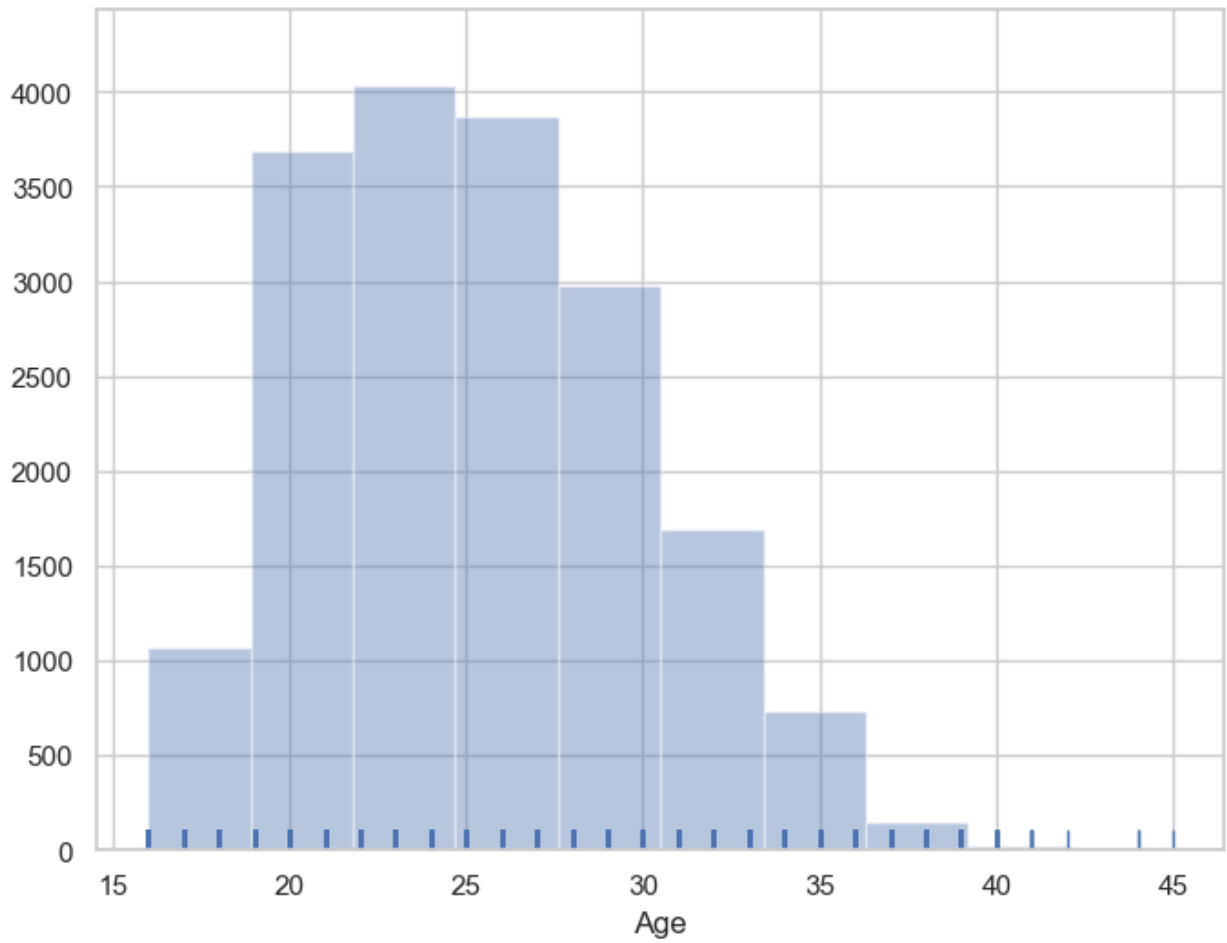
```
f, ax = plt.subplots(figsize=(8,6))
x = fifa19['Age']
x = pd.Series(x, name="Age variable")
ax = sns.kdeplot(x)
plt.show()
```



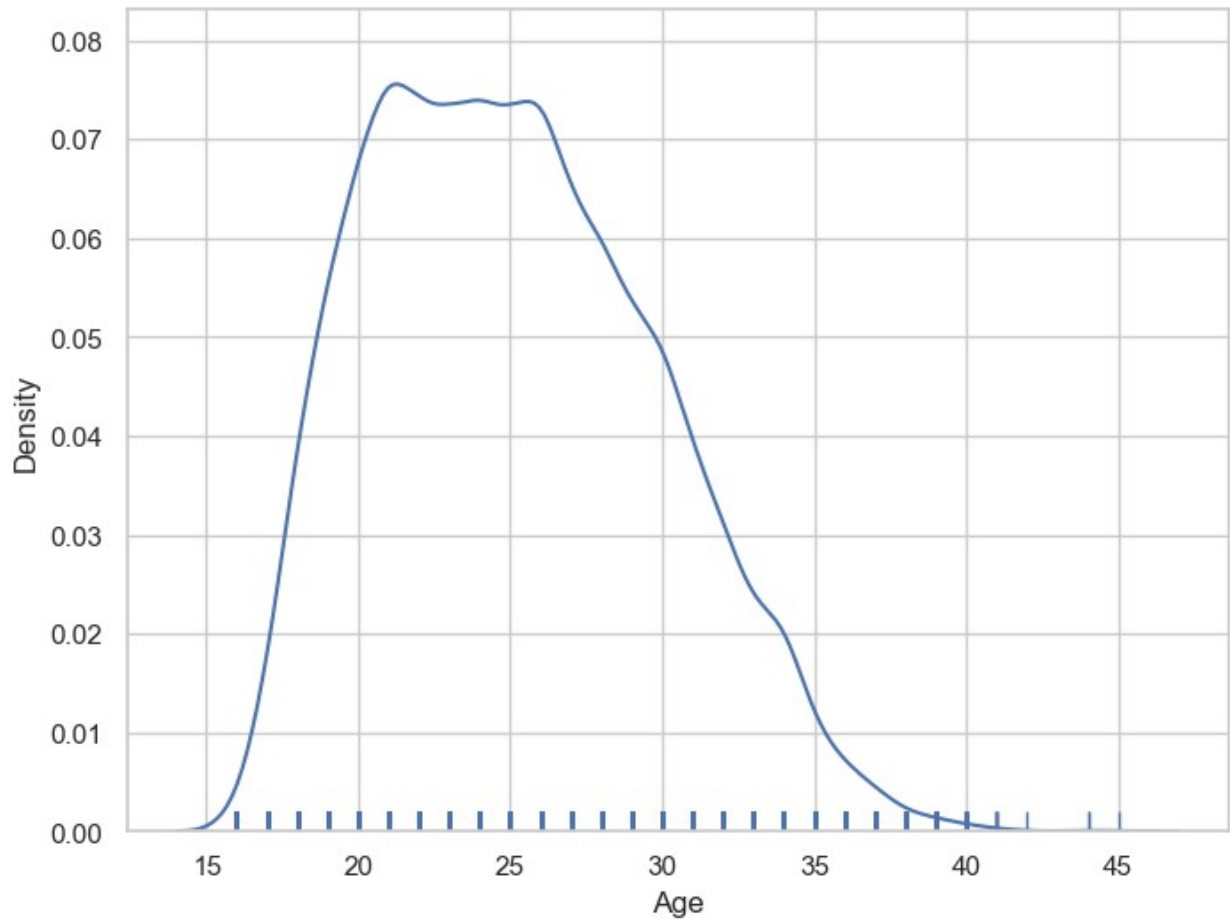
```
f, ax = plt.subplots(figsize=(8,6))
x = fifa19['Age']
x = pd.Series(x, name="Age variable")
ax = sns.kdeplot(x, shade=True, color='r')
plt.show()
```



```
f, ax = plt.subplots(figsize=(8,6))  
x = fifa19['Age']  
ax = sns.distplot(x, kde=False, rug=True, bins=10)  
plt.show()
```

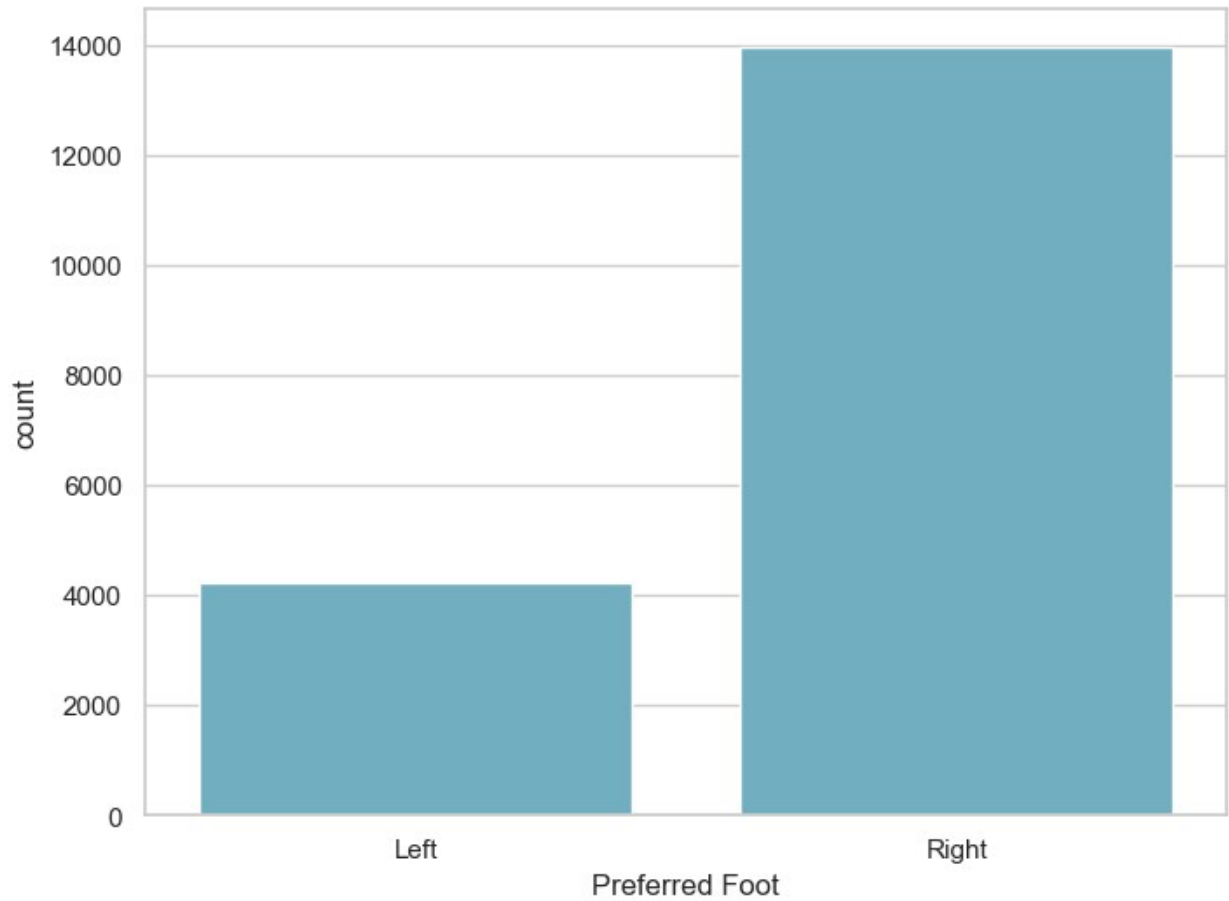


```
f, ax = plt.subplots(figsize=(8,6))
x = fifa19['Age']
ax = sns.distplot(x, hist=False, rug=True, bins=10)
plt.show()
```

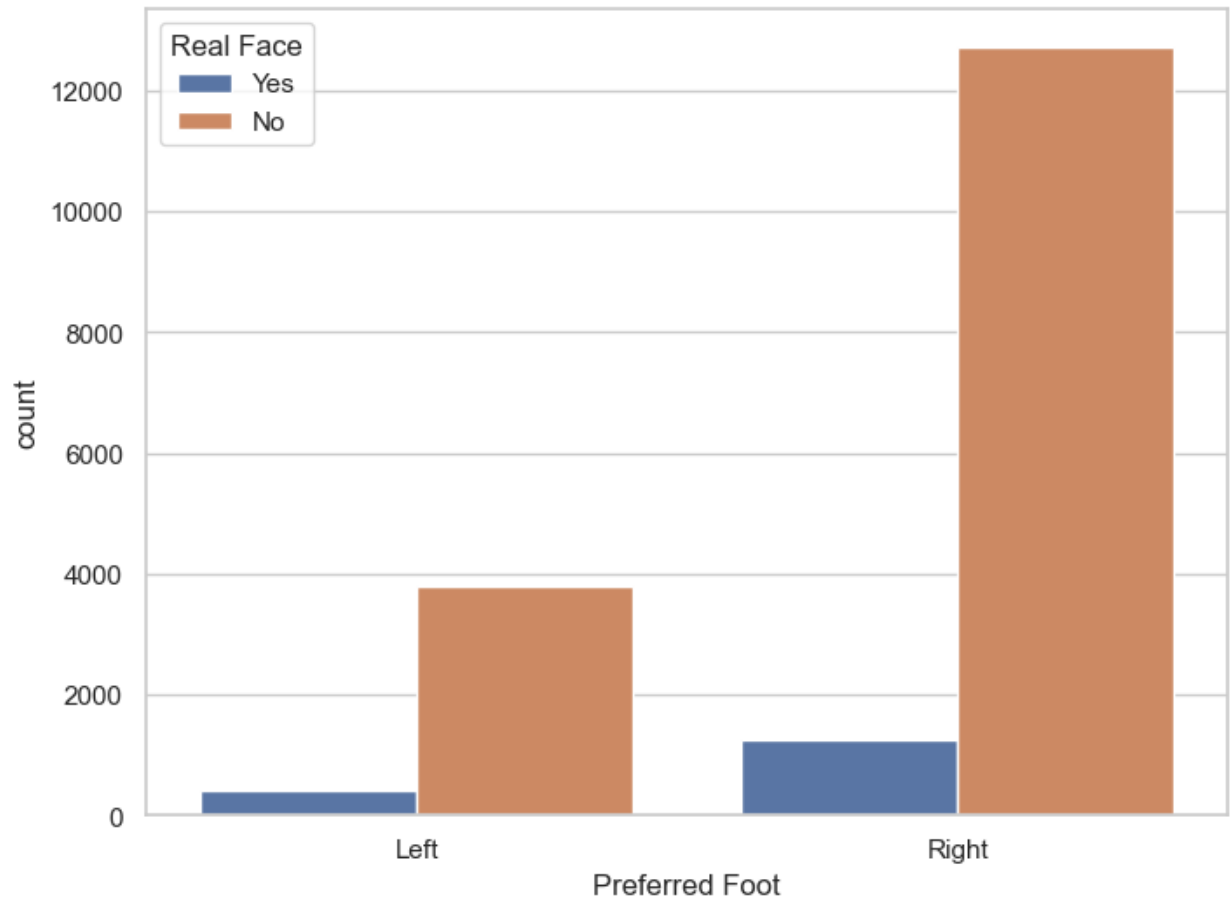


```
fifa19['Preferred Foot'].nunique()
2
fifa19['Preferred Foot'].value_counts()
Preferred Foot
Right    13948
Left     4211
Name: count, dtype: int64

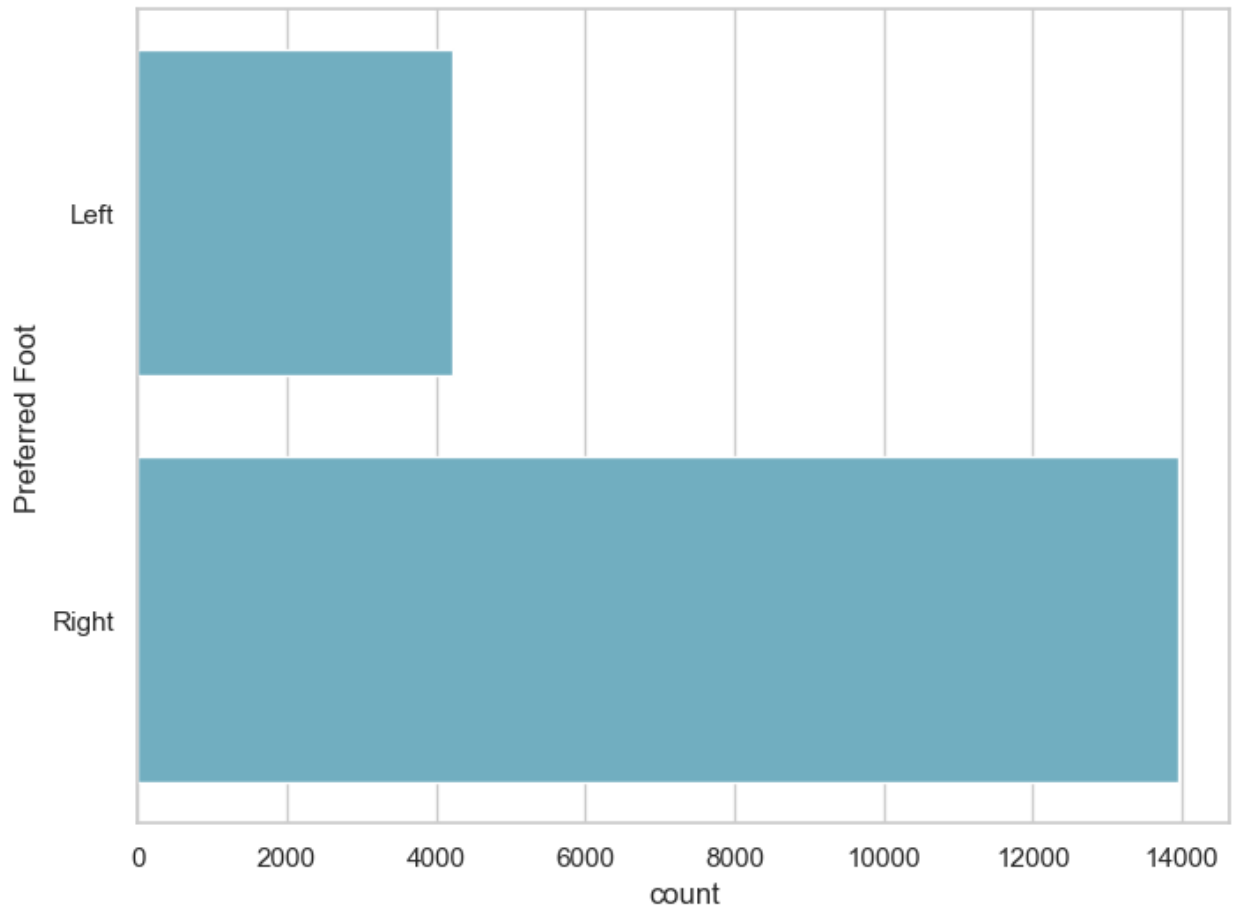
f, ax = plt.subplots(figsize=(8, 6))
sns.countplot(x="Preferred Foot", data=fifa19, color="c")
plt.show()
```



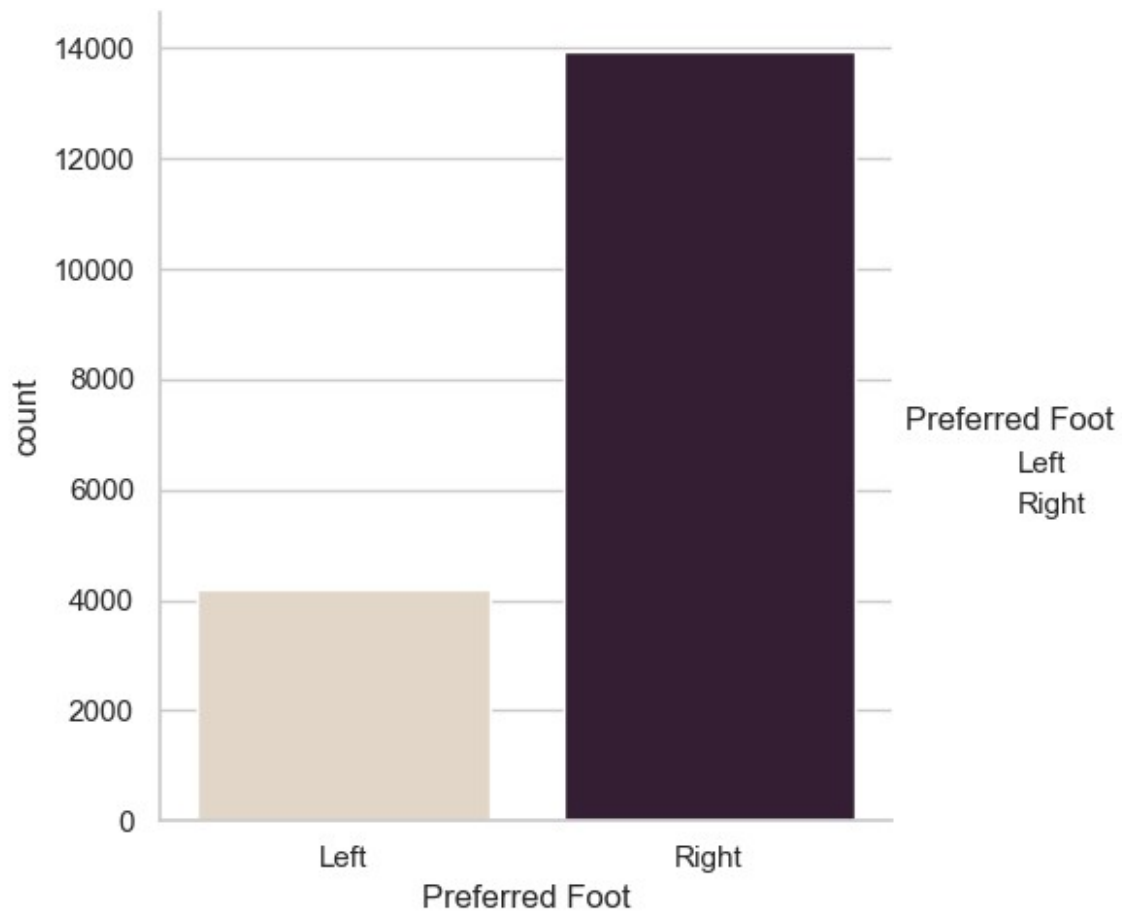
```
f, ax = plt.subplots(figsize=(8, 6))
sns.countplot(x="Preferred Foot", hue="Real Face", data=fifa19)
plt.show()
```



```
f, ax = plt.subplots(figsize=(8, 6))
sns.countplot(y="Preferred Foot", data=fifa19, color="c")
plt.show()
```

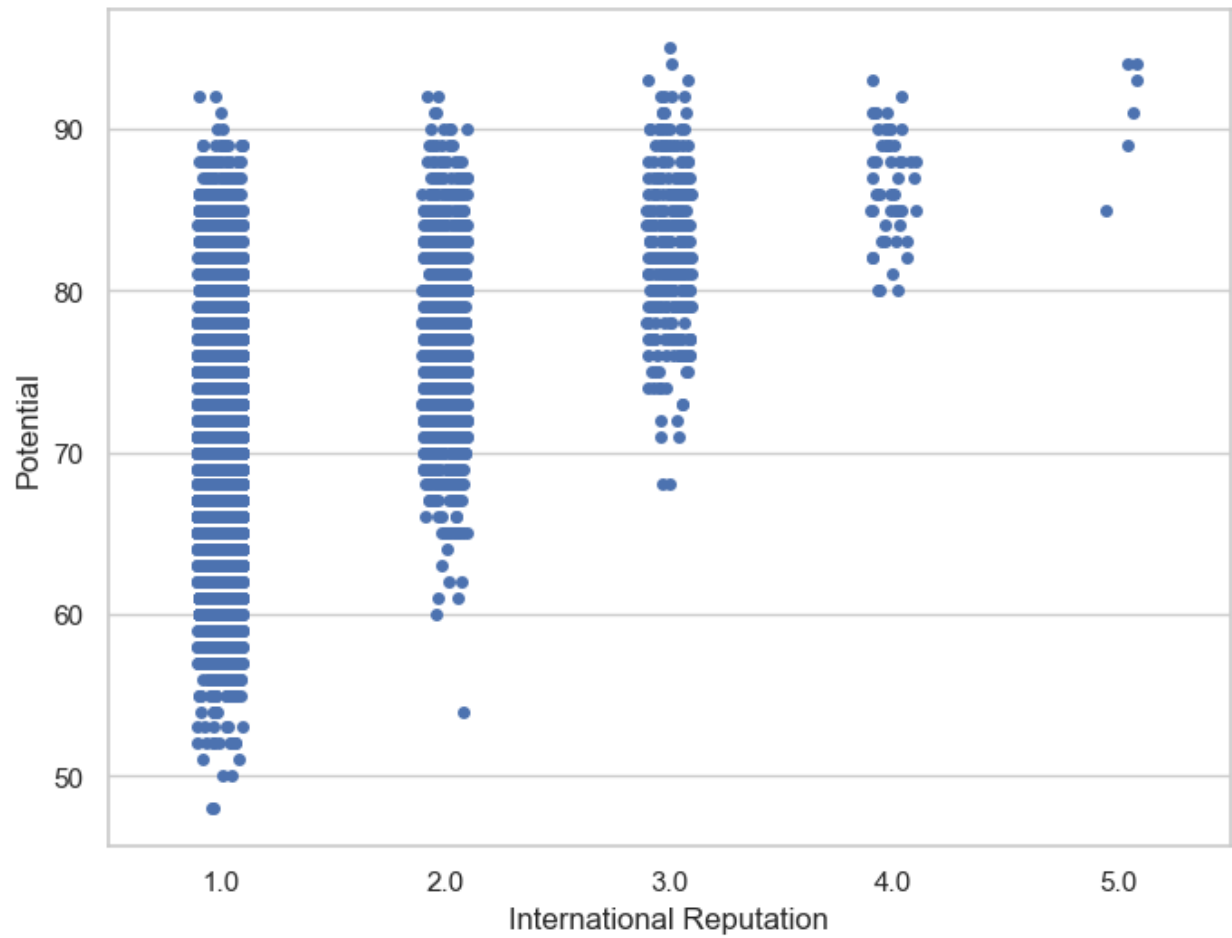


```
g = sns.catplot(x="Preferred Foot", kind="count", palette="ch:.25",  
data=fifa19)  
plt.show()
```

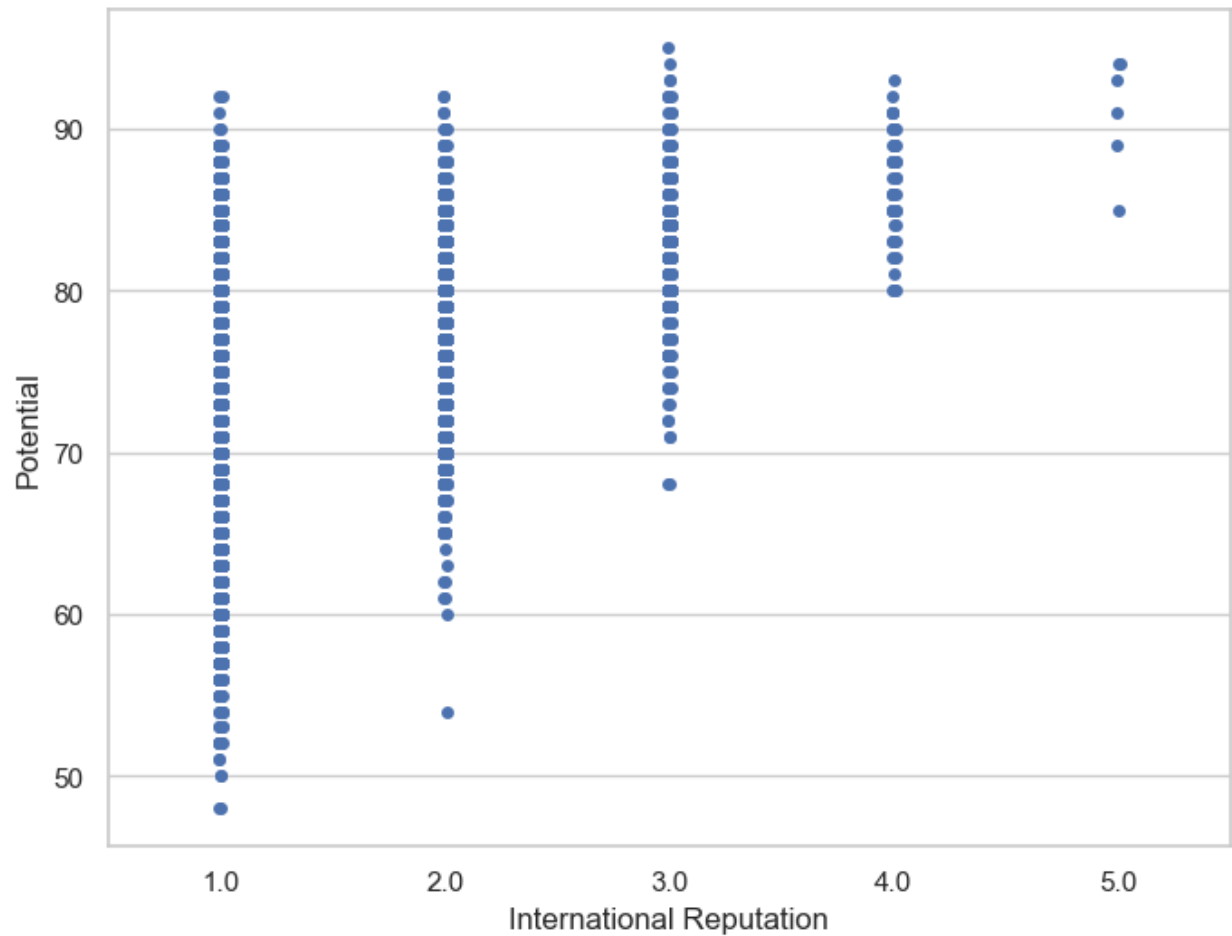



```
fifa19['International Reputation'].nunique()
5
fifa19['International Reputation'].value_counts()
International Reputation
1.0    16532
2.0    1261
3.0     309
4.0     51
5.0      6
Name: count, dtype: int64

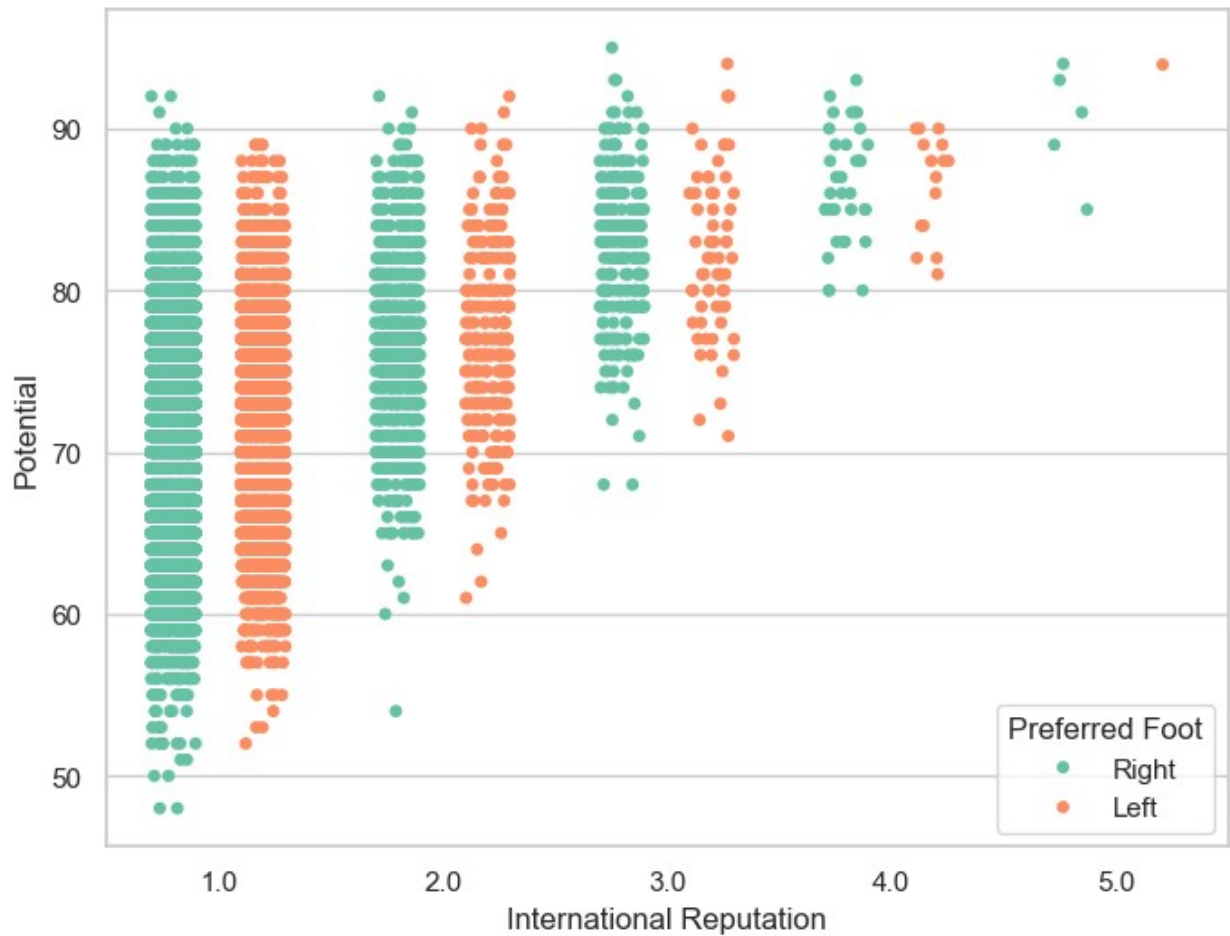
f, ax = plt.subplots(figsize=(8, 6))
sns.stripplot(x="International Reputation", y="Potential",
data=fifa19)
plt.show()
```



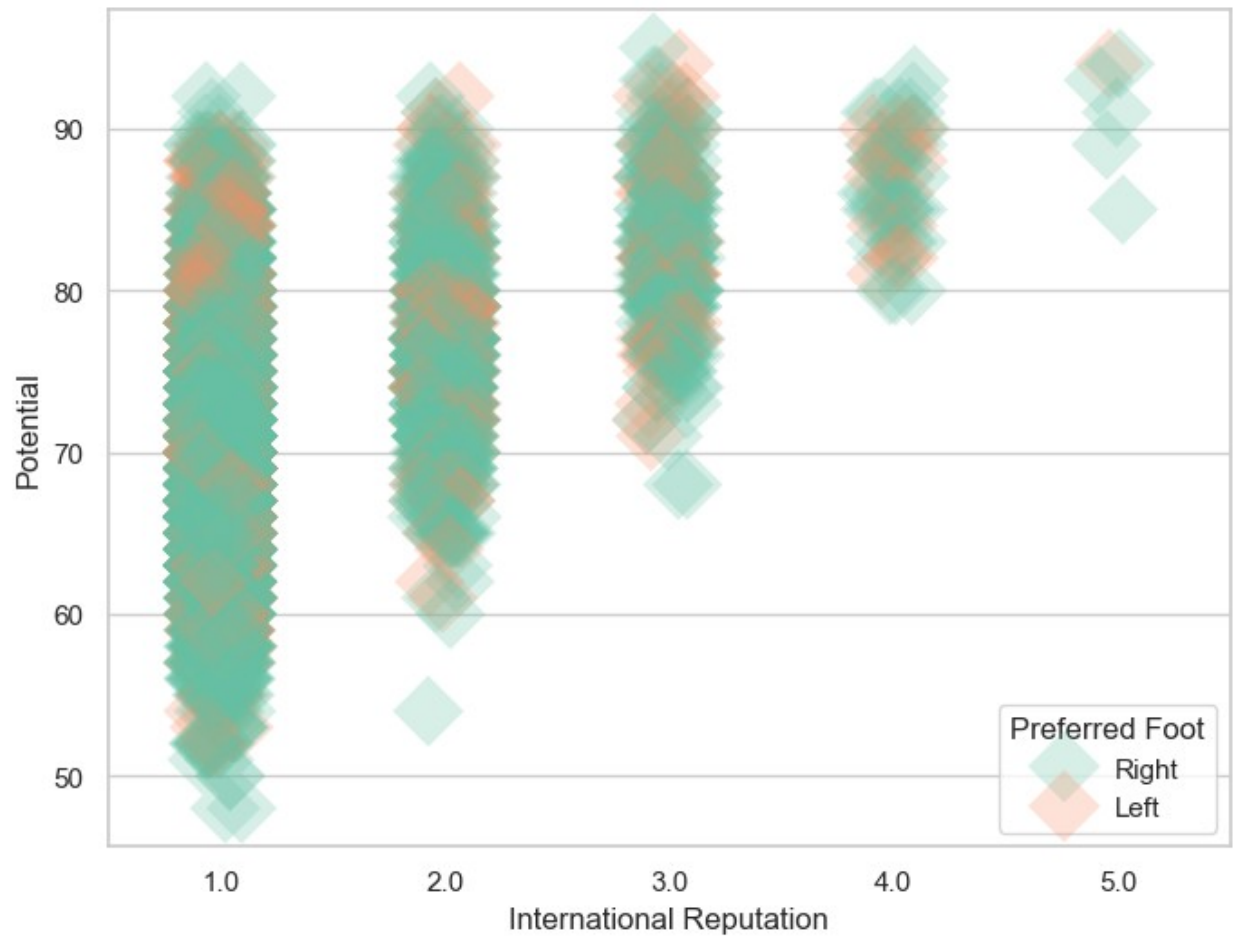
```
f, ax = plt.subplots(figsize=(8, 6))
sns.stripplot(x="International Reputation", y="Potential",
data=fifa19, jitter=0.01)
plt.show()
```



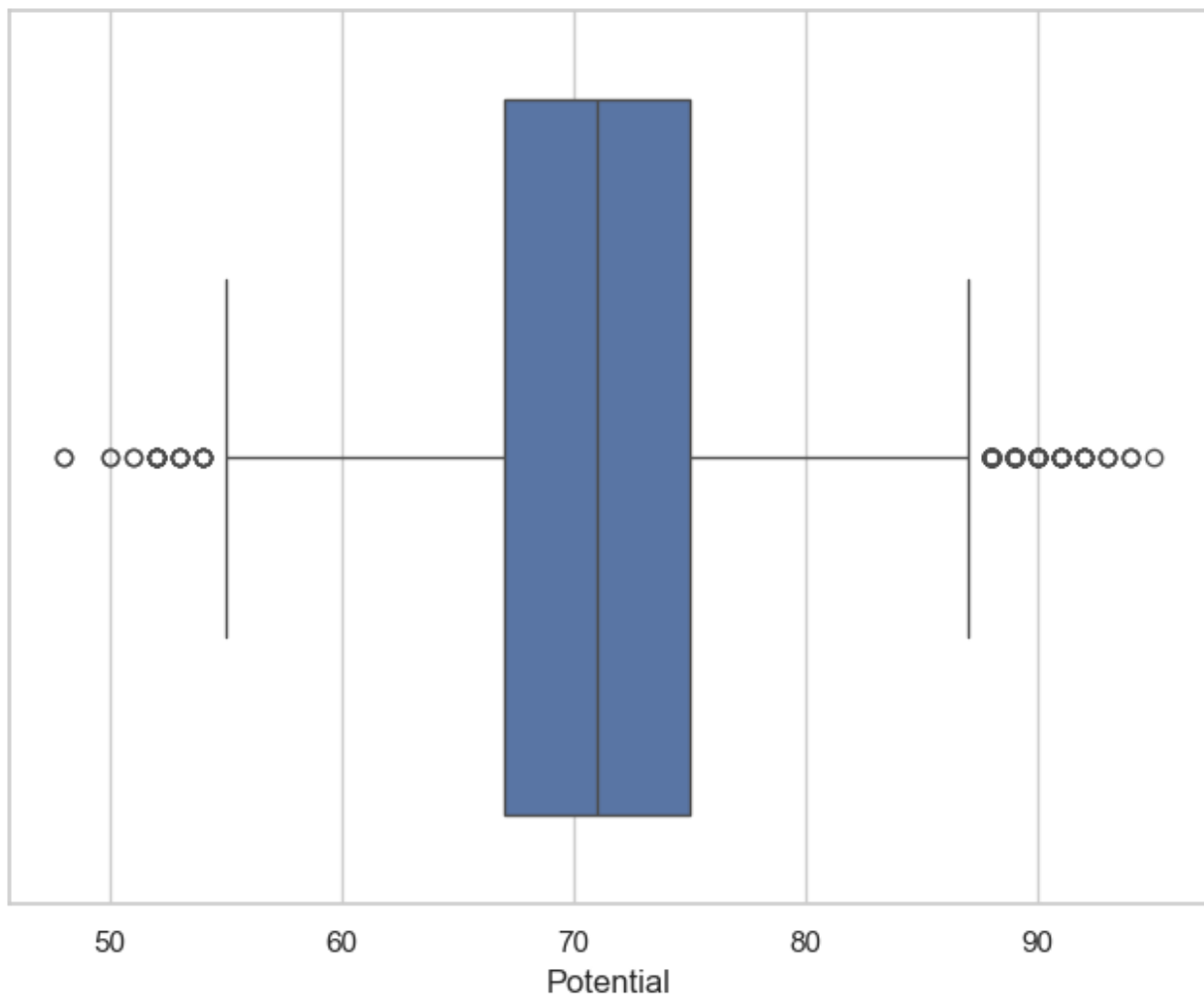
```
f, ax = plt.subplots(figsize=(8, 6))
sns.stripplot(x="International Reputation", y="Potential",
hue="Preferred Foot",
data=fifa19, jitter=0.2, palette="Set2",
dodge=True)
plt.show()
```



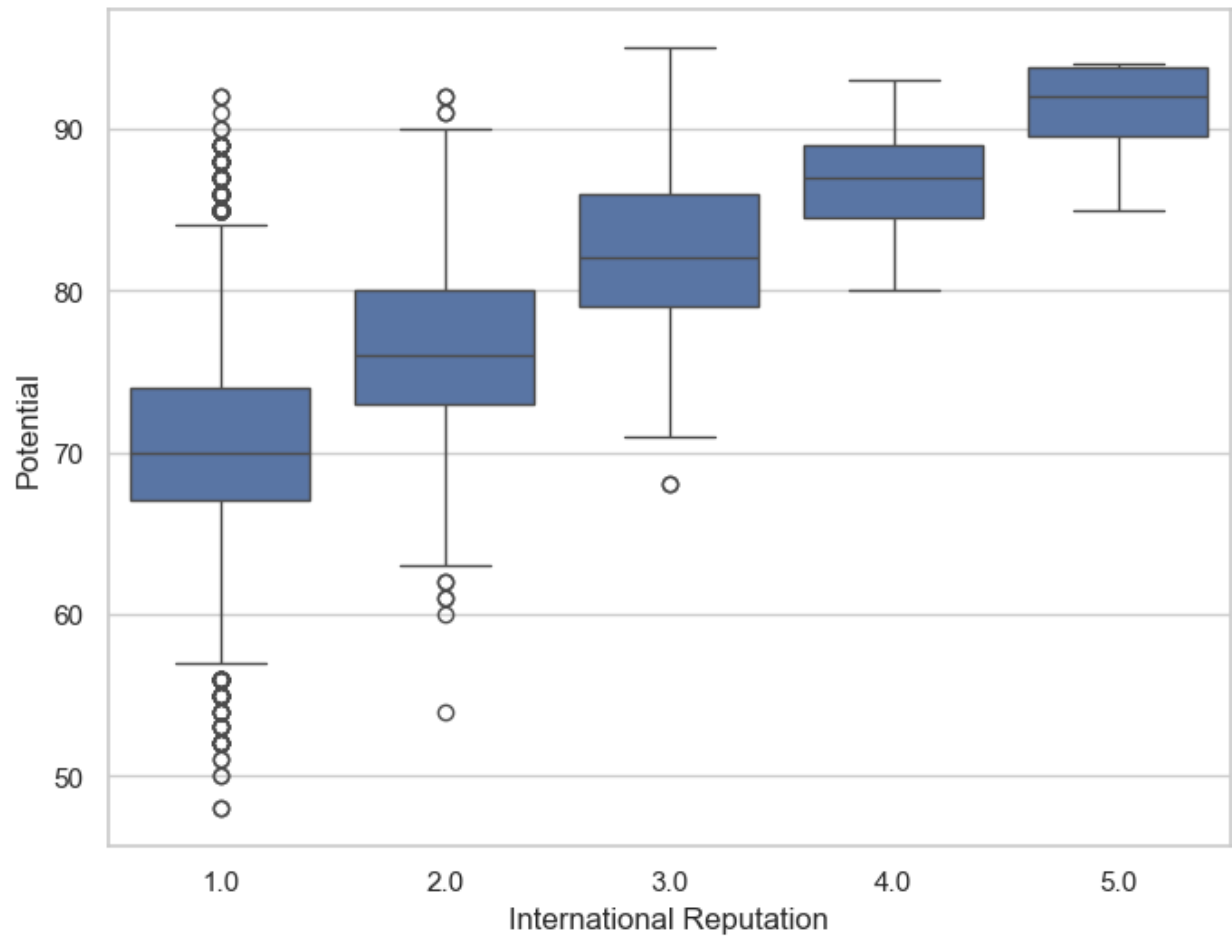
```
f, ax = plt.subplots(figsize=(8, 6))
sns.stripplot(x="International Reputation", y="Potential",
             hue="Preferred Foot",
             data=fifa19, palette="Set2", size=20, marker="D",
             edgecolor="gray", alpha=.25)
plt.show()
```



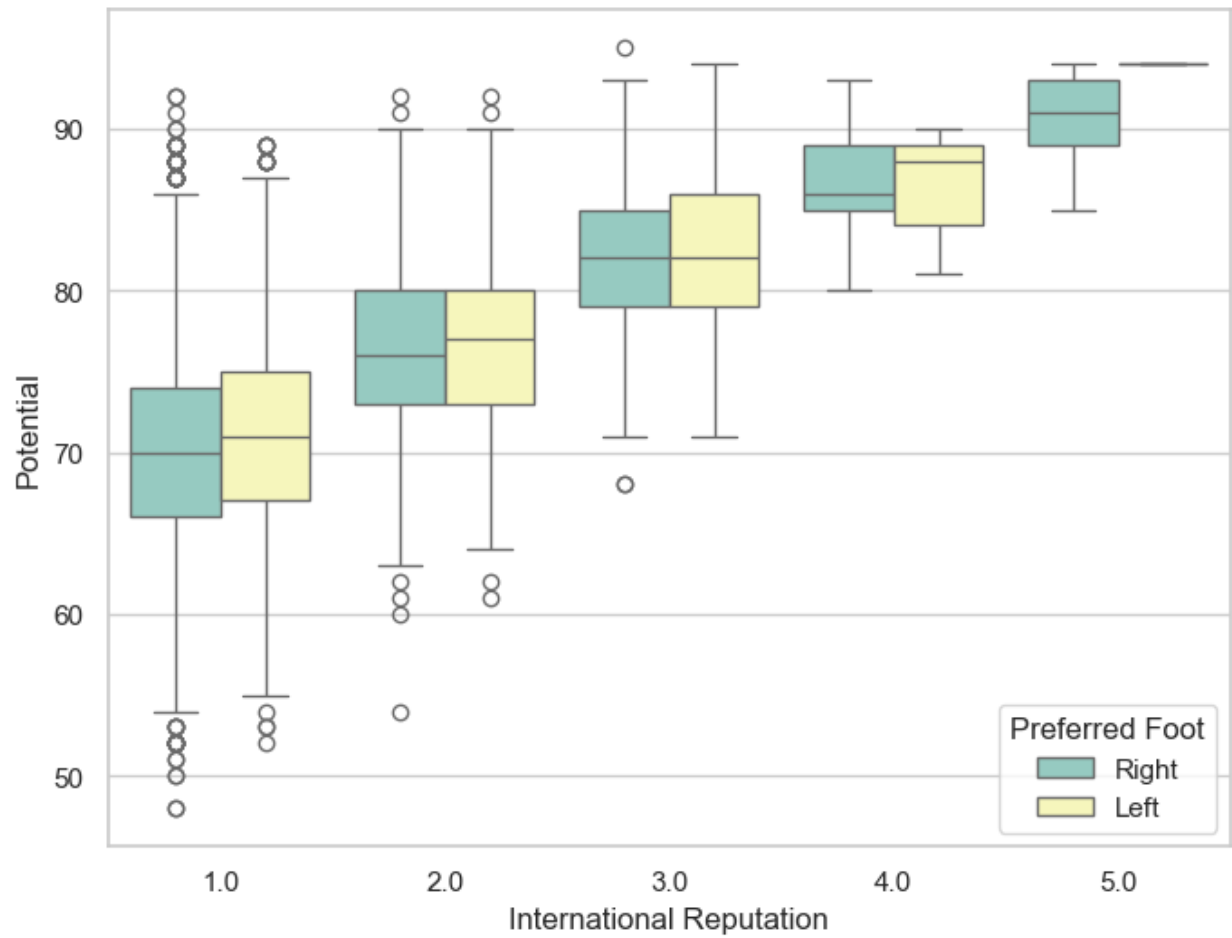
```
f, ax = plt.subplots(figsize=(8, 6))  
sns.boxplot(x=fifa19['Potential'])  
plt.show()
```



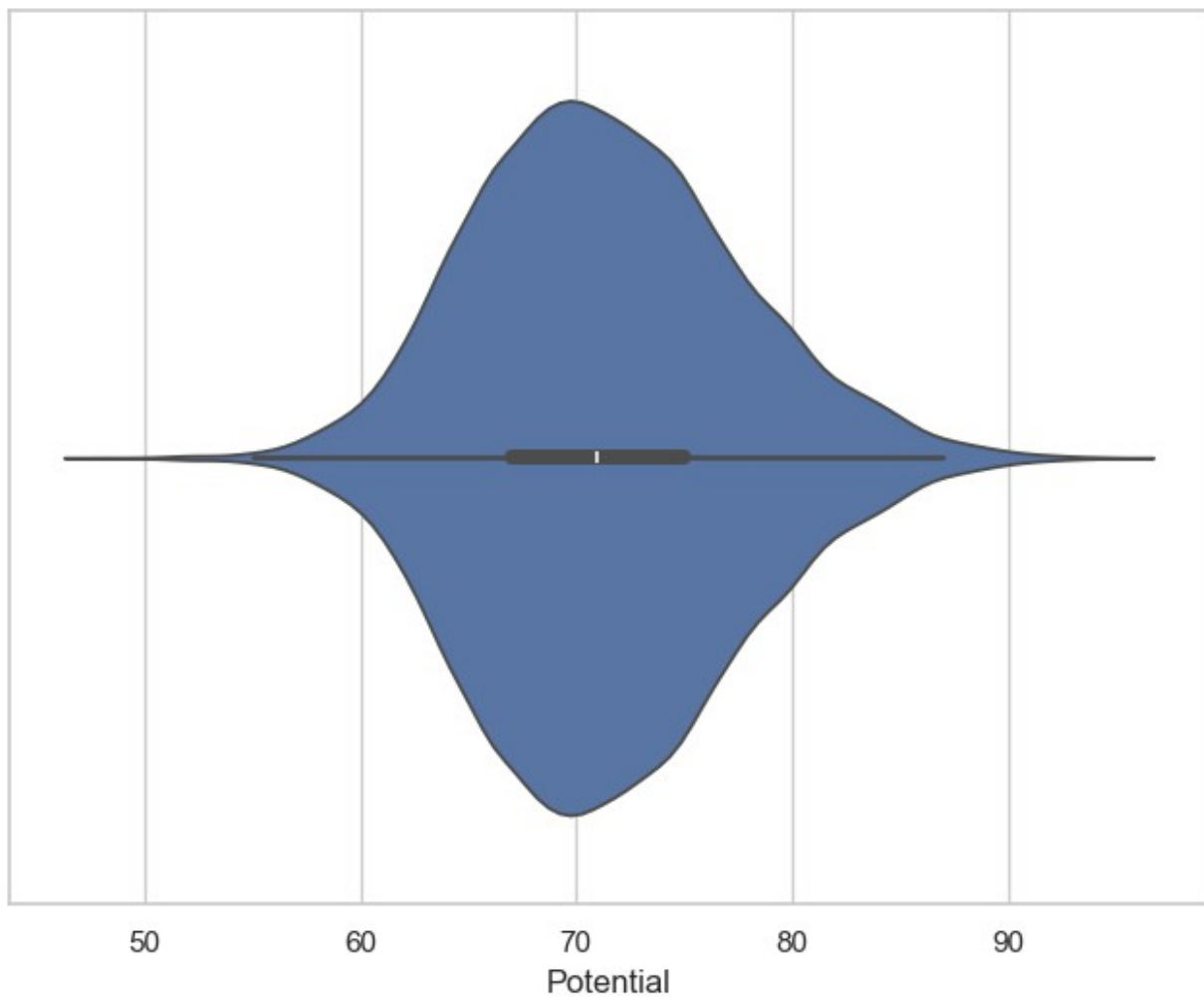
```
f, ax = plt.subplots(figsize=(8, 6))  
sns.boxplot(x="International Reputation", y="Potential", data=fifa19)  
plt.show()
```



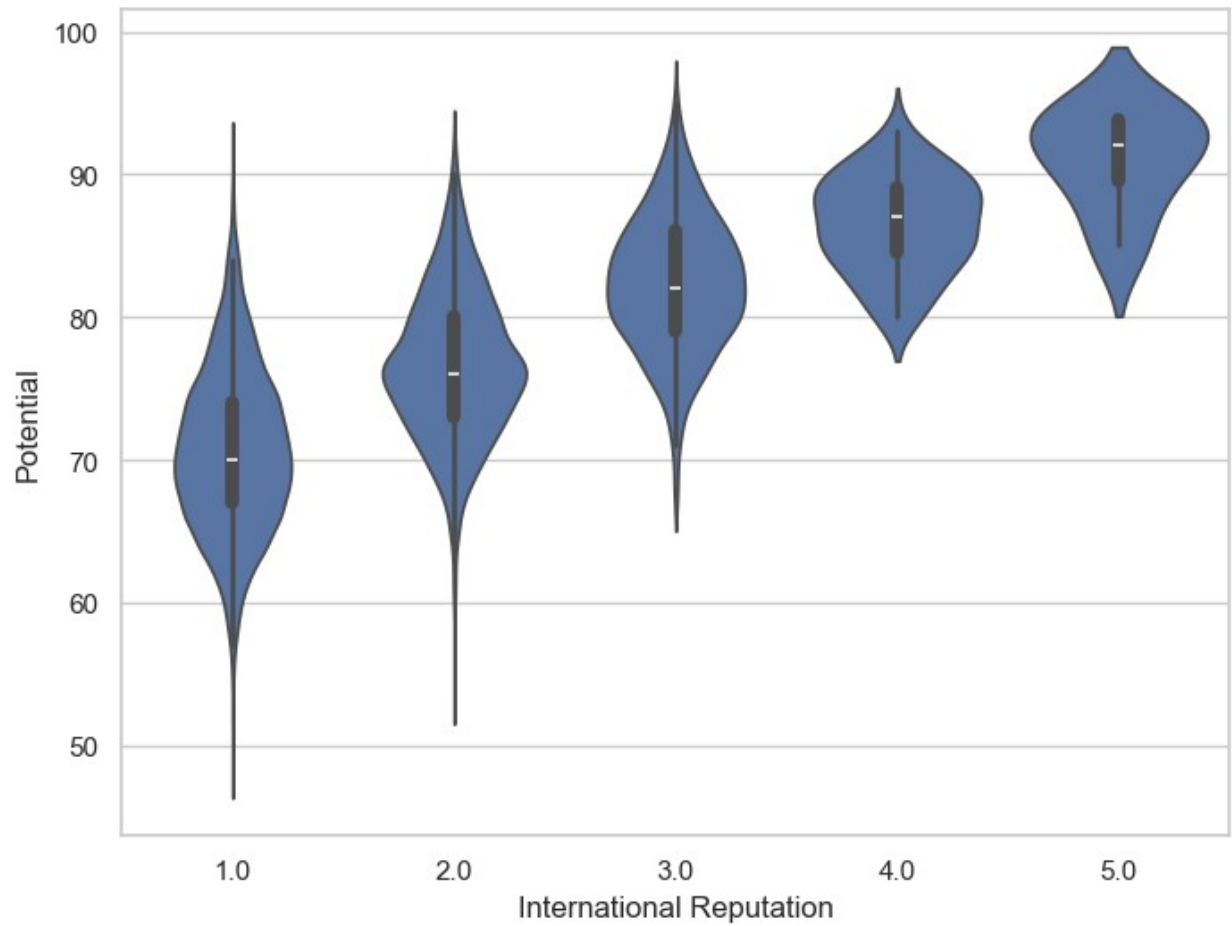
```
f, ax = plt.subplots(figsize=(8, 6))
sns.boxplot(x="International Reputation", y="Potential",
hue="Preferred Foot", data=fifa19, palette="Set3")
plt.show()
```



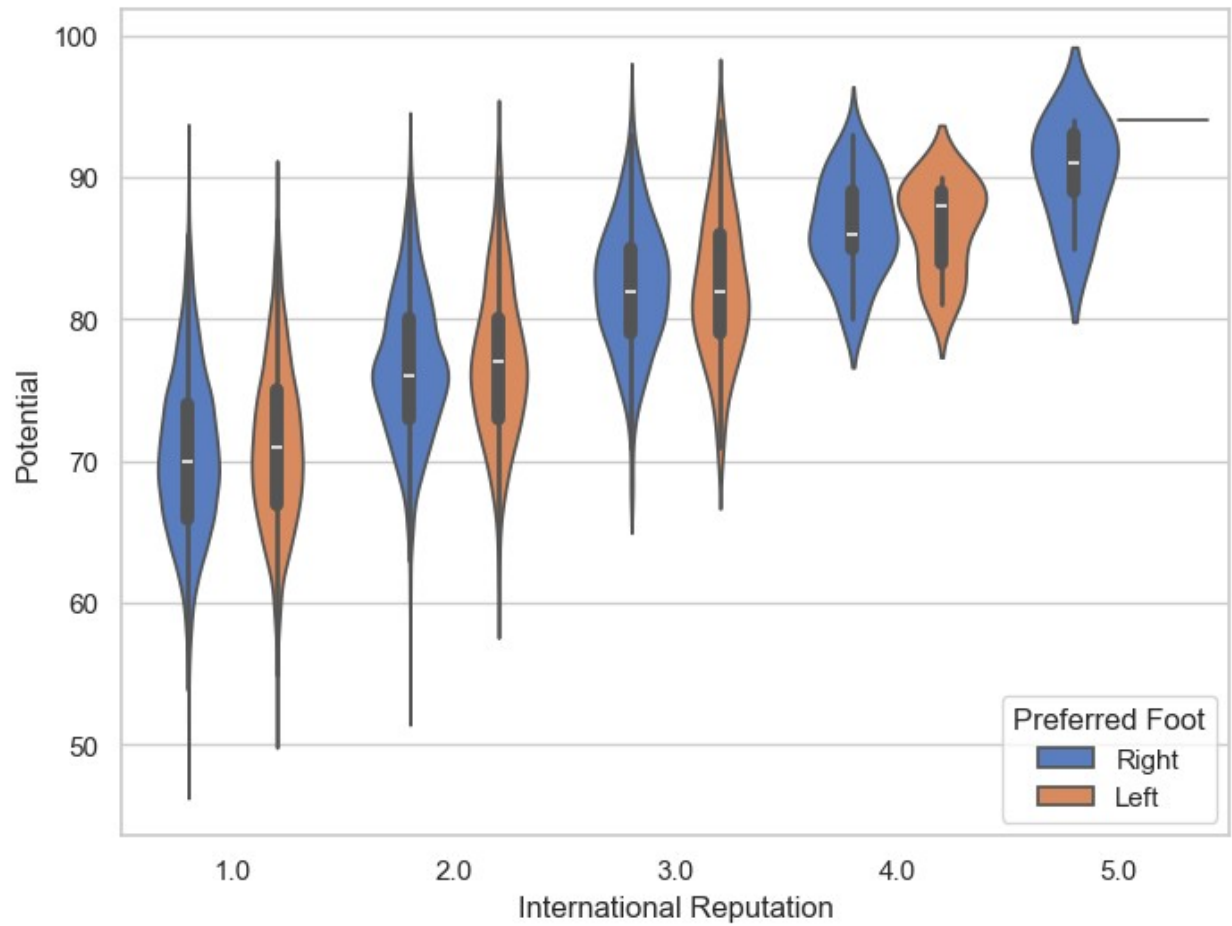
```
f, ax = plt.subplots(figsize=(8, 6))
sns.violinplot(x=fifa19["Potential"])
plt.show()
```

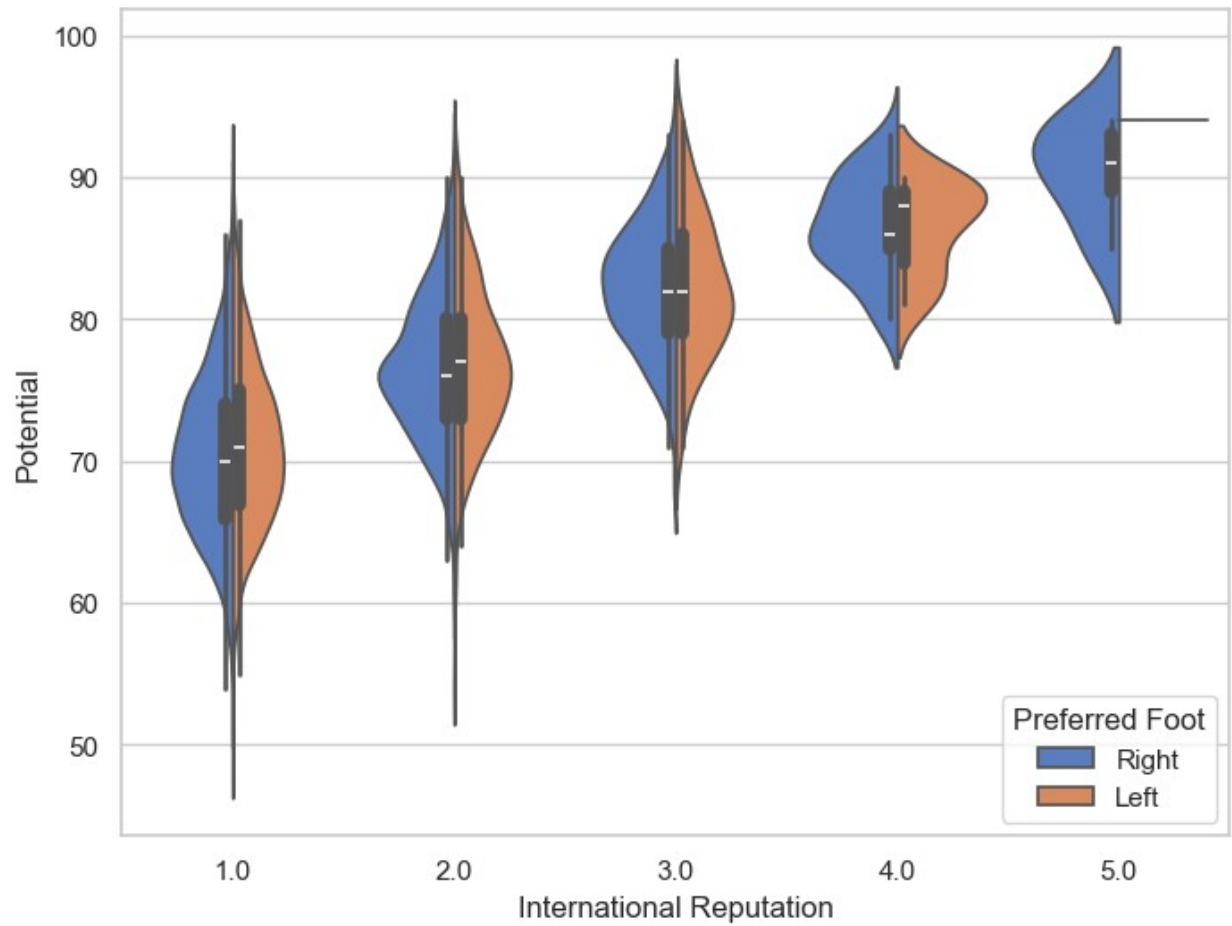
```
f, ax = plt.subplots(figsize=(8, 6))
sns.violinplot(x="International Reputation", y="Potential",
data=fifa19)
plt.show()
```



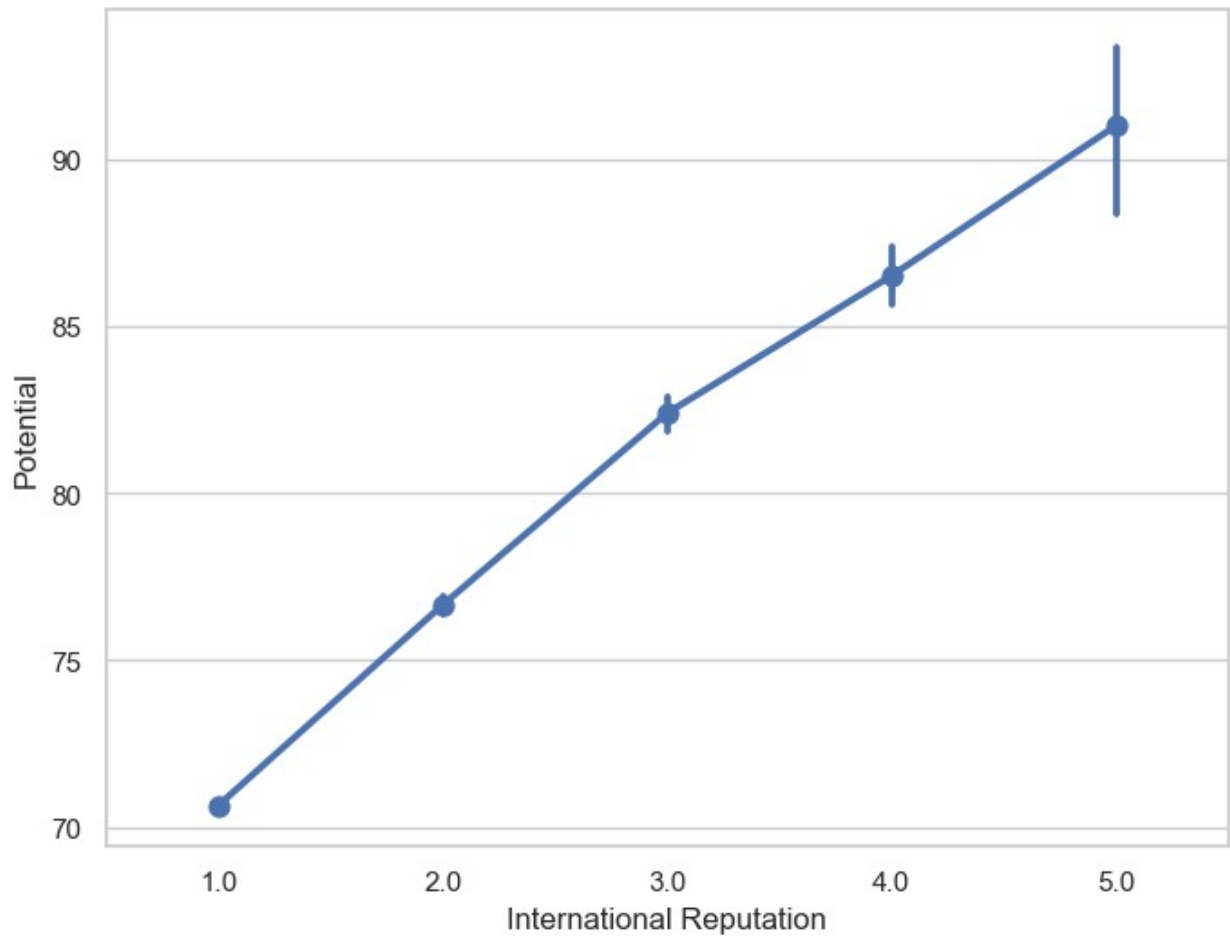
```
f, ax = plt.subplots(figsize=(8, 6))
sns.violinplot(x="International Reputation", y="Potential",
hue="Preferred Foot", data=fifa19, palette="muted")
plt.show()
```



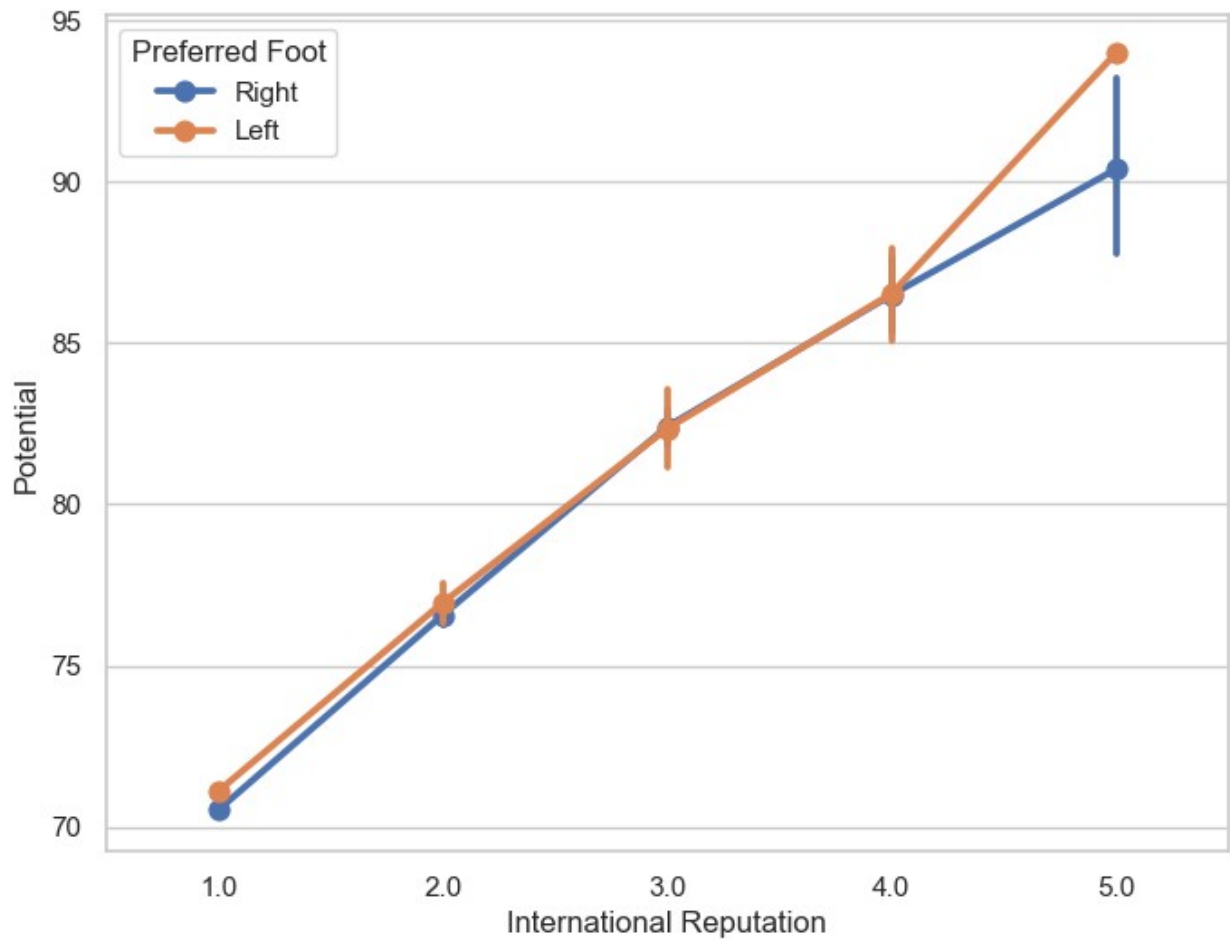
```
f, ax = plt.subplots(figsize=(8, 6))
sns.violinplot(x="International Reputation", y="Potential",
               hue="Preferred Foot",
               data=fifa19, palette="muted", split=True)
plt.show()
```



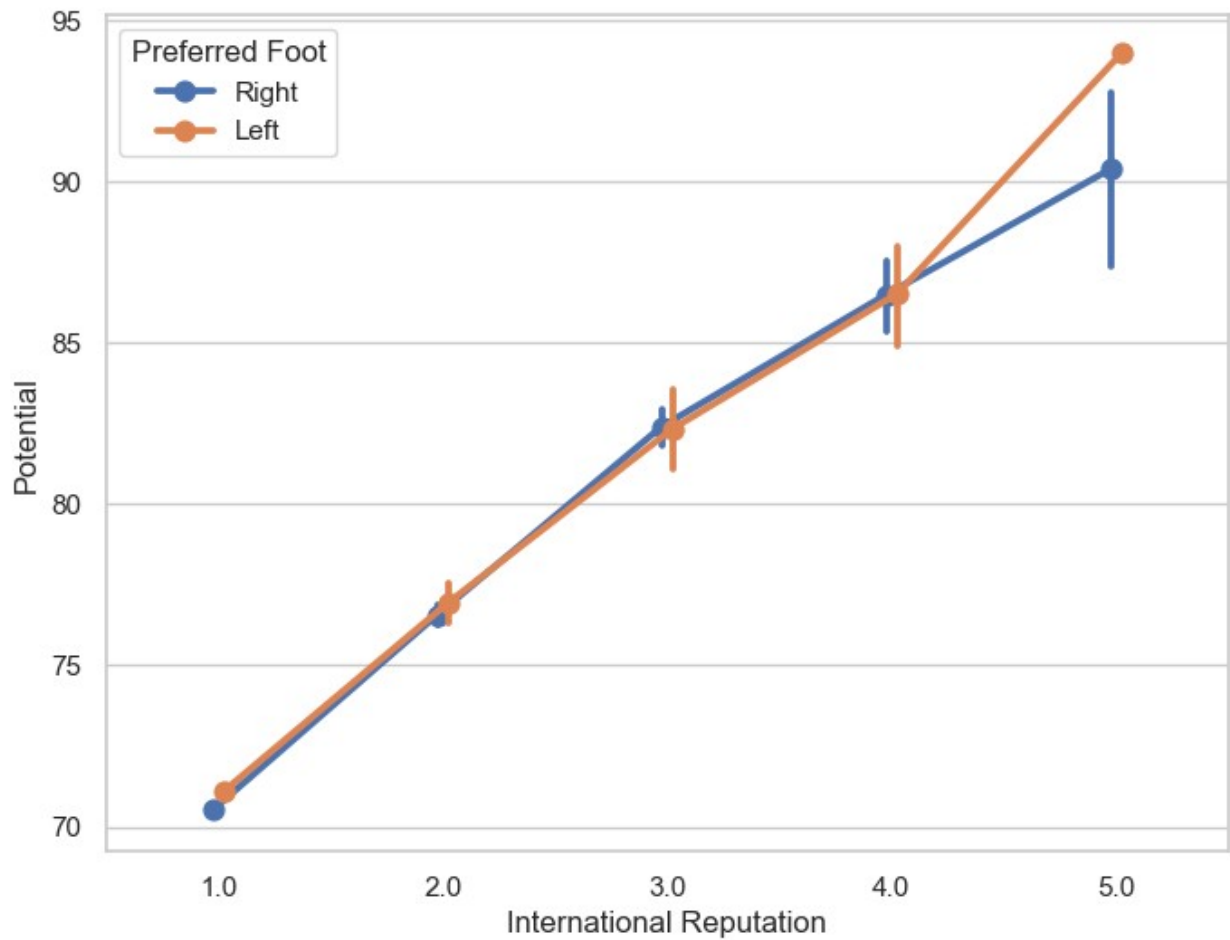
```
f, ax = plt.subplots(figsize=(8, 6))
sns.pointplot(x="International Reputation", y="Potential",
data=fifa19)
plt.show()
```



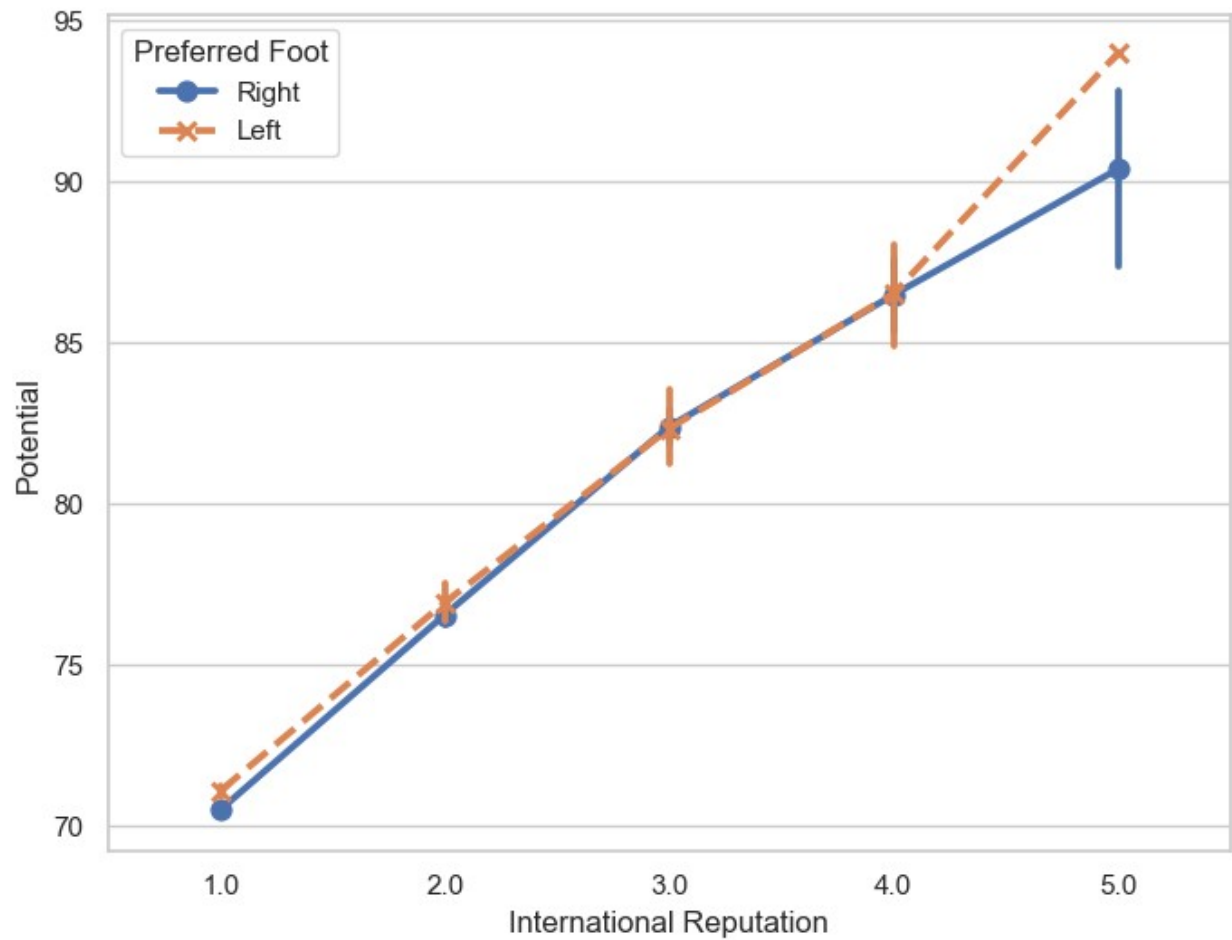
```
f, ax = plt.subplots(figsize=(8, 6))
sns.pointplot(x="International Reputation", y="Potential",
hue="Preferred Foot", data=fifa19)
plt.show()
```



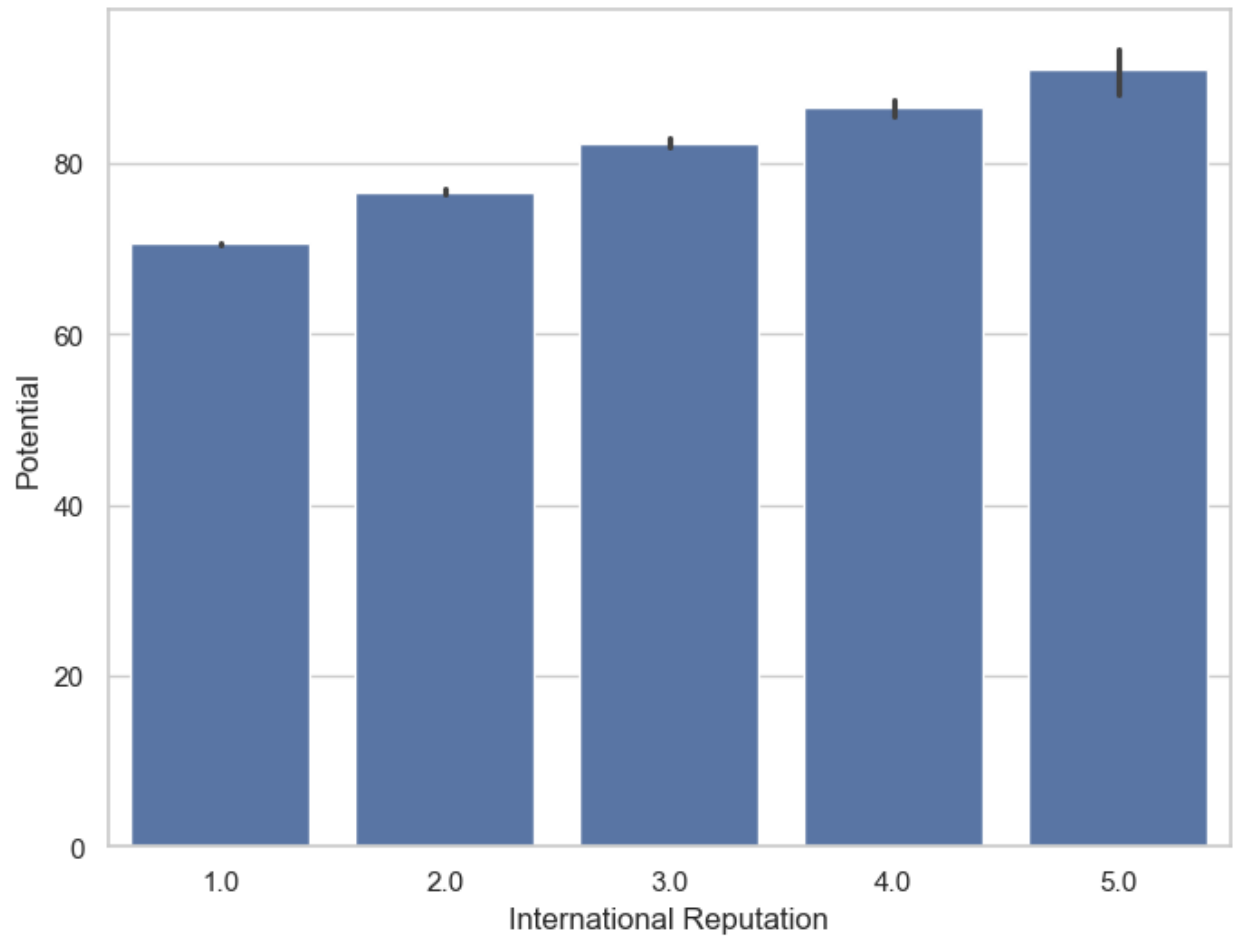
```
f, ax = plt.subplots(figsize=(8, 6))
sns.pointplot(x="International Reputation", y="Potential",
hue="Preferred Foot", data=fifa19, dodge=True)
plt.show()
```



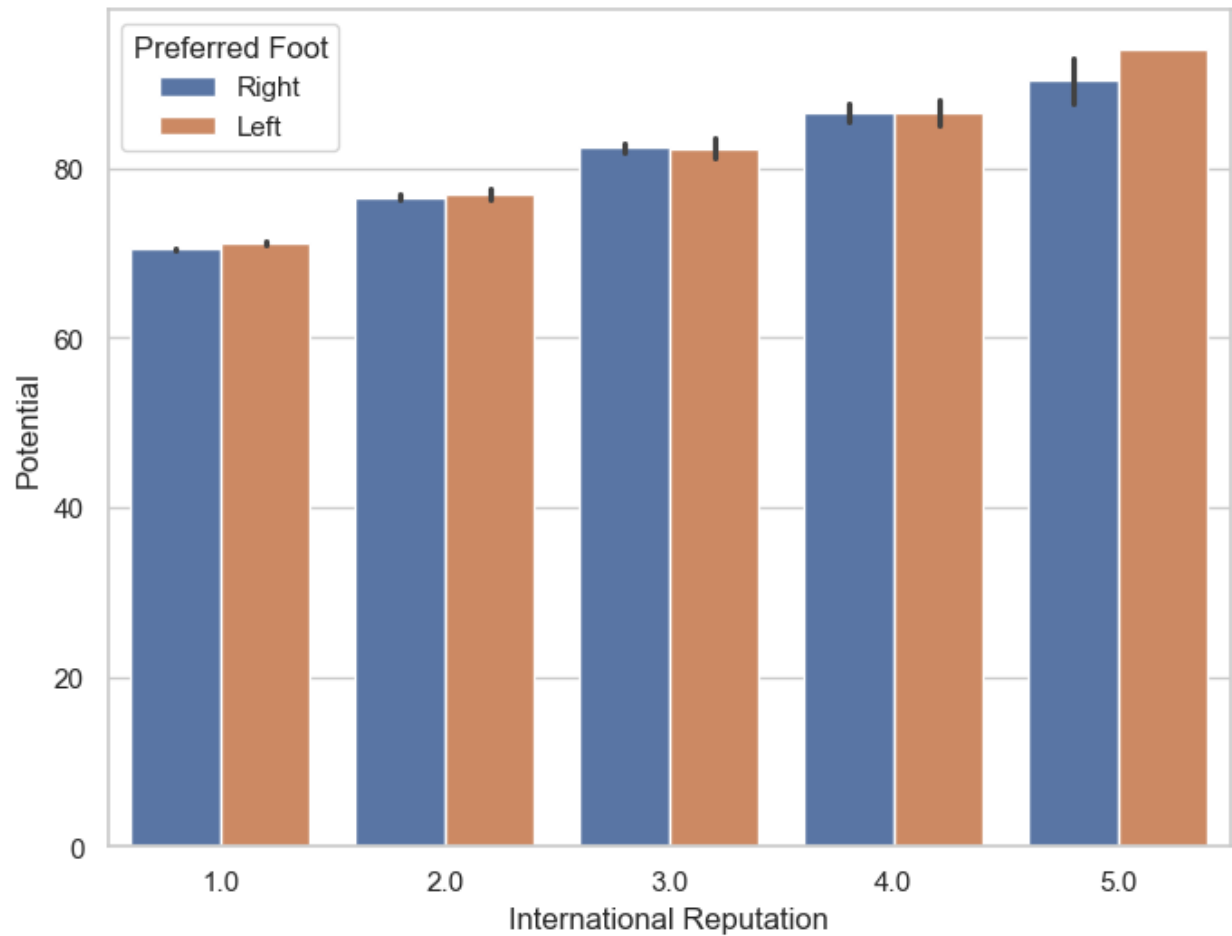
```
f, ax = plt.subplots(figsize=(8, 6))
sns.pointplot(x="International Reputation", y="Potential",
hue="Preferred Foot",
data=fifa19, markers=["o", "x"], linestyles=["-", "--"])
plt.show()
```



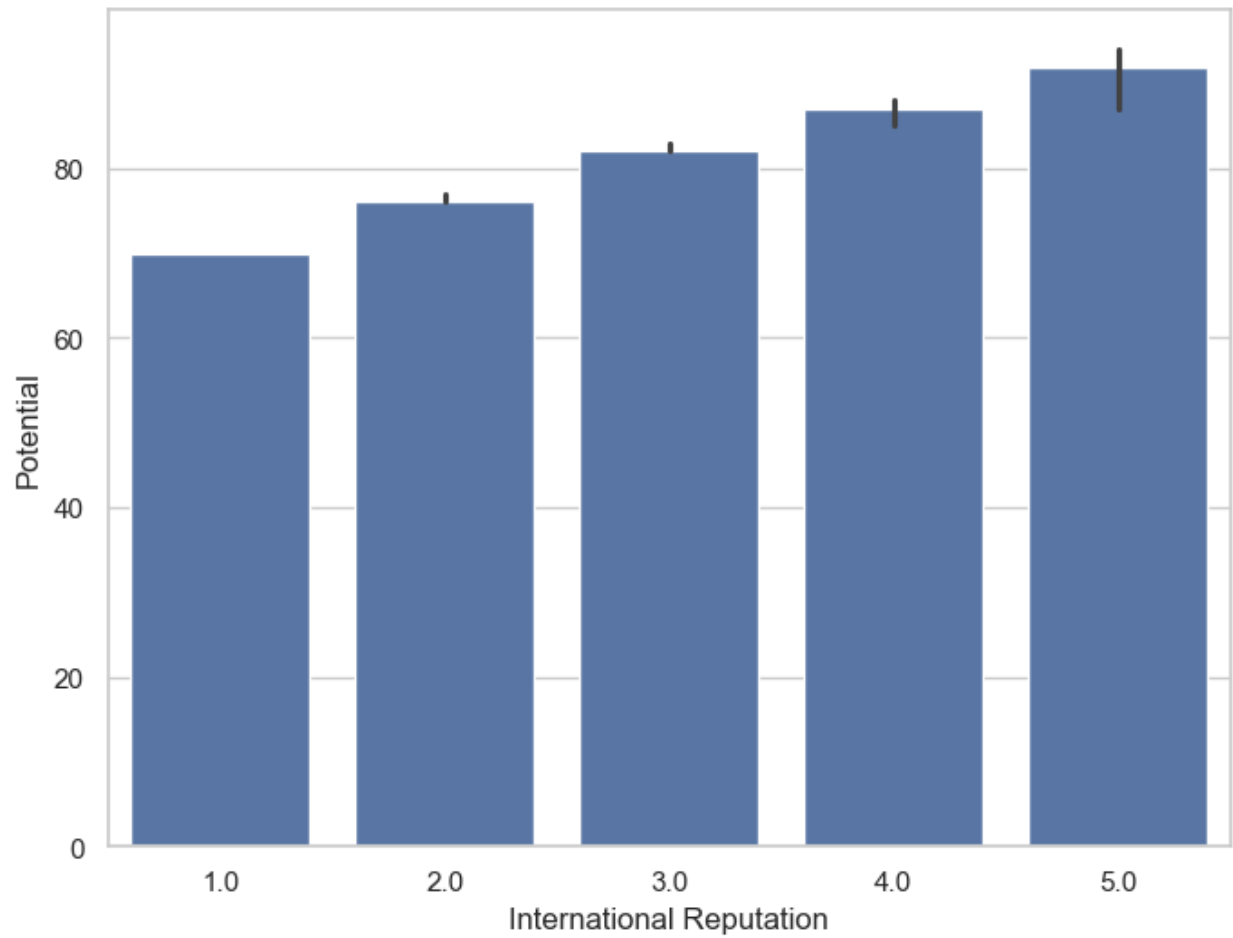
```
f, ax = plt.subplots(figsize=(8, 6))
sns.barplot(x="International Reputation", y="Potential", data=fifa19)
plt.show()
```

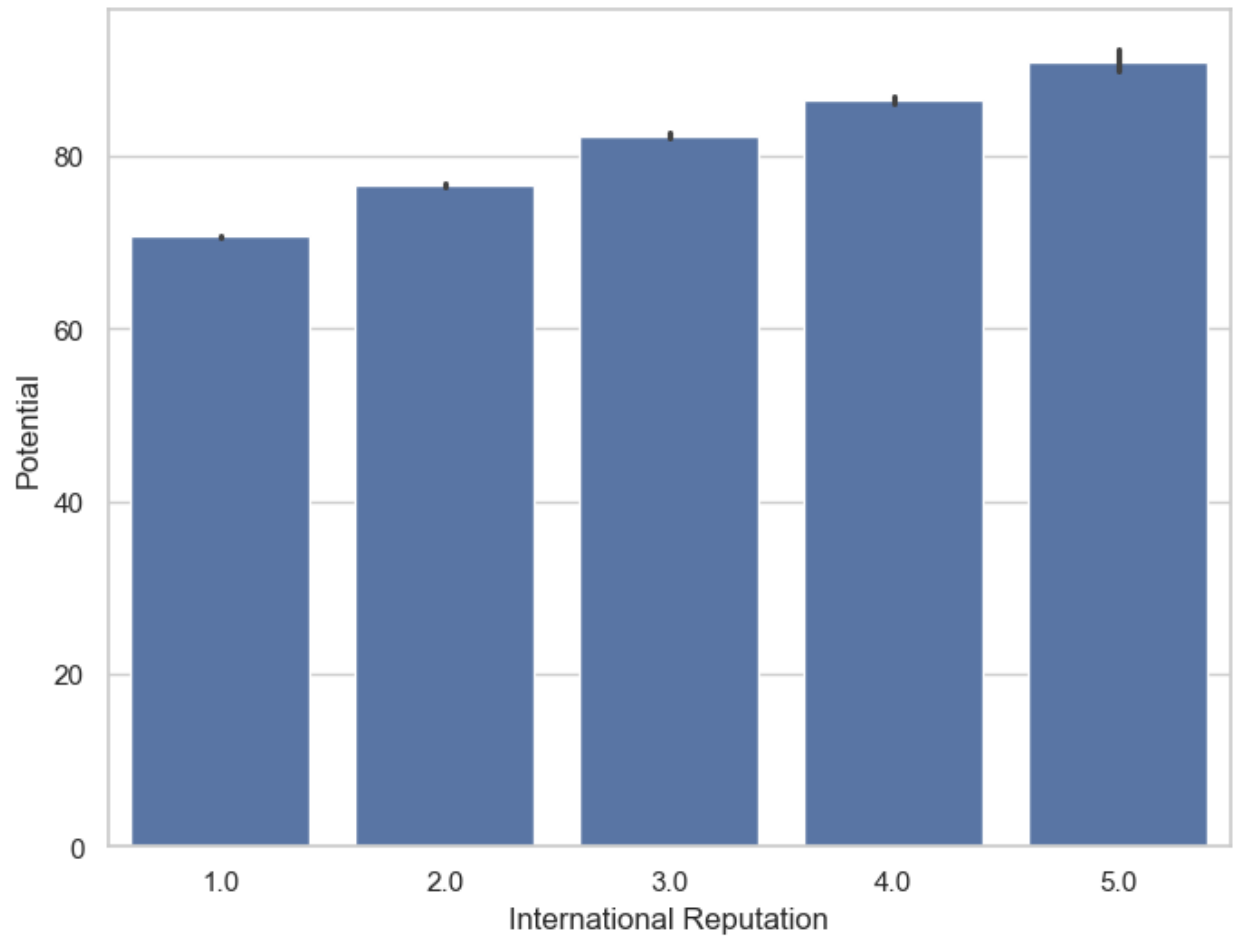
```
f, ax = plt.subplots(figsize=(8, 6))
sns.barplot(x="International Reputation", y="Potential",
            hue="Preferred Foot", data=fifa19)
plt.show()
```



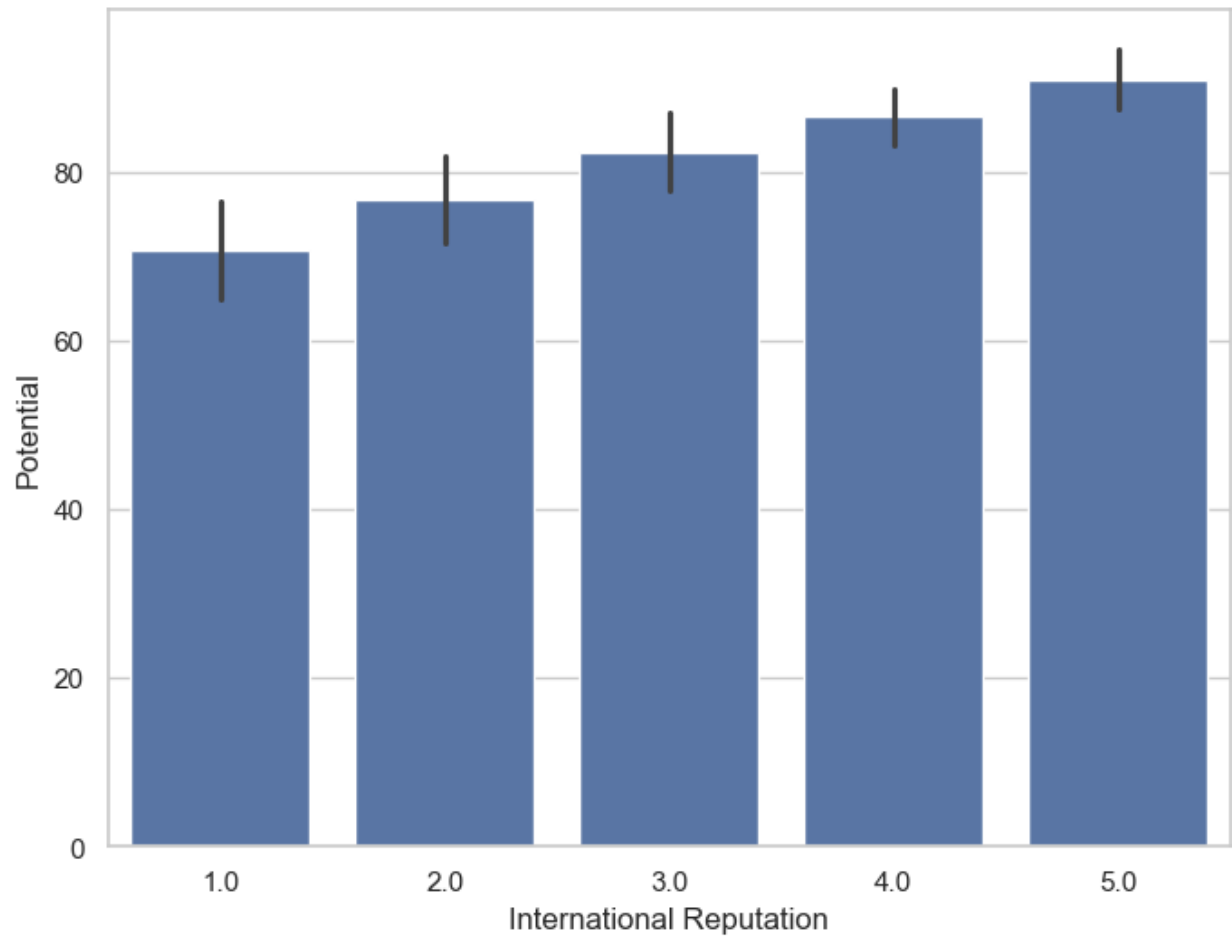
```
from numpy import median
f, ax = plt.subplots(figsize=(8, 6))
sns.barplot(x="International Reputation", y="Potential", data=fifa19,
            estimator=median)
plt.show()
```



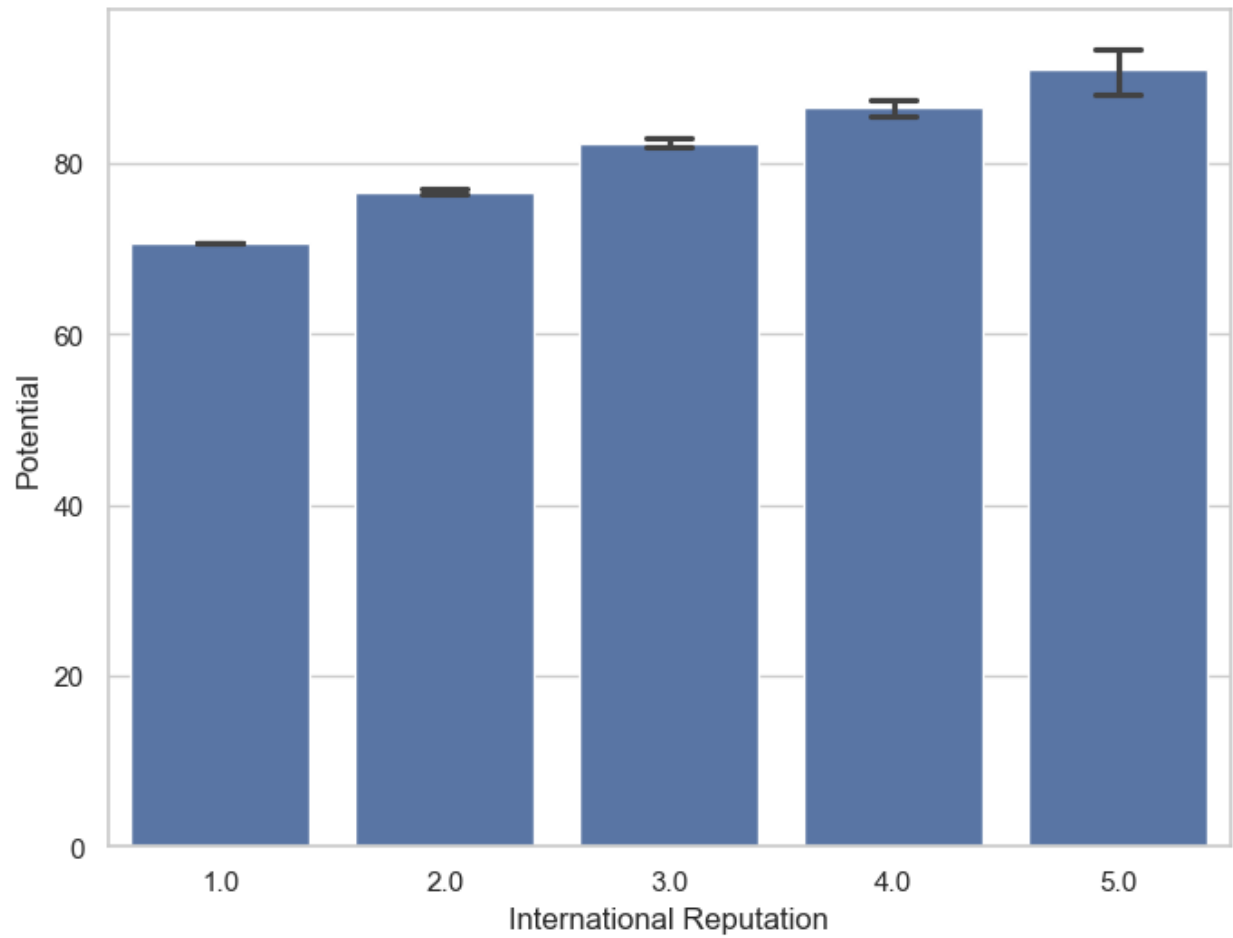
```
f, ax = plt.subplots(figsize=(8, 6))  
sns.barplot(x="International Reputation", y="Potential", data=fifa19,  
ci=68)  
plt.show()
```



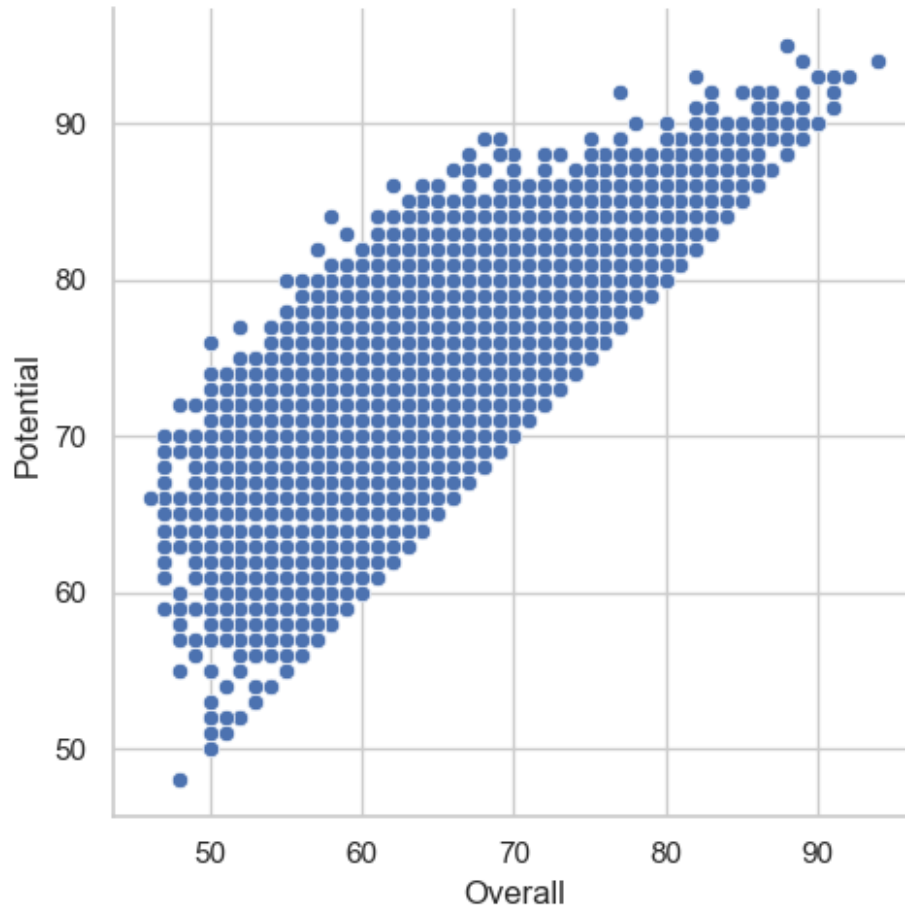
```
f, ax = plt.subplots(figsize=(8, 6))  
sns.barplot(x="International Reputation", y="Potential", data=fifa19,  
ci="sd")  
plt.show()
```



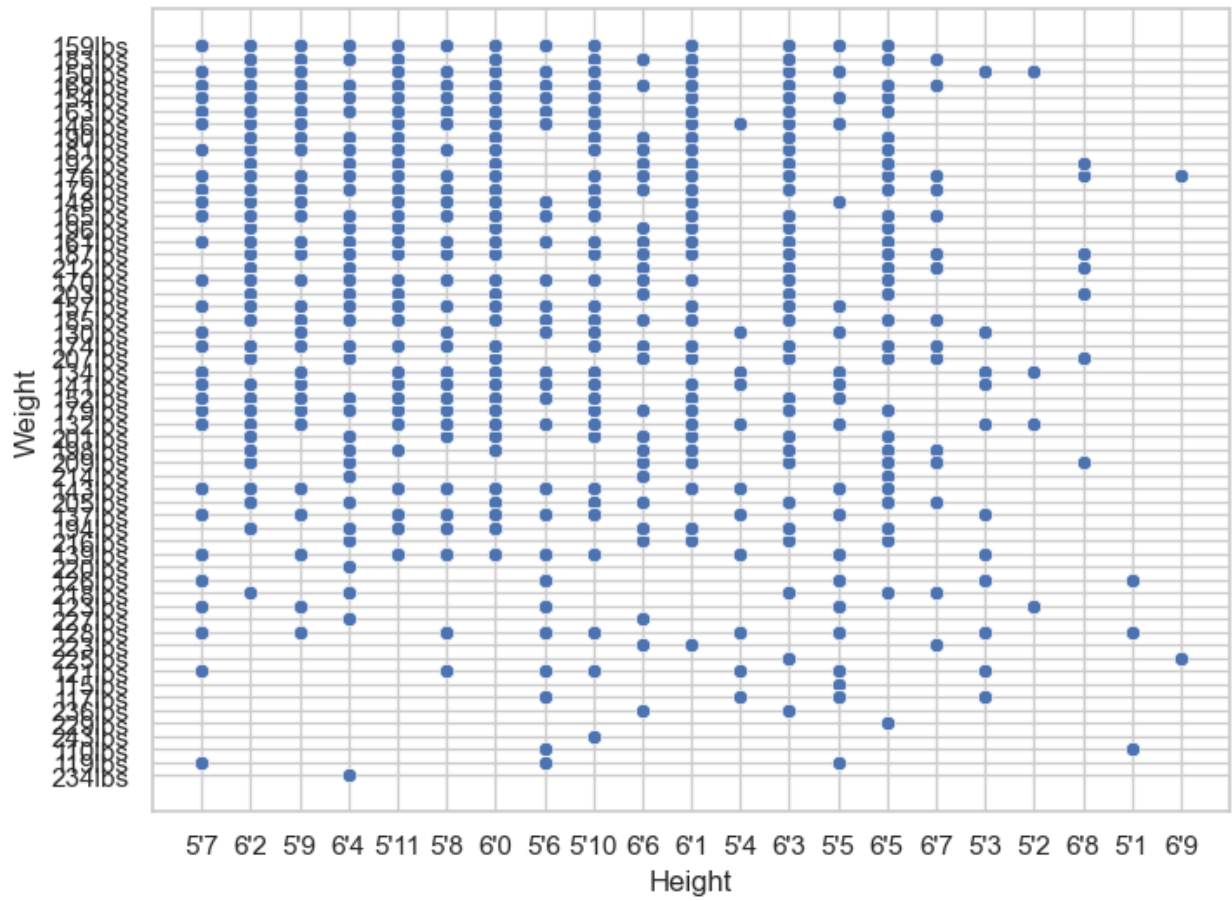
```
f, ax = plt.subplots(figsize=(8, 6))
sns.barplot(x="International Reputation", y="Potential", data=fifa19,
capsize=0.2)
plt.show()
```



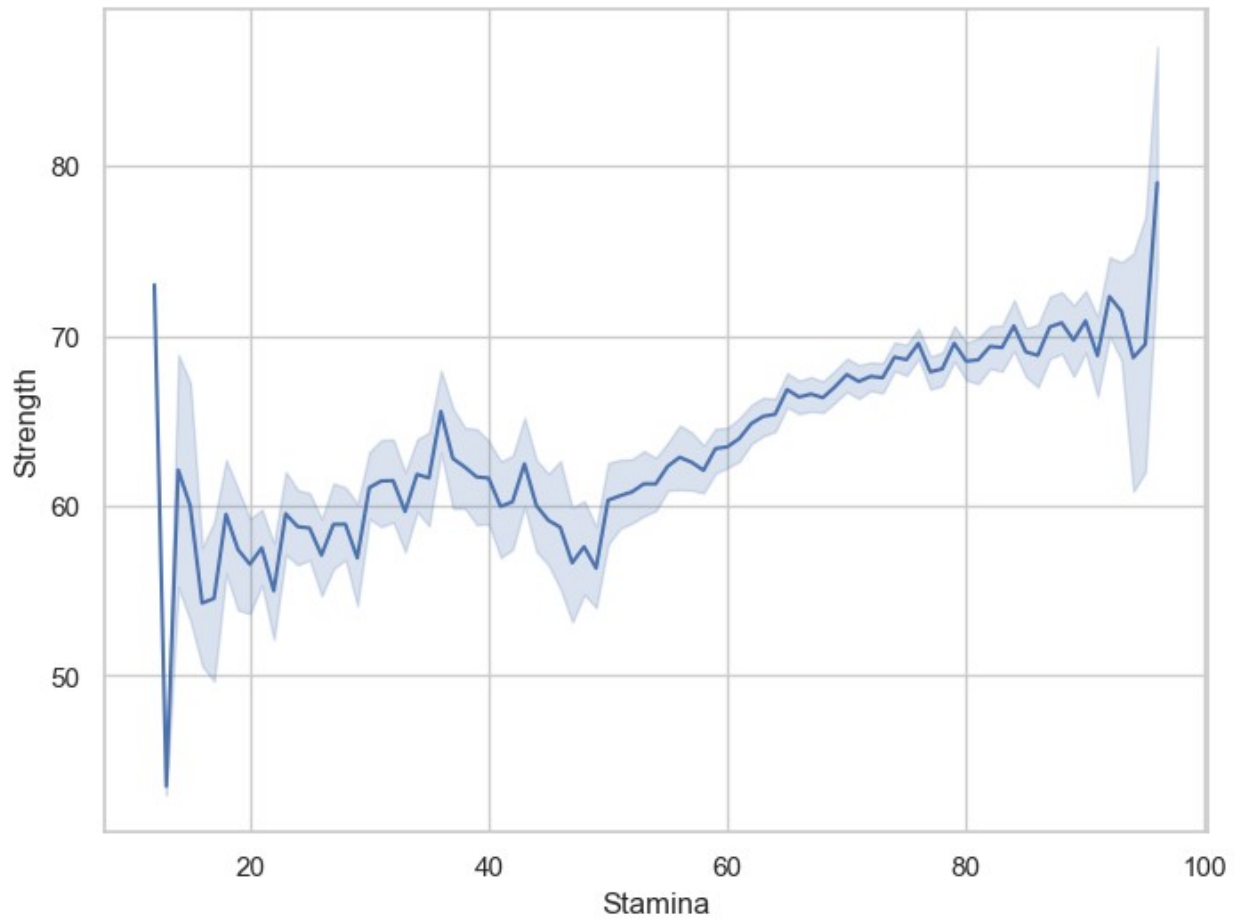
```
g = sns.relplot(x="Overall", y="Potential", data=fifa19)  
plt.show()
```



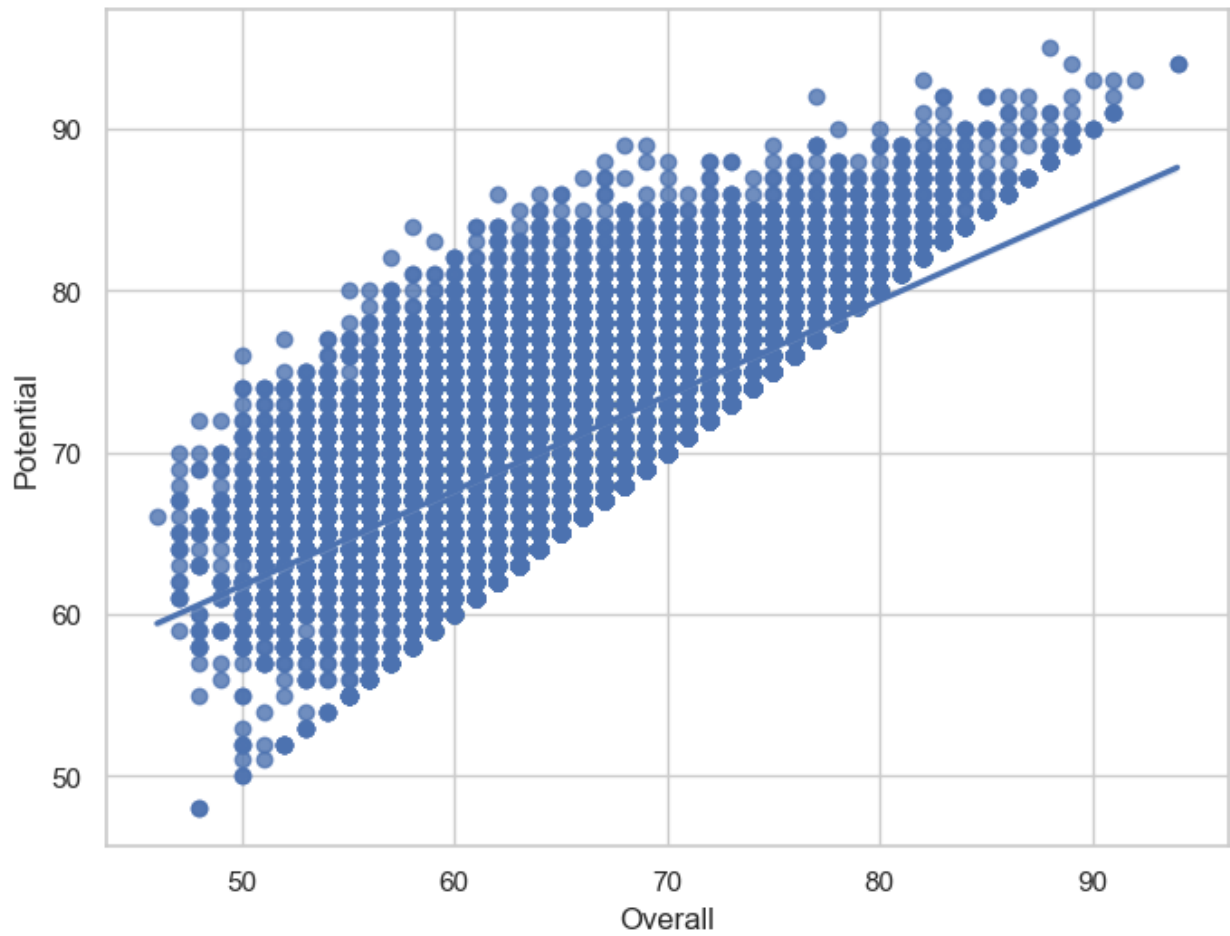
```
f, ax = plt.subplots(figsize=(8, 6))
sns.scatterplot(x="Height", y="Weight", data=fifa19)
plt.show()
```



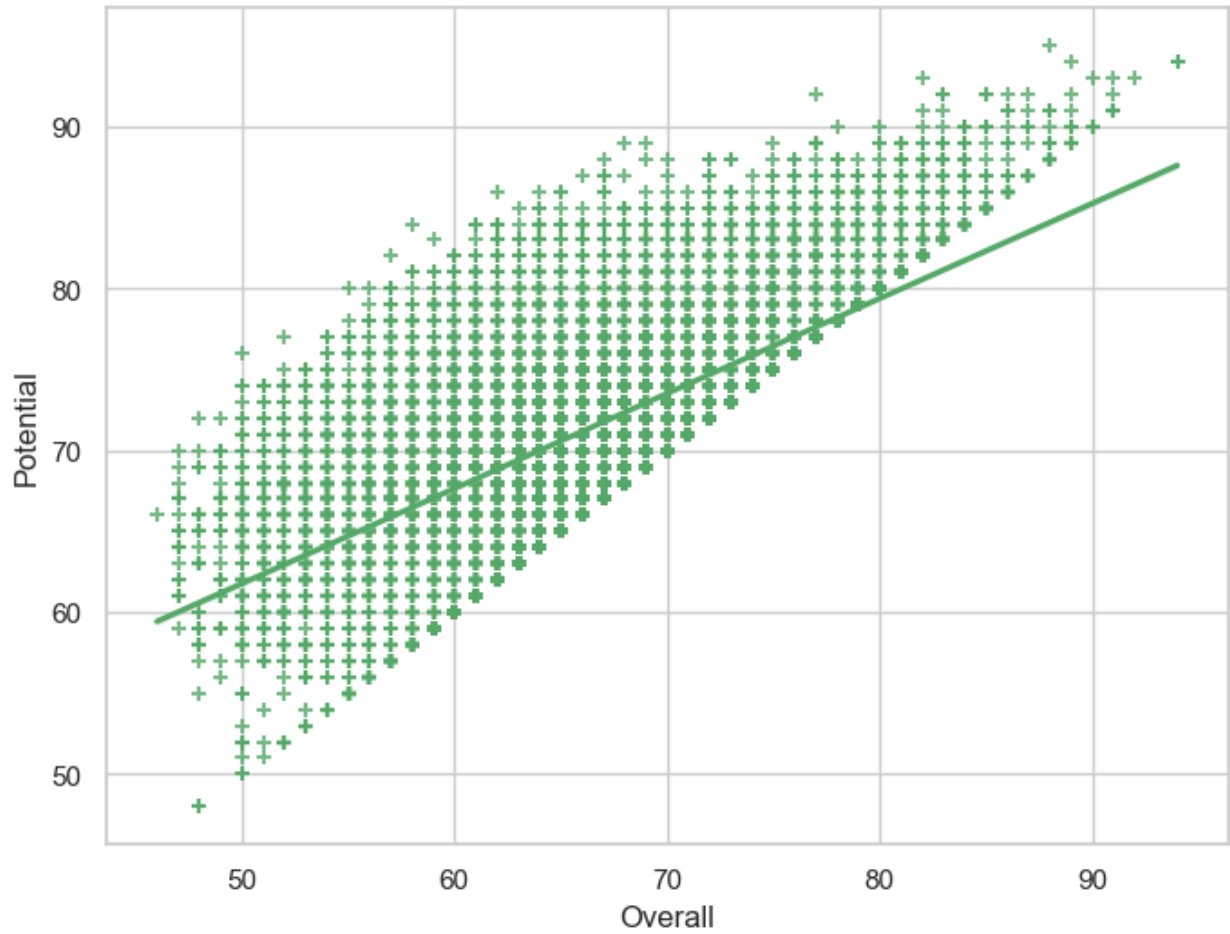
```
f, ax = plt.subplots(figsize=(8, 6))
ax = sns.lineplot(x="Stamina", y="Strength", data=fifa19)
plt.show()
```

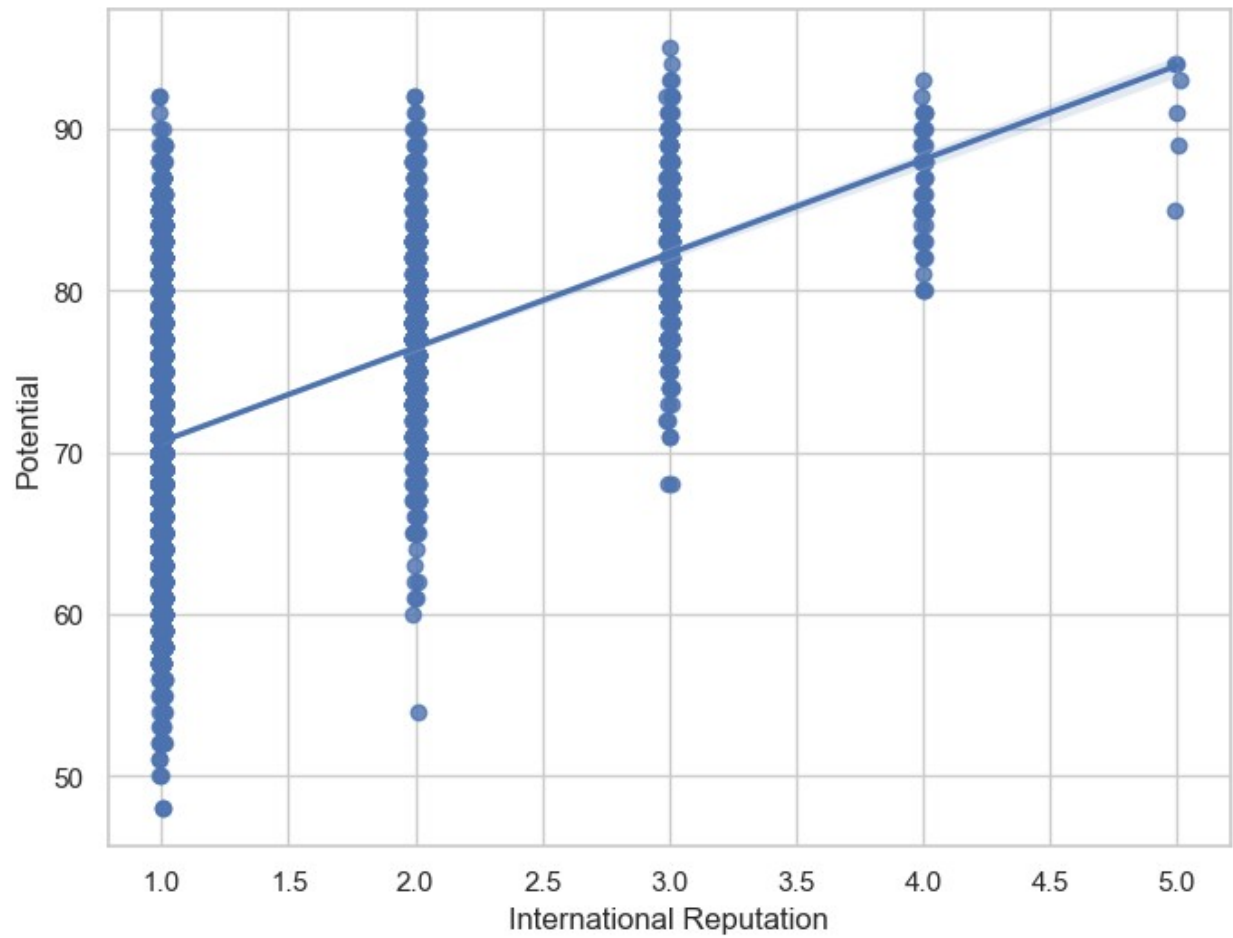
```
f, ax = plt.subplots(figsize=(8, 6))
ax = sns.regplot(x="Overall", y="Potential", data=fifa19)
plt.show()
```



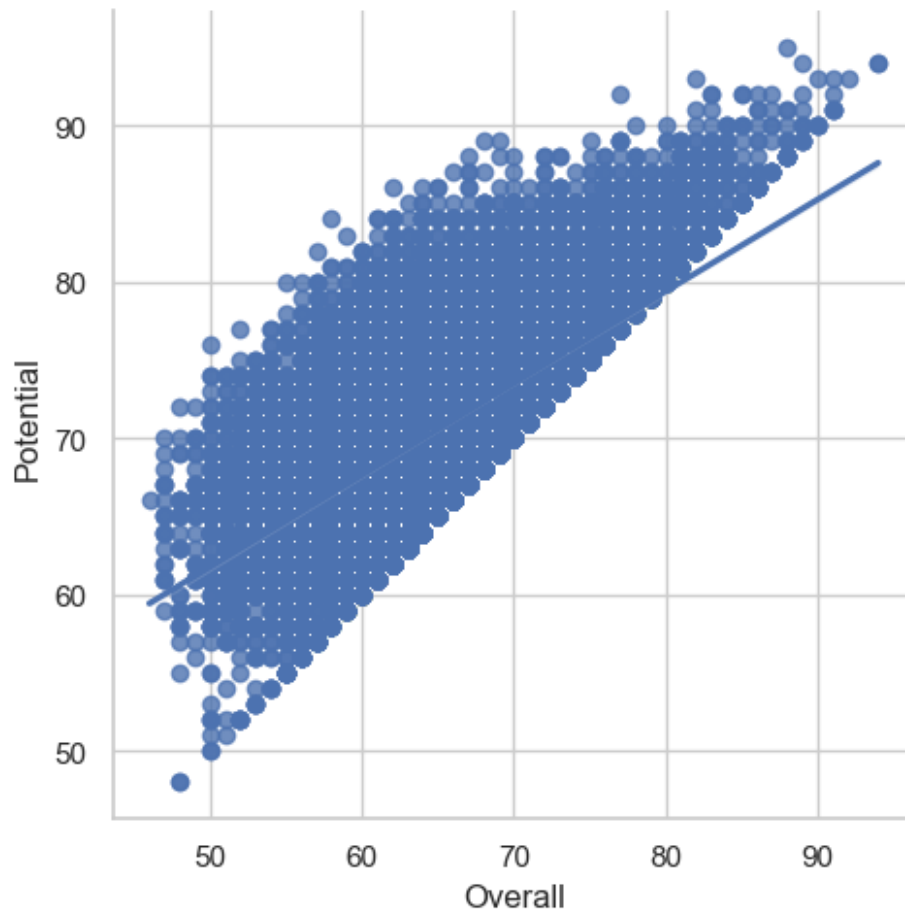
```
f, ax = plt.subplots(figsize=(8, 6))
ax = sns.regplot(x="Overall", y="Potential", data=fifa19, color="g",
marker="+")
plt.show()
```



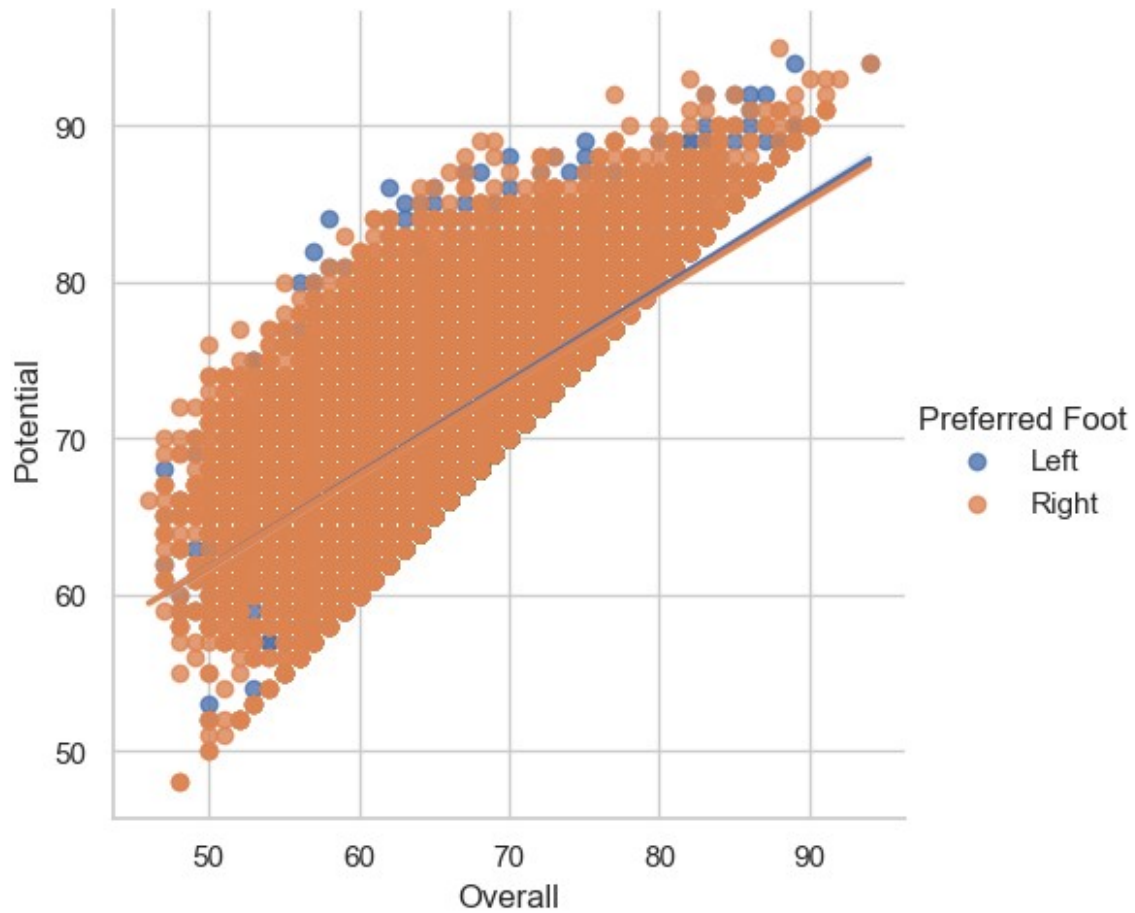
```
f, ax = plt.subplots(figsize=(8, 6))
sns.regplot(x="International Reputation", y="Potential", data=fifa19,
x_jitter=.01)
plt.show()
```



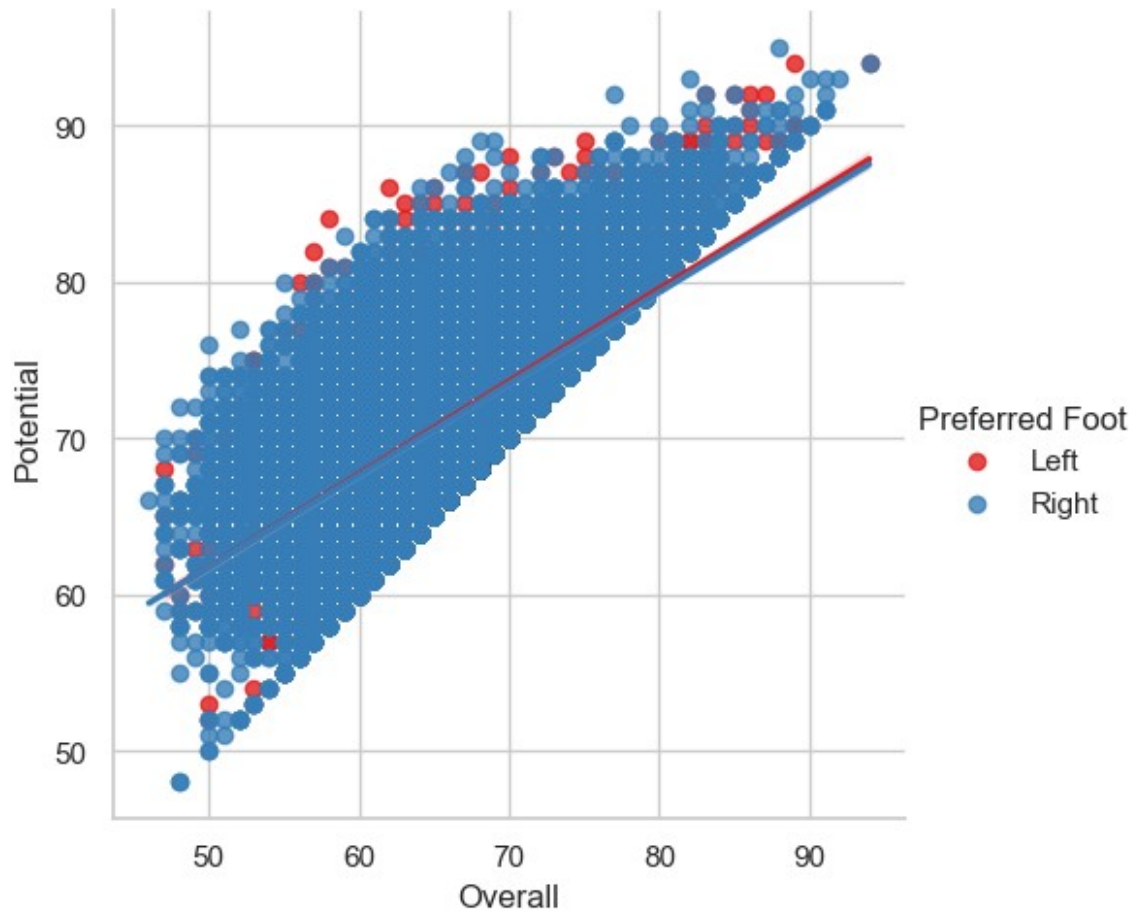
```
g= sns.lmplot(x="Overall", y="Potential", data=fifa19)
plt.show()
```



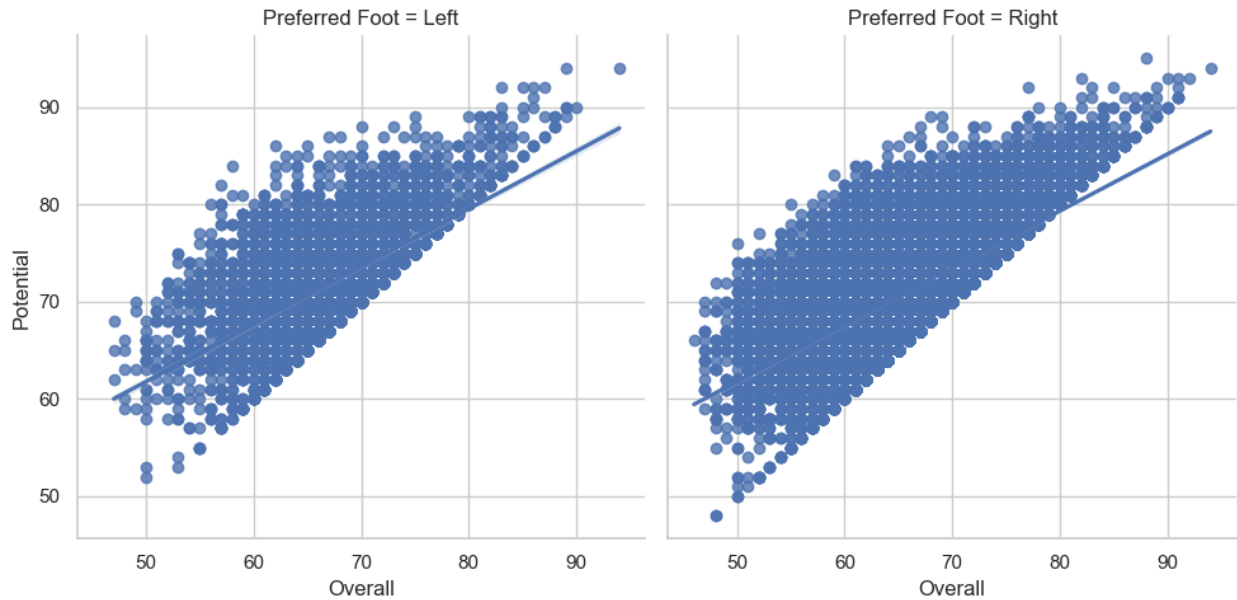
```
g= sns.lmplot(x="Overall", y="Potential", hue="Preferred Foot",  
data=fifa19)  
plt.show()
```



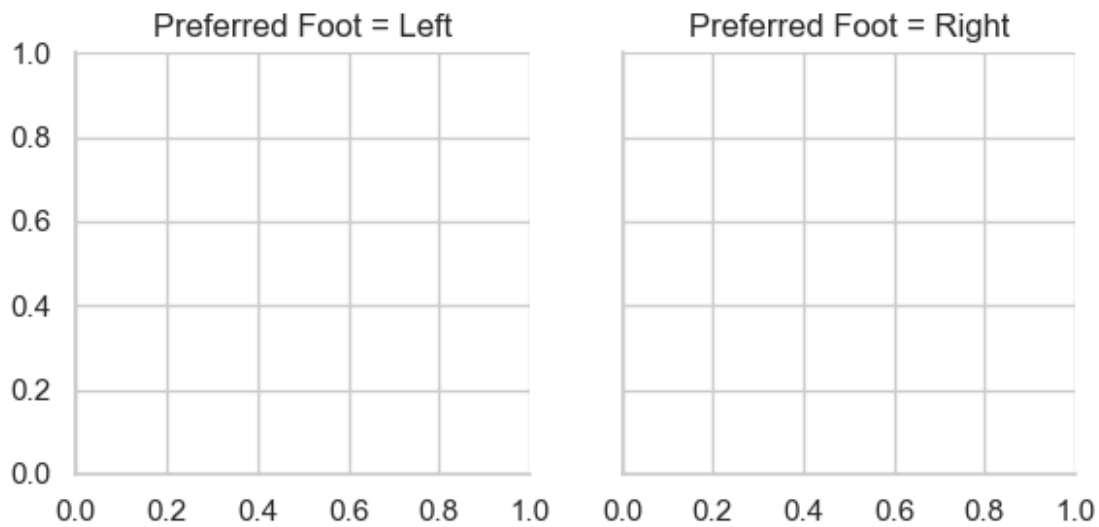
```
g= sns.lmplot(x="Overall", y="Potential", hue="Preferred Foot",  
data=fifa19, palette="Set1")  
plt.show()
```



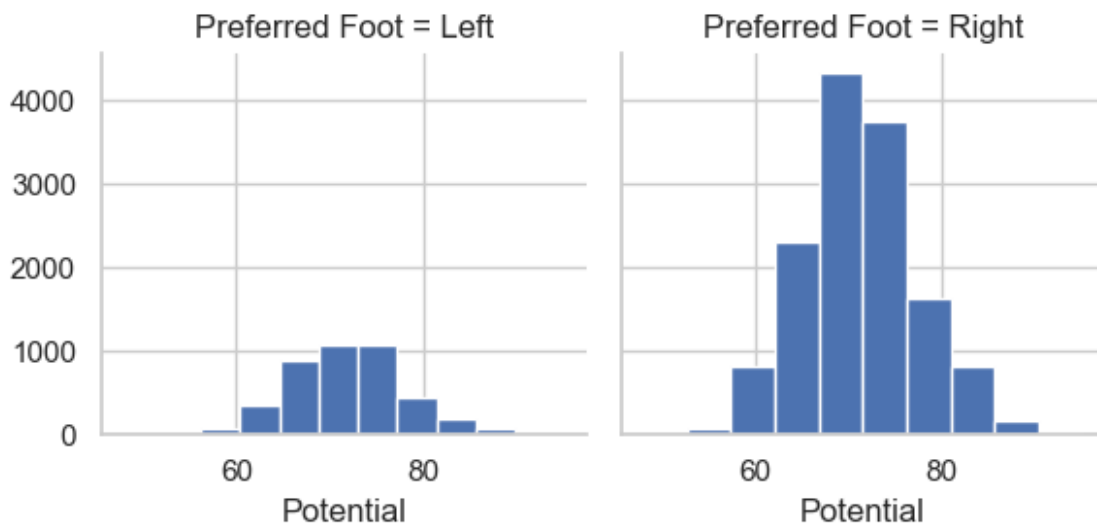
```
g= sns.lmplot(x="Overall", y="Potential", col="Preferred Foot",  
data=fifa19)  
plt.show()
```



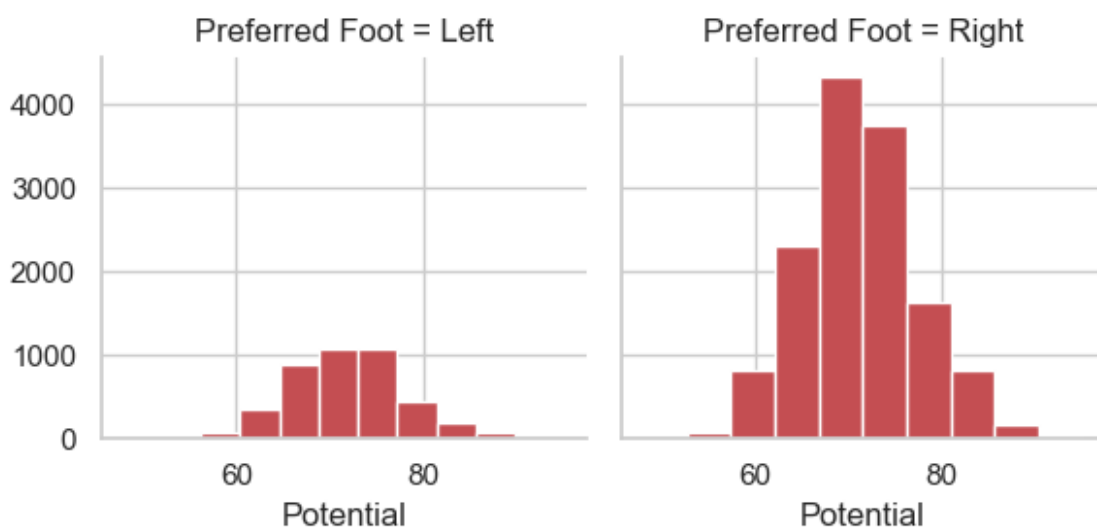
```
g = sns.FacetGrid(fifa19, col="Preferred Foot")
plt.show()
```



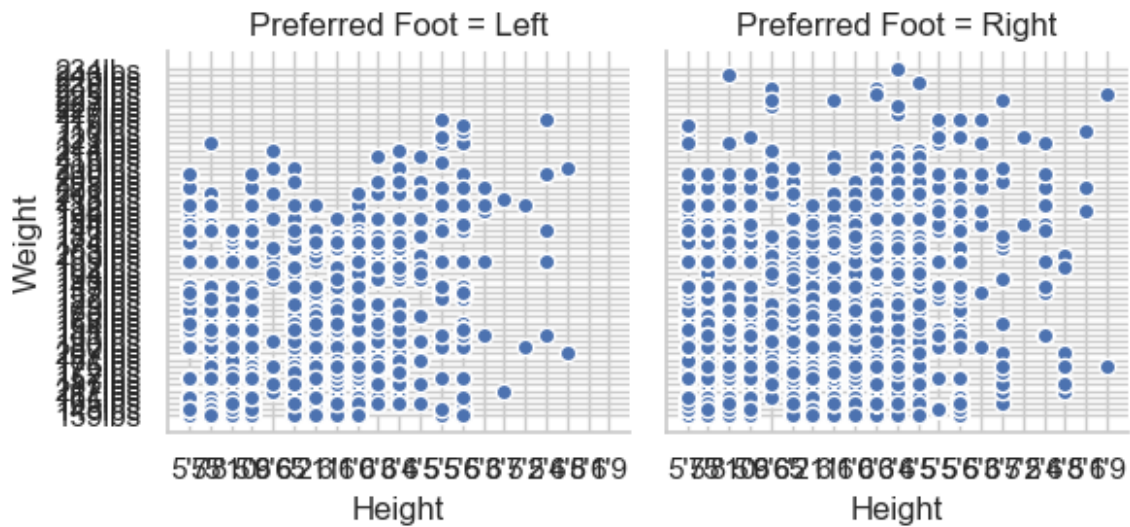
```
g = sns.FacetGrid(fifa19, col="Preferred Foot")
g = g.map(plt.hist, "Potential")
plt.show()
```

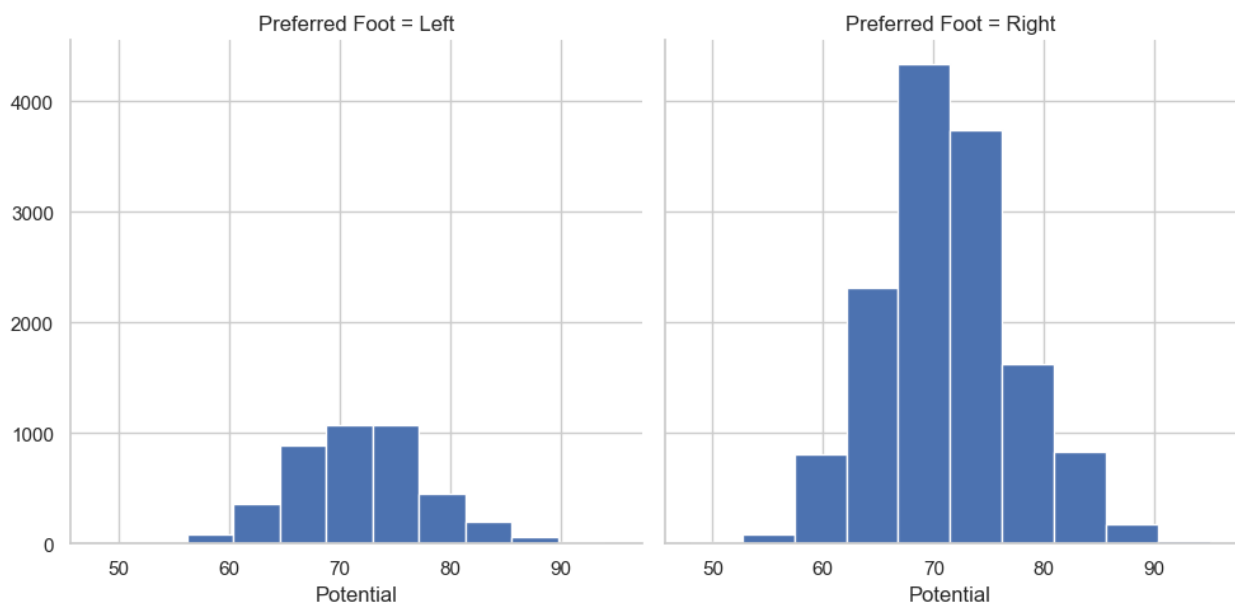
```
g = sns.FacetGrid(fifa19, col="Preferred Foot")
g = g.map(plt.hist, "Potential", bins=10, color="r")
plt.show()
```



```
g = sns.FacetGrid(fifa19, col="Preferred Foot")
g = (g.map(plt.scatter, "Height", "Weight",
edgecolor="w").add_legend())
plt.show()
```

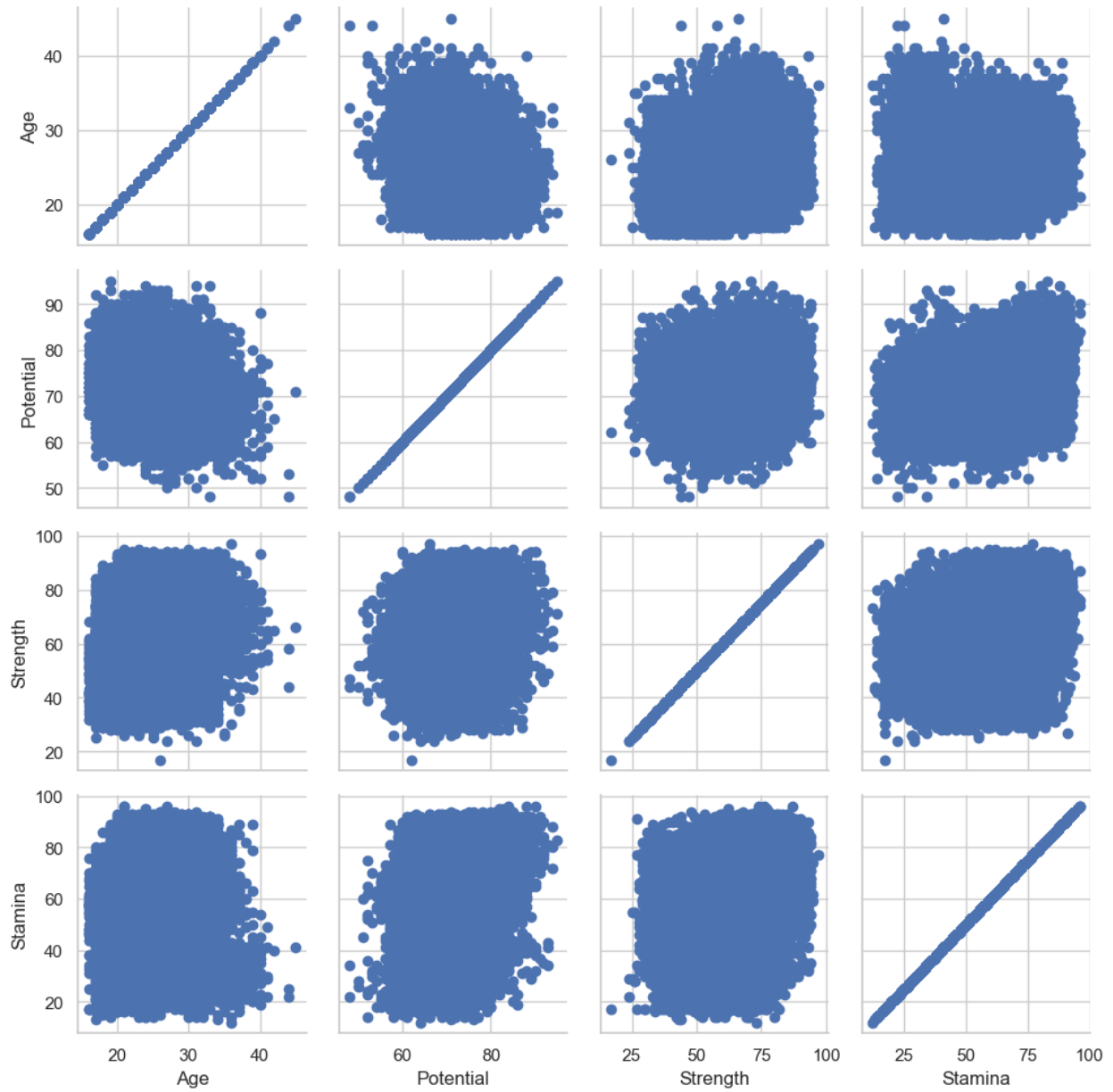


```
g = sns.FacetGrid(fifa19, col="Preferred Foot", height=5, aspect=1)
g = g.map(plt.hist, "Potential")
plt.show()
```

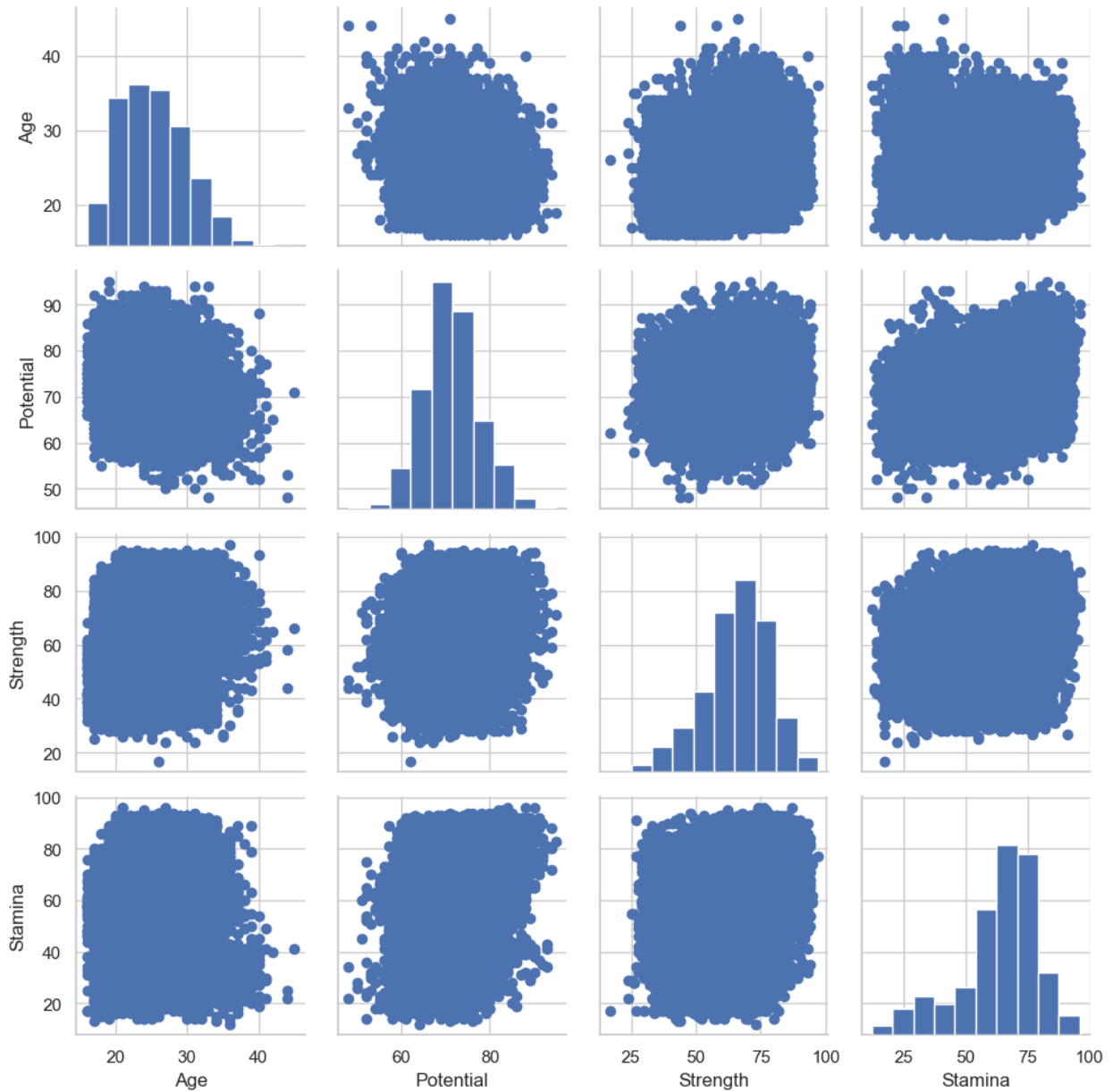


```
fifa19_new = fifa19[['Age', 'Potential', 'Strength', 'Stamina',
'Preferred Foot']]

g = sns.PairGrid(fifa19_new)
g = g.map(plt.scatter)
plt.show()
```

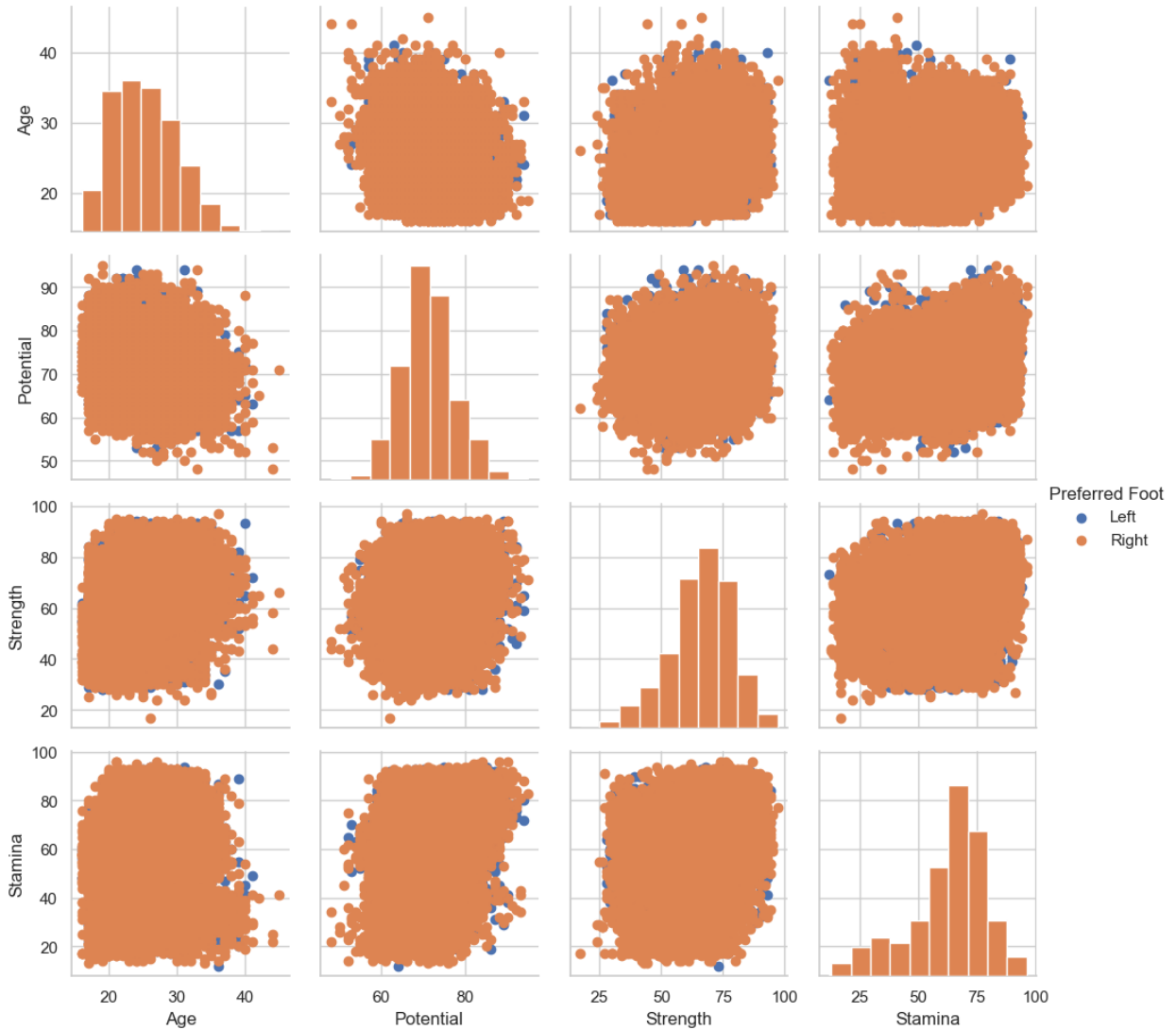


```
g = sns.PairGrid(fifa19_new)
g = g.map_diag(plt.hist)
g = g.map_offdiag(plt.scatter)
plt.show()
```

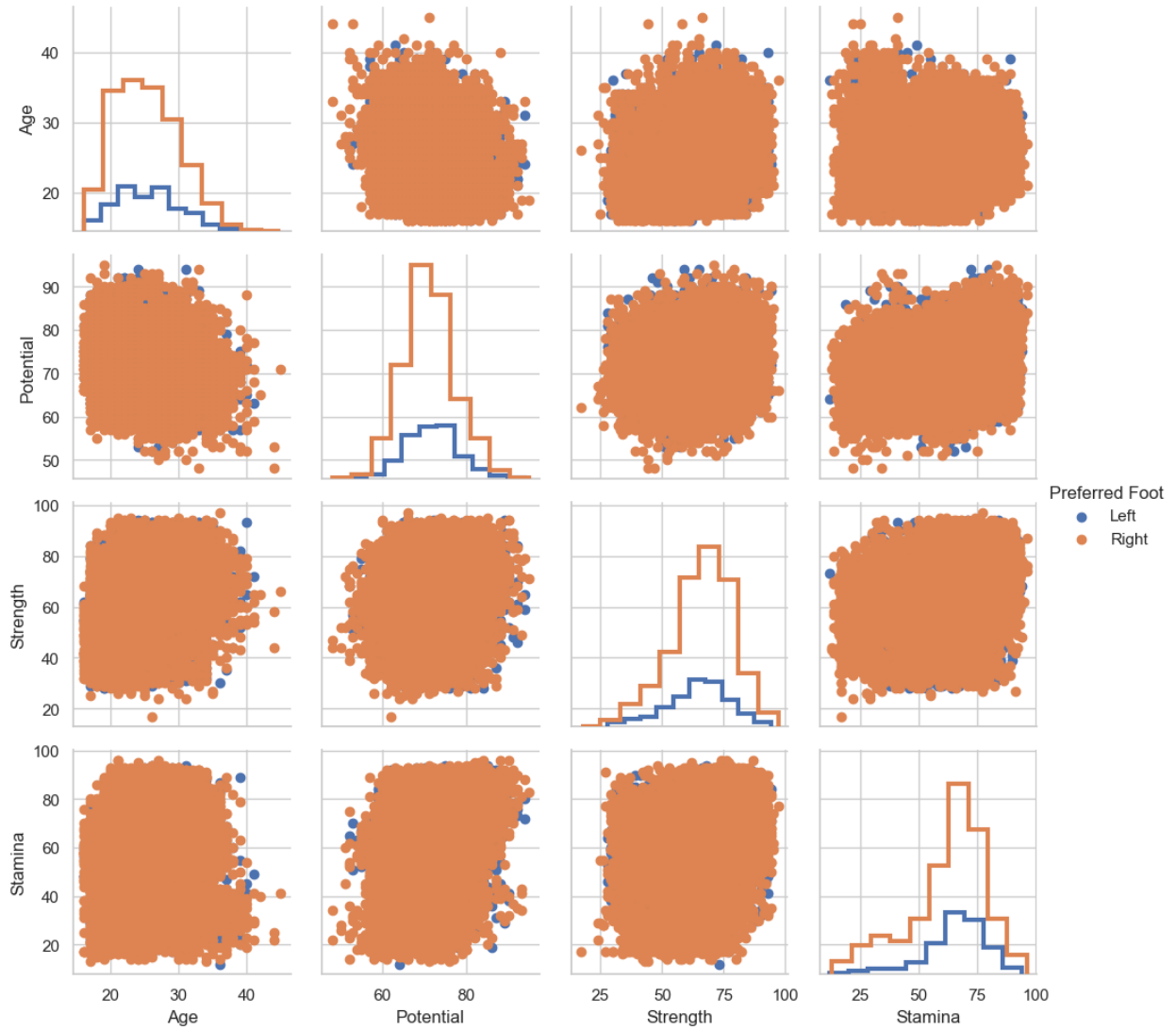


```
g = sns.PairGrid(fifa19_new, hue="Preferred Foot")
g = g.map_diag(plt.hist)
g = g.map_offdiag(plt.scatter)
g = g.add_legend()

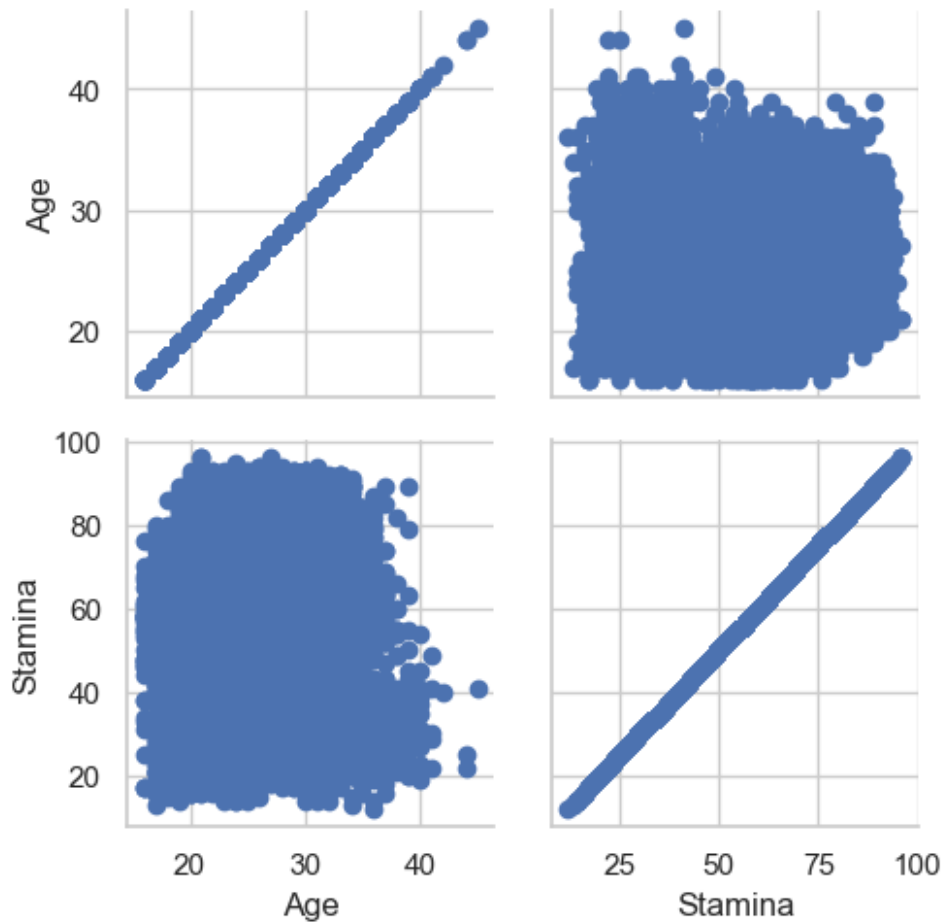
plt.show()
```



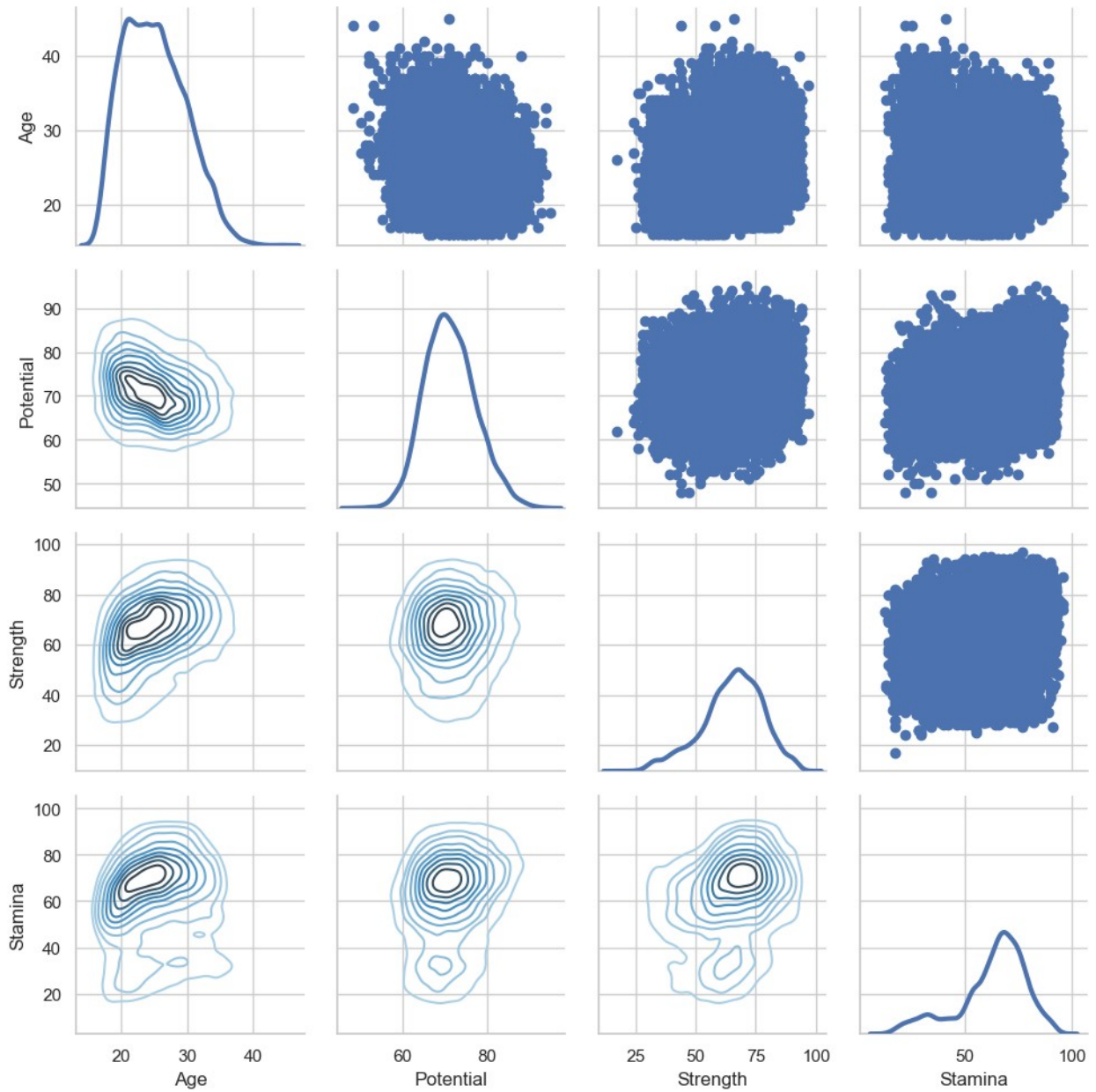
```
g = sns.PairGrid(fifa19_new, hue="Preferred Foot")
g = g.map_diag(plt.hist, histtype="step", linewidth=3)
g = g.map_offdiag(plt.scatter)
g = g.add_legend()
plt.show()
```



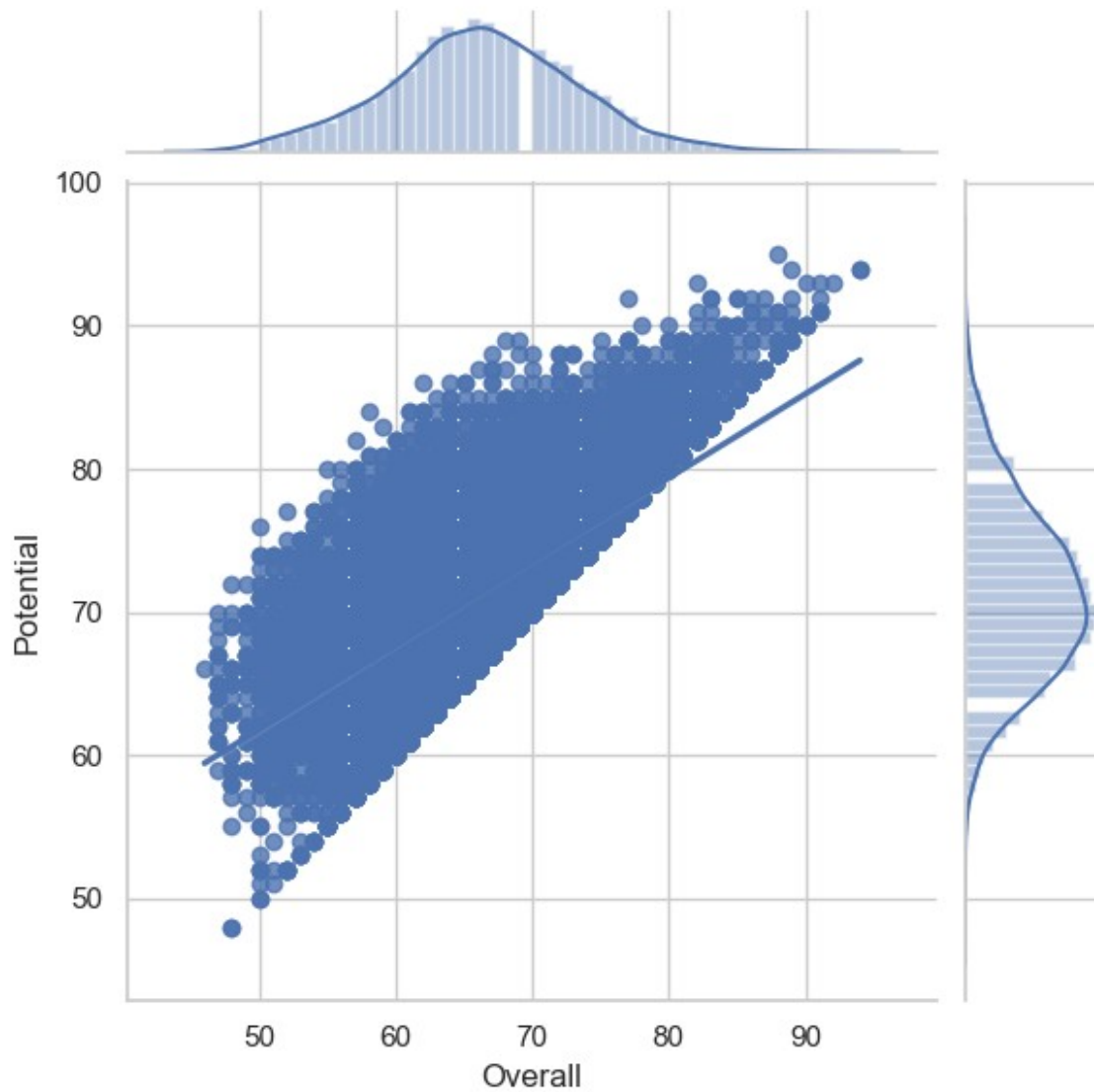
```
g = sns.PairGrid(fifa19_new, vars=['Age', 'Stamina'])
g = g.map(plt.scatter)
plt.show()
```



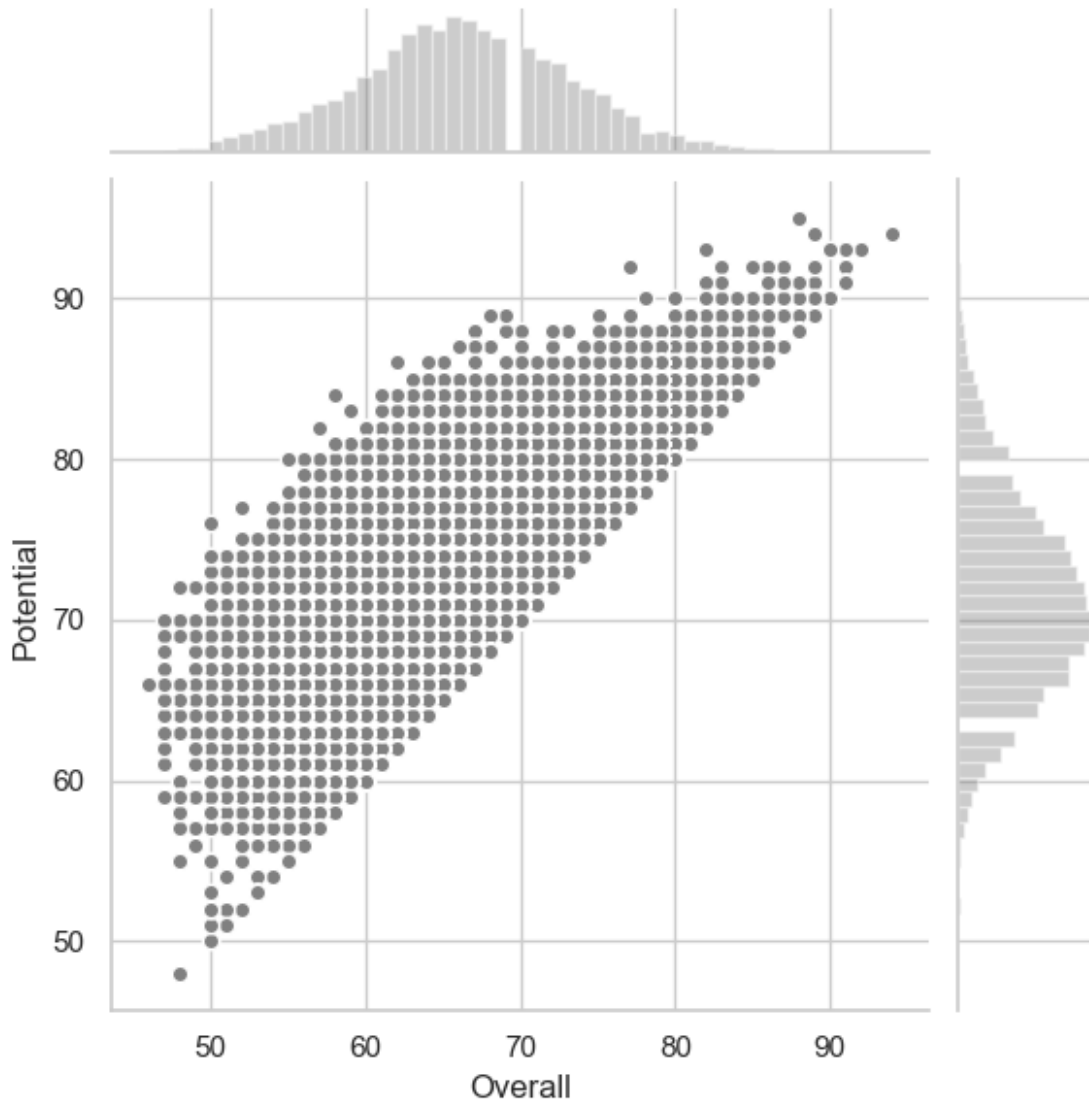
```
g = sns.PairGrid(fifa19_new)
g = g.map_upper(plt.scatter)
g = g.map_lower(sns.kdeplot, cmap="Blues_d")
g = g.map_diag(sns.kdeplot, lw=3, legend=False)
plt.show()
```



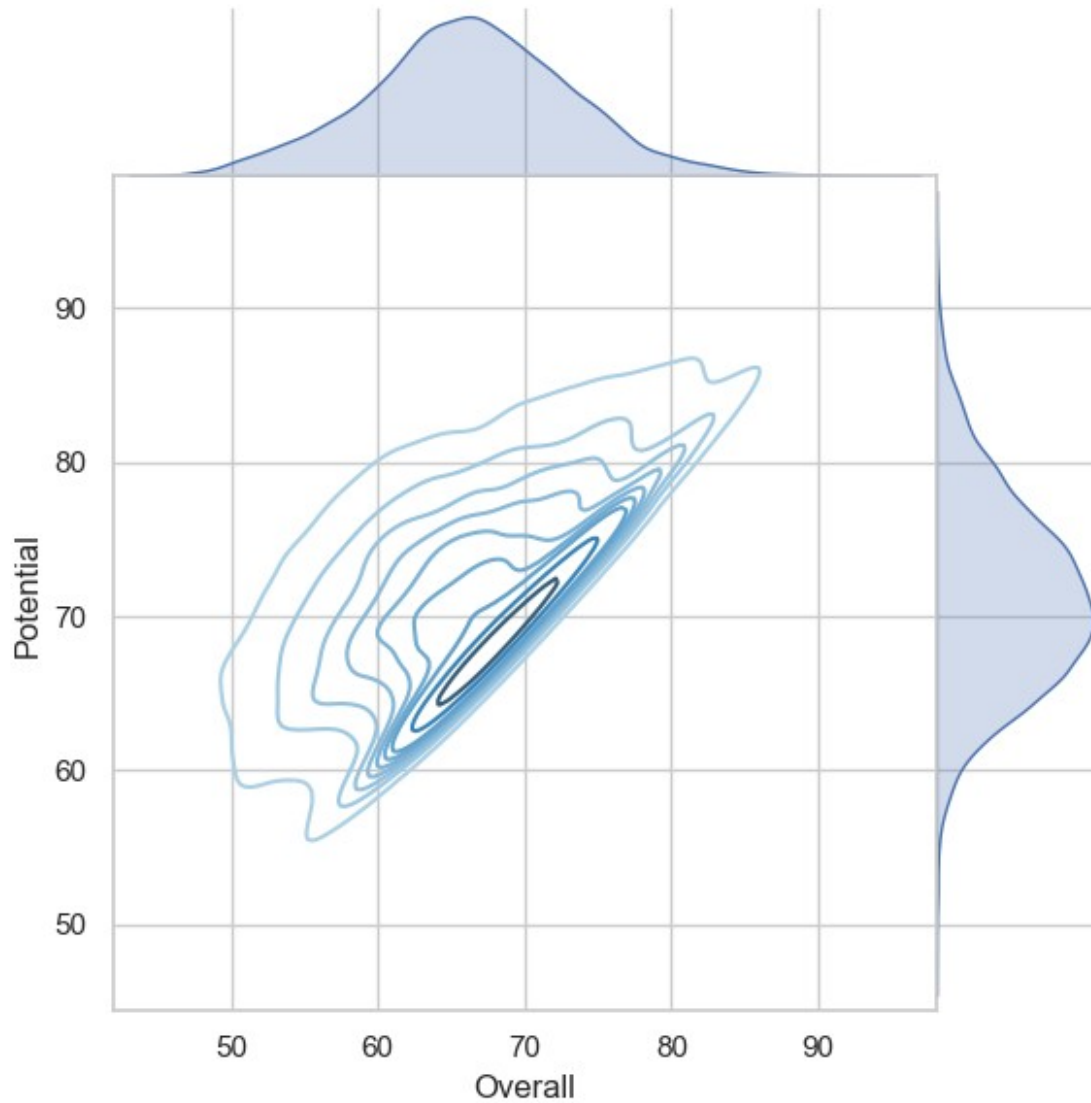
```
g = sns.JointGrid(x="Overall", y="Potential", data=fifa19)
g = g.plot(sns.regplot, sns.distplot)
plt.show()
```

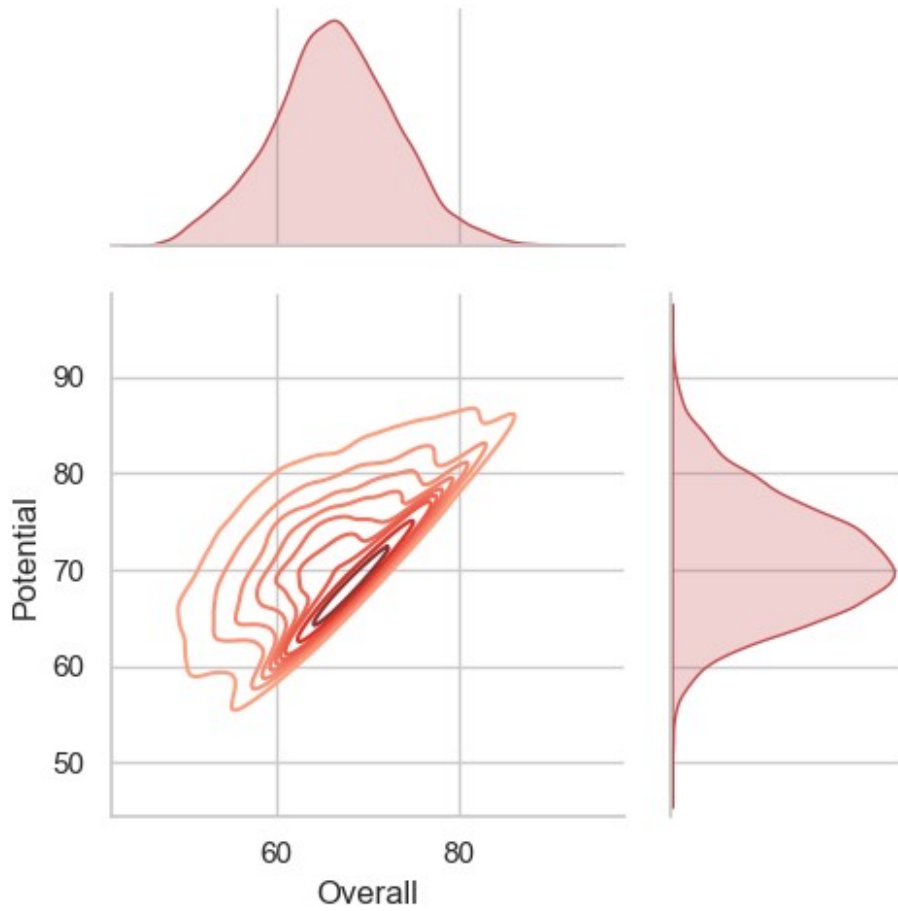
```
g = sns.JointGrid(x="Overall", y="Potential", data=fifa19)
g = g.plot_joint(plt.scatter, color=".5", edgecolor="white")
g = g.plot_marginals(sns.histplot, kde=False, color=".5")
plt.show()
```



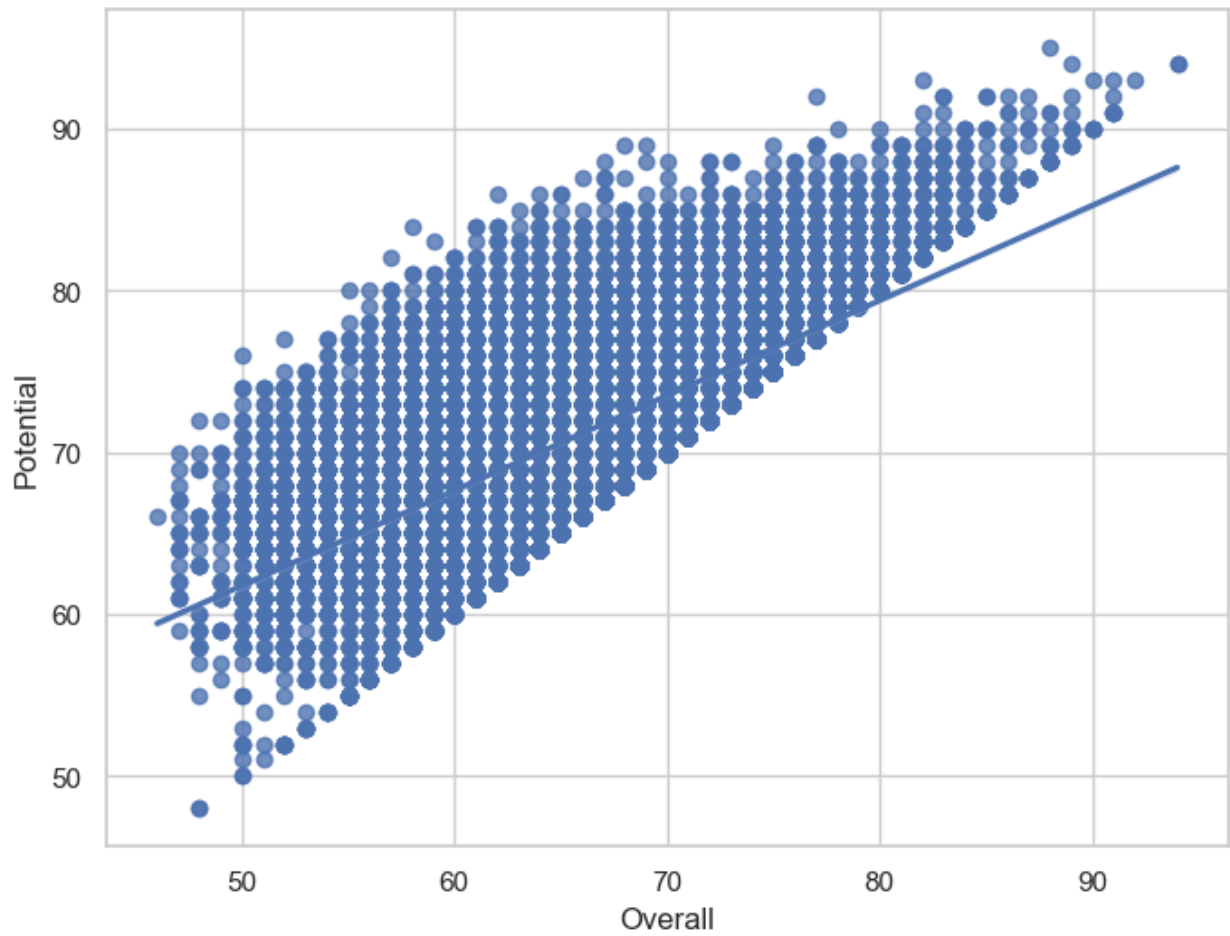
```
g = sns.JointGrid(x="Overall", y="Potential", data=fifa19, space=0)
g = g.plot_joint(sns.kdeplot, cmap="Blues_d")
g = g.plot_marginals(sns.kdeplot, shade=True)
plt.show()
```



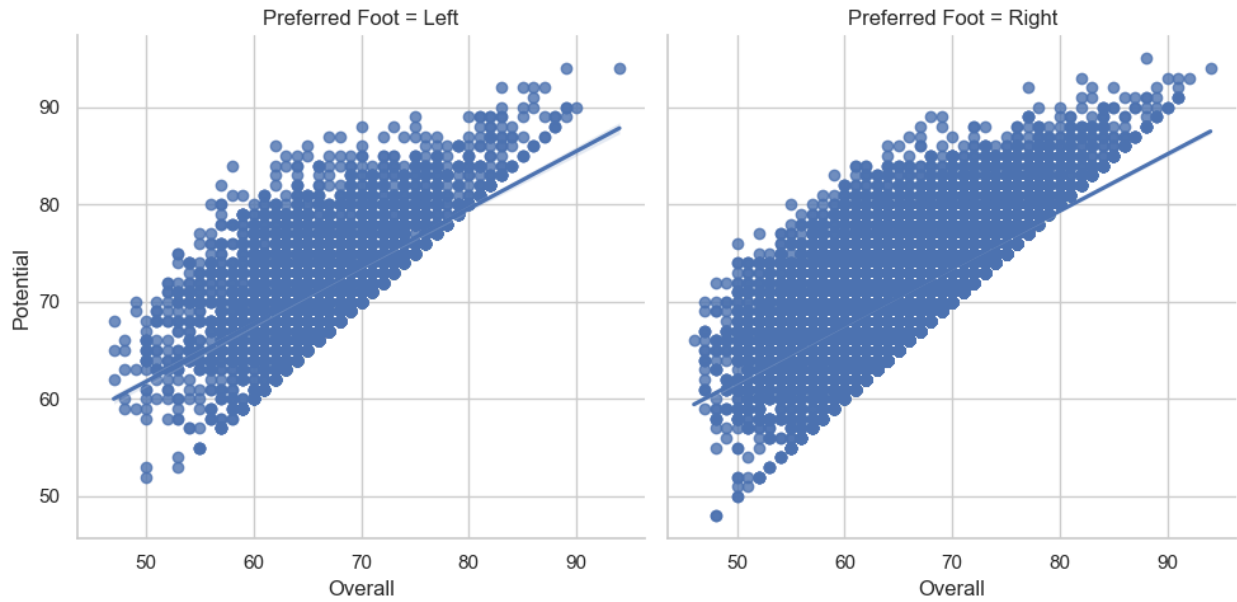
```
g = sns.JointGrid(x="Overall", y="Potential", data=fifa19, height=5,  
ratio=2)  
g = g.plot_joint(sns.kdeplot, cmap="Reds_d")  
g = g.plot_marginals(sns.kdeplot, color="r", shade=True)  
plt.show()
```



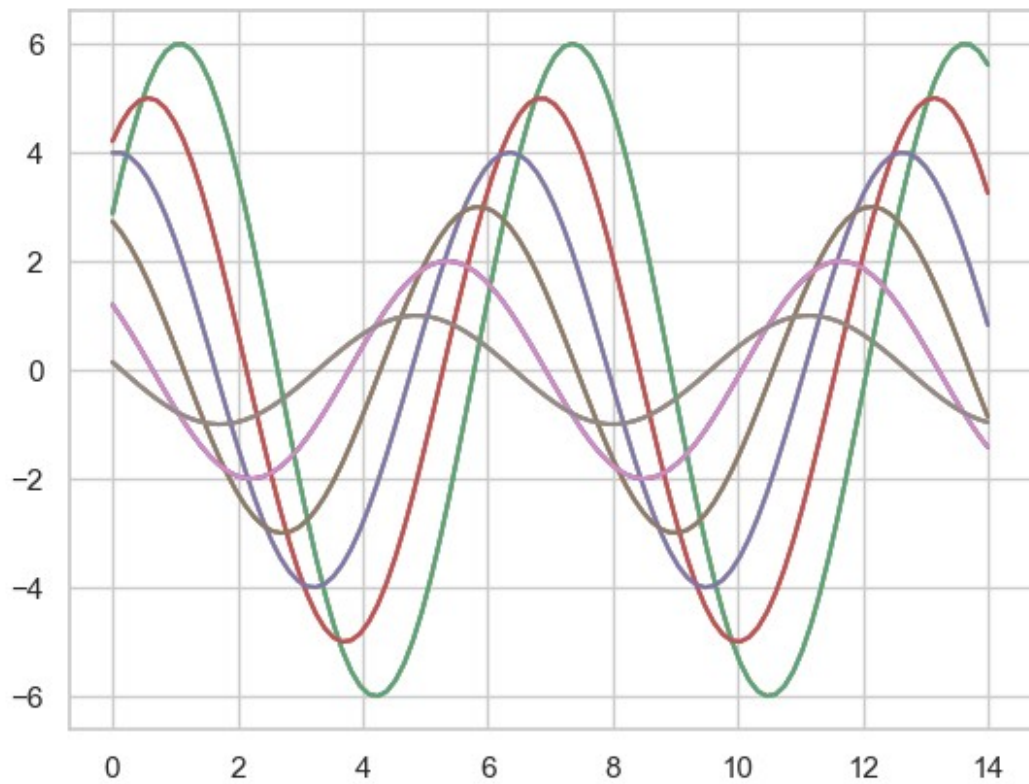
```
f, ax = plt.subplots(figsize=(8, 6))
ax = sns.regplot(x="Overall", y="Potential", data=fifa19);
plt.show()
```



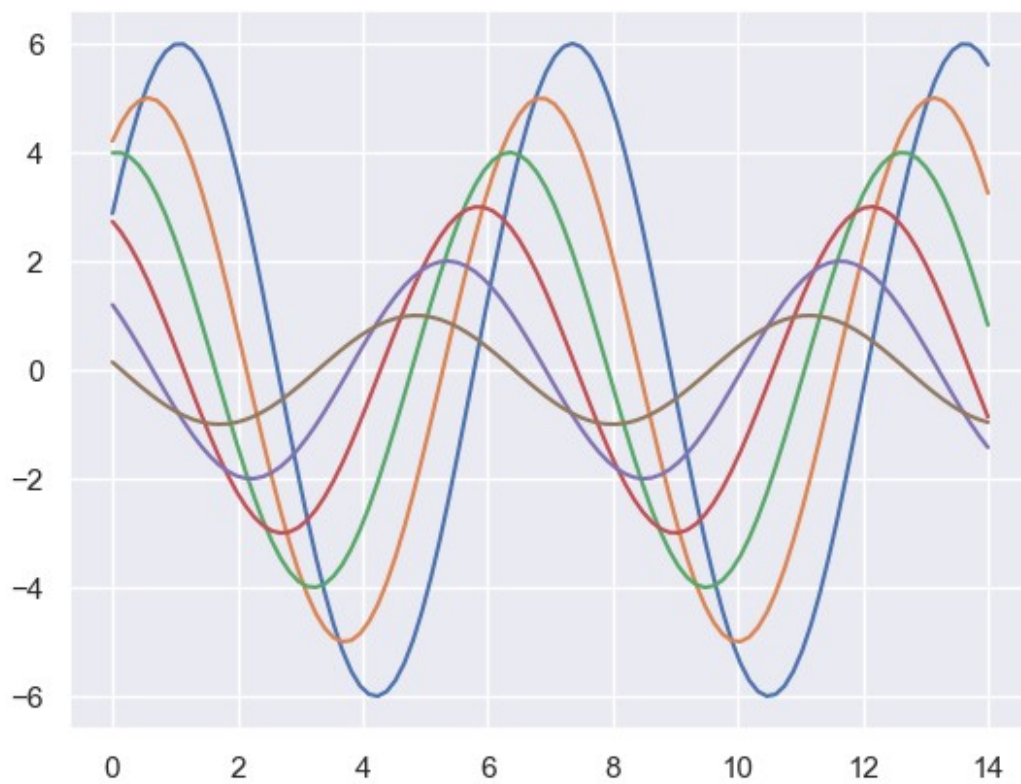
```
sns.lmplot(x="Overall", y="Potential", col="Preferred Foot",  
data=fifa19, col_wrap=2, height=5, aspect=1)  
plt.show()
```



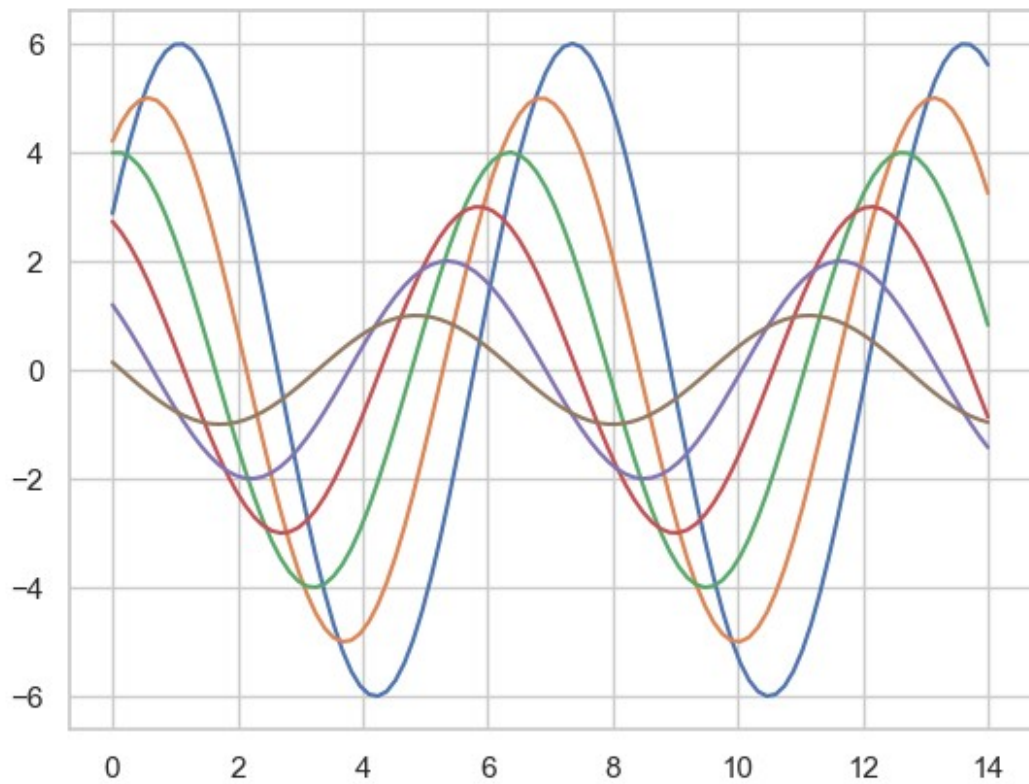
```
def sinplot(flip=1):  
    x = np.linspace(0, 14, 100)  
    for i in range(1, 7):  
        plt.plot(x, np.sin(x + i * .5) * (7 - i) * flip)  
  
plt.show()  
  
sinplot()  
plt.show()
```



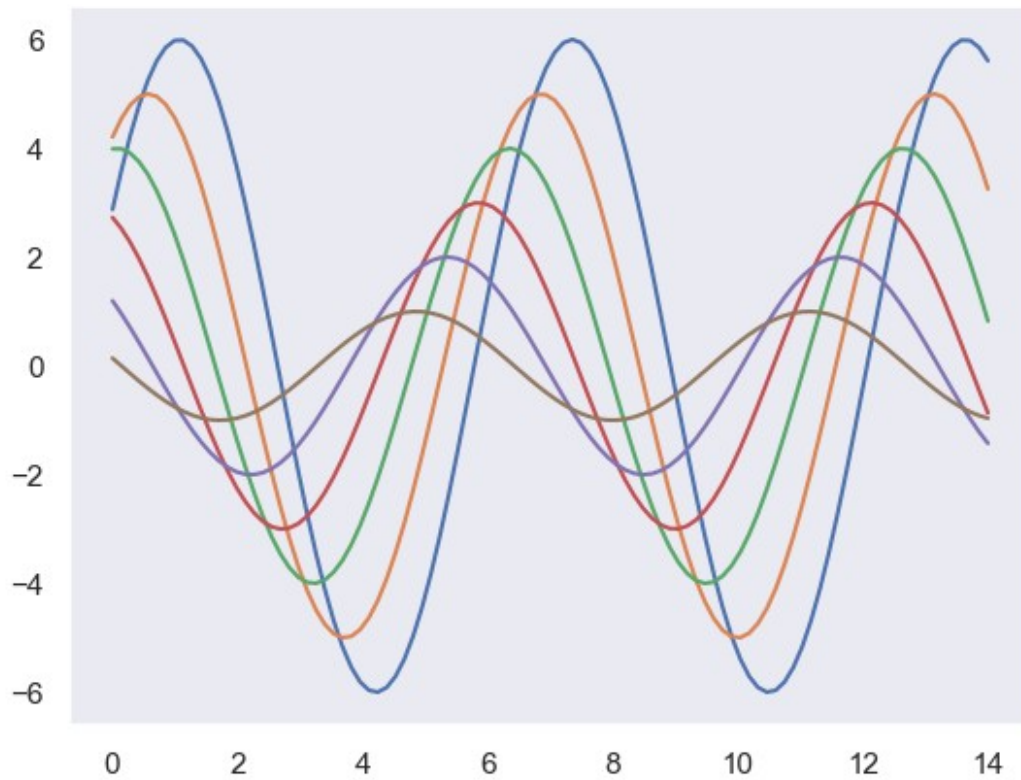
```
sns.set()  
sinplot()  
plt.show()
```



```
sns.set_style("whitegrid")  
sinplot()  
plt.show()
```

```
sns.set_style("dark")  
sinplot()  
plt.show()
```



```
sns.set_style("ticks")  
sinplot()  
plt.show()
```

